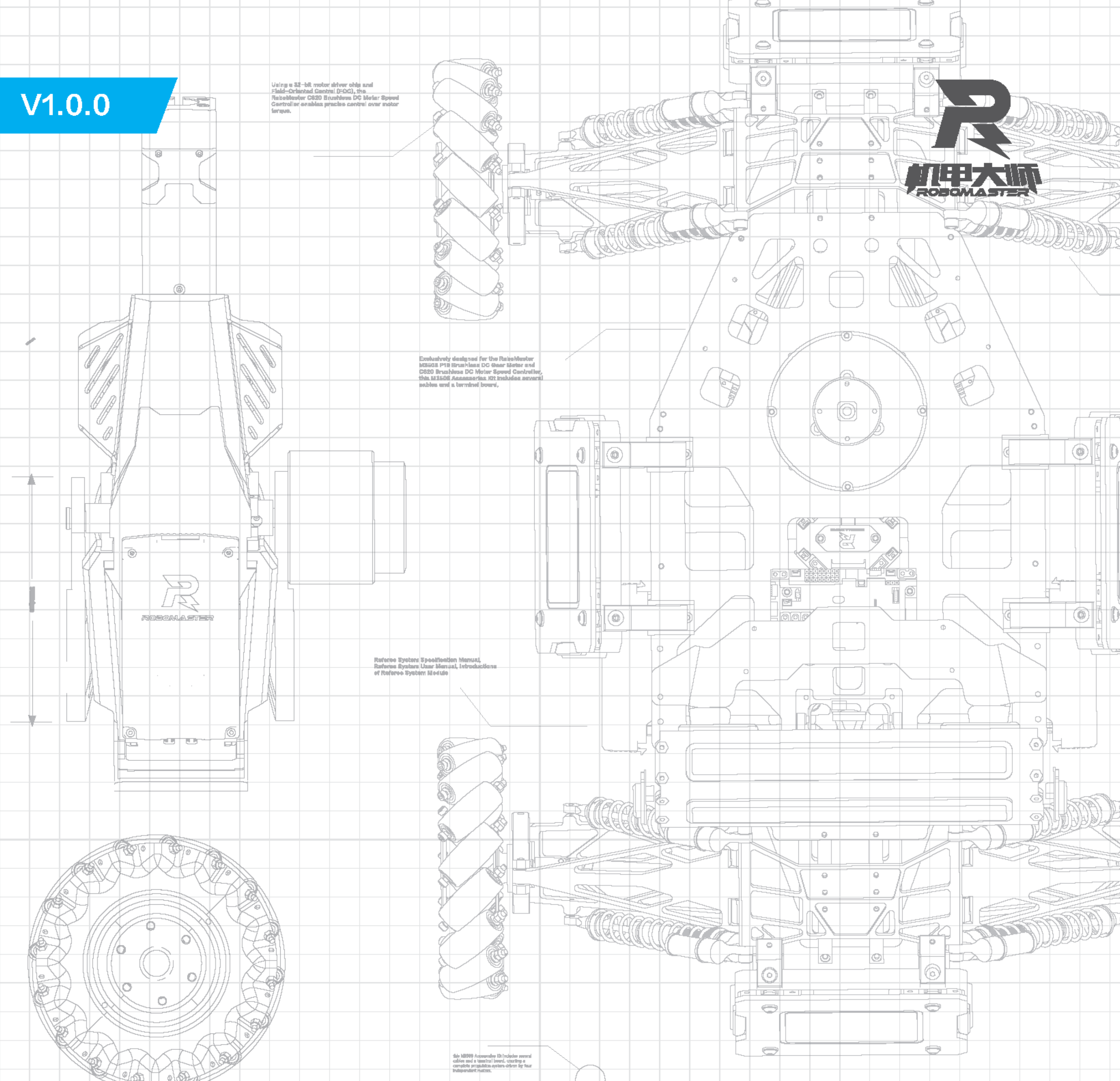
V1.0.0



The 25th China University Robot Competition

ROBOMASTER 2026

University Championship

Rule Manual

Prepared by RoboMaster Organizing Committee

Released in October, 2025

Intellectual Property Statement

The RoboMaster Organizing Committee (RMOC) encourages and promotes technical innovation and open-source technology and respects the intellectual property rights of participating teams. All rights related to the intellectual property developed during the competition are owned by individual teams. The RMOC will not participate in the handling of intellectual property disputes within the team. Participating teams must properly handle all aspects of intellectual property rights among internal school members, company members, and other members of the team.

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For suggestions regarding open-source materials, please refer to: https://bbs.robomaster.com/article/4820?source=1.

Using this Manual

The RoboMaster competition is a robotics event aimed at cultivating youth engineers. Documents related to the competition may contain sensitive terms, names, operations, and technical content. These do not reflect any existing technologies or actions in reality, nor do they intend to involve any specific country, organization, or individual. This manual is intended solely to explain the rules of the RoboMaster competition and must not be used for any other purposes. It should not be interpreted as being associated with any real events or entities.

Version Number Description

The Rules Manual adopts semantic versioning in the format of X.Y.Z, the meanings of which are as follows:

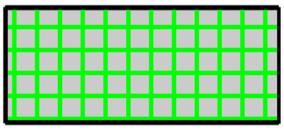
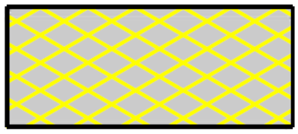
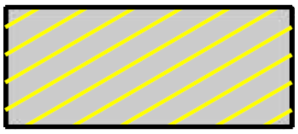
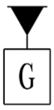
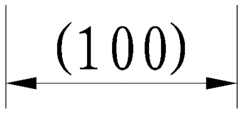
- X: Updates for different competition stages are indicated as follows: “1” for the Regional Competition and “2” for Final Tournament. The specific year is displayed in the file name and on the cover.
- Y: Updates to major content such as the Mechanism and Battlefield, for example, changes to the Competition Mechanism or to the properties of Robots.
- Z: Corrections of errors without altering the original Mechanism or Battlefield, such as fixing a typo, clarifying ambiguous wording, or replacing an incorrect figure.

Example:

- **The first version of the manual for a season (for the Regional Competition) is numbered V1.0.0.**
- **Some competition mechanisms were adjusted based on the first version (for the Regional Competition), and the updated manual is numbered V1.1.0.**

- **Three typos were corrected based on the second version (for the Regional Competition), and the updated manual is numbered V1.1.1.**
- **Some competition mechanisms were adjusted based on the third version (for the Final Tournament), and the updated manual is numbered V2.0.0.**

Legend for Battlefield Drawings

		
Buff point for both sides	Buff point for one side	Penalty zone for both sides
		
Penalty zone for one side	The battlefield is located on the lowest plane.	Dimensions are for reference only.

Release Notes

Date	Version	Revisions
2025.10.21	V1.0.0	First release

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1. Foreword

1.1 About the Competition

The RoboMaster University Championship ("RMUC"), under the China University Robot Competition, is a shooting sport-style robotics competition. Participating teams are required to design, develop and create a fleet of robots in compliance with the specifications. During a seven-minute round, each team strives to destroy the other's Base through tactical combat to win the match.

RMUC is dedicated to cultivate excellent youth engineers. Team members are encouraged to put what they have learned into practice and achieve continued growth during the process.

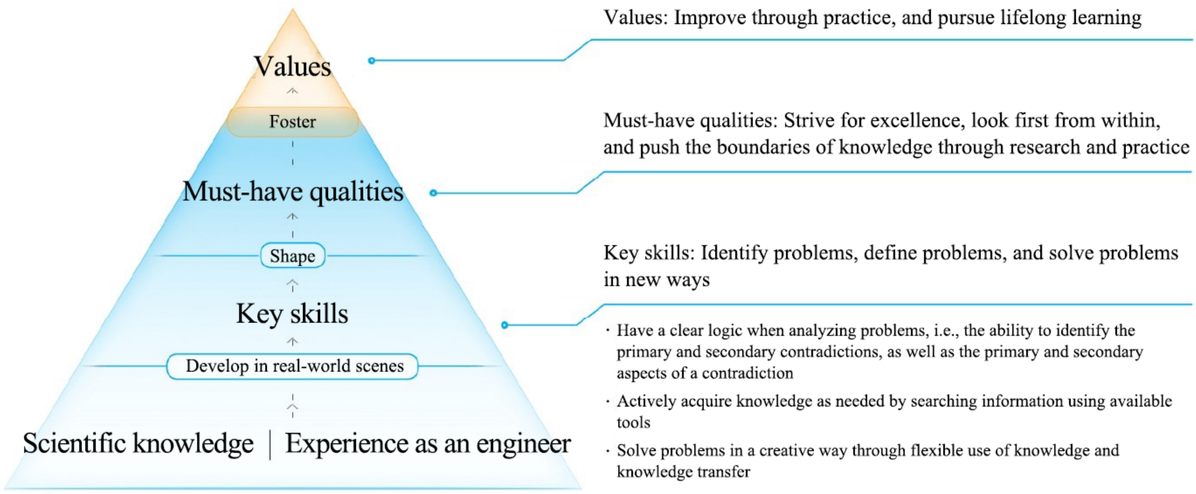


Figure 1-1 Talent Development Philosophy

1.2 Main Changes to New Season

Compared with RMUC 2025, RMUC 2026 introduces the following key changes:

Robots

- Canceled the semi-automatic control mode for Hero, Engineer, and Infantry robots, and introduced a semi-automatic control mode for Sentry robots.
- Added a new custom client.

Competition Area

- Canceled the Elevated Ground Tunnel, added a new tunnel to the Road Zone, and adjusted the location of the Trapezoid-Shaped Elevated Ground.

- Adjusted the location of Bumpy Roads.
- Removed Large Resource Island, Small Resource Island, Exchange Station, and Ore; introduced new Resource Zones, Tech Cores, and Energy Units.
- Adjusted the location of the Radar Foundation.
- Canceled the pilot headset.

Competition Mechanism

- Adjusted the mechanisms related to the Deployment Mode of Hero.
- Canceled the Engineer's task of exchanging Ores, and introduced a new task for Engineers to assemble Energy Units.
- Adjusted the Drone-related Mechanism, and introduced an Experience System for Drones.
- Adjusted the mechanisms related to Sentry.
- Adjusted gains for successful Dart hits, and added terminal moving targets.
- Adjusted the mechanisms related to Radar, and introduced the new features of countering Drones and analyzing Radar signals.
- Adjusted the experience and the performance systems.
- Adjusted the mechanisms related to Base and Outpost.
- Adjusted the Power Rune mechanism.
- Adjusted the Battlefield Buff mechanism.
- Adjusted the wireless charging mechanism.

Competition Process

- Adjusted the number of Backup Robots.

Violation Penalty

- Adjusted the maximum number of Yellow Cards for Robots.

1.3 About the Competition Rule Documents

The competition rule documents include the Rule Manual, Participant Manual, Robot Specification Manual, Referee System Serial Protocol, and supplements to these documents. The competition rule documents are applicable to all participating teams, referees, competition staff, and other partners.

- During the preparation period, the RMOC reserves the right to update the competition rules as necessary.

- During the competition period, to allow teams to fully showcase their technical achievements, the RMOC may adjust the competition rules in the following phases, but such adjustments will not involve direct structural changes to the robots.
  - During Regional Competitions: after the end of the competition for a single division.
  - During the Wild Card Competition and Final Tournament: after the end of a competition phase (e.g., Group Stage or 8/16 Knockout Stage)

The RMOC reserves the right of final interpretation over the competition rule documents. During the event, only the Chief Referee can answer queries on the competition rule documents on behalf of the RMOC. Any questions related to the competition rule documents may only be directed to the Chief Referee for consultation.

For more reference materials on RMU, please go to the Rules/Resource Hub, a site built and maintained by the RMOC that provides all the official materials required for participation to help teams efficiently access information. Materials on this site include:

- Rulebooks: rule manual, robot specification manual, participant manual, Referee System serial protocol
- Event Products: Referee System (including on-board modules and the RoboMaster Champion), robot kits & accessories, RM Assistant product details, downloads, purchase & after-sales policies
- Technical Assessment: assessment criteria, instructions for each stage
- Event Promotion Kit: promotional material package
- Event Sponsorship Guide: sponsorship guidelines
- Learning Labs: participation guide, course salon, open-source materials

1.4 Q&A

Any participating team or other relevant personnel who have questions about the competition rule documents may submit them through our official channel. The RMOC will reply to them through the following Q&A process.

1. To submit questions about the competition rule documents, please complete a questionnaire available at: https://qingflow.com/f/c5rf6rkkbs02
2. The RMOC will respond at: https://qingflow.com/appView/c5rf6rkkbs02/shareView/c5rf6slgbs02

Notes about timeliness:

- Questions submitted during the competition period will be answered by the RMOC within one calendar day.
- Questions submitted during the preparation Period will be answered by the RMOC within five business days.

- Competition Period: From the check-in day of the event until the end of the final competition day.
- Preparation Period: Any time outside of the competition period.

Note: The preparation period and competition period are calculated separately for different events and stages.

The Rules Q&A is considered as authoritative as the competition rule documents. In the case of any discrepancy between the Q&A and competition rule documents, the latest document will prevail. The Q&A for a given season applies only to that season.

2. Key Terms

In this chapter, we will provide a list of commonly used terms in the competition rules. For details on each term, please refer to the relevant section using associated keywords.

Table 2-1 Overview of Key Terms

<table>
<tr><th>Term</th><th>Definition</th></tr>
<tr><td colspan="2">Robots</td></tr>
<tr><td>Main States of Robots</td><td>Alive: The Referee System Main Control Module is normally connected to the Referee System server and the robot's HP is not zero.
Out of Combat: An alive robot has not launched a projectile or suffered any HP loss for six consecutive seconds. By default, robots are out of combat at the start of the competition.
Defeated: The robot's HP drops to zero. (Reasons include: Armor Module suffering damage from attacks or collision, Referee System Onboard Module disconnection, robot ejection, and HP losses caused by a Dart.)
Ejected: A robot is issued a Red Card.
Temporarily Activated: A robot's chassis and gimbal are powered on temporarily after the robot has been defeated. The Launching Mechanism of the robot is powered off in this period.
Abnormal Disconnection: The Referee System Main Control Module is disconnected from the Referee System server during the competition, due to a power outage on the robot or other reasons.
Note: After a robot is defeated, the Referee System will cut off the power supply to the robot (except for the Mini PC).</td></tr>
<tr><td>Ground Robot</td><td>An umbrella name for Hero, Engineer, Infantry, and Sentry.</td></tr>
<tr><td>Referee System</td><td>An electronic penalty system integrating computation, communication, and control features. It includes the onboard module, as well as the RoboMaster Champion (Server, Referee's Client, Player's Client) installed on the computer. It can monitor robots' power, projectile launches, and damage, and automatically determine the outcome according to competition rules.</td></tr>
<tr><td>Inter-robot Communication</td><td>An interaction method for robots to communicate with one another using the Referee System serial protocol.</td></tr>
</table>

Term	Definition
Robot Chassis	A mechanism that carries the robot propulsion system and its accessories, and supports the body of a robot.
Chassis Power	The power of the system that enables robots to move horizontally and rotate. For details, refer to the definition of Chassis Power in the "Referee System Mounting Specifications" section of the RoboMaster 2026 University Series Robot Specification Manual.
Launching Mechanism	A mechanism capable of launching a projectile from a robot on a fixed trajectory and at a certain initial speed.
Locking of the Launching Mechanism	A locked Launching Mechanism is powered off.
Unlocking of the Launching Mechanism	If the Launching Mechanism is unlocked, it will be powered on or off depending on its Projectile Allowance.
Launch	A robot's action of releasing projectiles or Darts. "Launch" mentioned in the following "Competition Mechanism" section is considered valid when a launch action is detected by the Referee System.
Initial Launching Speed	The initial speed measured by the relevant modules of the Referee System after a projectile or Dart has completed its acceleration using the Launching Mechanism.
Barrel Heat	Barrel heat accumulates as a robot launches projectiles. It is used to limit continuous projectile firing.
Projectile Allowance	The quantity of projectiles each robot is allowed to launch currently.
Initial HP	The HP value set by the Referee System for a robot at the start of the competition.
Current HP	A robot's real-time HP.
Maximum HP	The maximum value to which a robot's HP can be restored.
Experience Point	The accumulated points required for a robot to level up, which can be obtained in various ways.
Invincible	A state in which a robot is immune to a projectile or Dart attack or impact.
Attack	A robot's action of launching projectiles or Darts.
Destroy	Reducing the opponent's Base, Outpost, or robot to zero HP.

Term	Definition
Occupy	It refers to a situation where an alive robot has reached a Buff Point, its RFID Interaction Module has detected the RFID Interaction Card in the area or has met other conditions, and it has obtained certain special effects.
Entanglement	Mechanisms of robots are entangled with one another during the competition, i.e., one robot remains connected to the other robot and is pulled with said robot whichever direction it moves.
Collision	An active act of collision by a robot during the competition.
Battlefield	
Buff Point	A zone that, once occupied by a robot during the competition, will generate a special effect.
Penalty Zone	An area into which a robot's entry is forbidden.
Battlefield Component	Composite elements of the Battlefield, including but not limited to: Base, Outpost, and Power Rune.
Personnel	
Arbitration Commission	A body consisting of the Chief Referee and other members of the RMOC, responsible for handling appeals.
Referee	Personnel responsible for maintaining the order of the competition and enforcing its rules.
Chief Referee	The person with the final right of interpretation over the Competition Rules documents during the competition.
Head Referee	The lead referee responsible for maintaining the order of the competition and enforcing its rules.
Head Inspector	The referee responsible for leading and assigning pre-match inspection tasks, with the final right of interpretation over the inspection standards.
Participating Team	A team that has registered and been recorded in the registration system for the current competition season.
Participant	An individual who has registered and been recorded in the registration system for the current competition season.

Term	Definition
Pit Crew	Regular Members, Team Advisors and Supervisors who have registered for this Season and have been entered into the registration system, and can enter the Staging Area and Competition Area.
Operator	The Pit Crew responsible for controlling robots during the competition, including Ground Robot Operators, Gimbal Operators, and Pilots.
Offending Team	A participating team that violates the competition rules.
Offender	A participant that violates the competition rules.
Competition Process	
Competition Time	"X minutes and Y seconds" in this manual refers to a countdown of (6-X) : (59-Y). "From X minutes Y seconds to M minutes N seconds" refers to a countdown of (6-X) : (59-Y) to (7-M):(60-N). **Example:** ● **Five minutes into the round, the countdown is 1:59.** ● **Two to five minutes into the round, the countdown is 4:59–2:00.**
Round	A complete competition that includes the Setup Period, the Referee System Initialization Period, a Five-Second Countdown Period, and the competition round.
Match	Depending on the Competition System, a match may contain several rounds.
Official Technical Timeout	A technical timeout initiated by the Head Referee during the Setup Period or Referee System Initialization Period.
Team Technical Timeout	A technical timeout requested by a participating team during the Setup Period.
Regular Battle Damage	During a Three-Minute Setup Period or a Seven-Minute Round of a match, a Referee System Onboard Module experiences a fault.
Victory Determination	
Attack Damage	HP lost when a robot, Base, or Outpost is struck by projectiles or Darts. Note: HP loss caused by the imposition of penalty for the violation of one side's robot, Base, or Outpost will be counted as a part of the opponent's Attack Damage.

Term	Definition
Non-attack Damage	HP lost as a result of collision on an Armor Module or Referee System Module disconnection.
Net Base HP	It is equal to the remaining HP of one side's Base minus the remaining HP of the opponent's Base at the end of a round.
Total Remaining HP	The total remaining HP of one side's alive robots at the end of each round.

3. Robot and Operator

RoboMaster requires robots to fight together as a team with good coordination and teamwork. For robot building specifications, please refer to the RoboMaster 2026 University Series Robot Specification Manual.

The required robot line-up is as follows:

Table 3-1 Robot Line-up

Type	No.	Full Roster Quantity (sets)	Competition Stage
Hero	1	1	Regional Competition, Wild Card, and Final Tournament
Engineer	2	1	
Infantry	3/4	2	
Drone	6	1	
Sentry	7	1	
Dart System	8	1	
Radar	9	1	

Minimum line-up for the first round of each match: at least four robots, excluding Radar and Dart System.

The operator line-up is as follows:

Table 3-2 Operator Line-up

Type	Robots Operated	Full Roster Size
Ground Robot Operator	Hero	1
	Engineer	1
	Infantry	2
	Sentry	0
Gimbal Operator	Drone (optional), Dart System (optional)	1
Pilot	Drone	1

- Operators must be Regular Members of a participating team for the current season or Team Advisors.
- After the end of each round, the Operator can be replaced by a Regular Member or Team Advisor among the Pit Crew members for the current match.
- Whether the Drone and Dart System are fielded or not, the Gimbal Operator may control the Referee System player's client for the Drone and Dart System and wear the corresponding headphone, but they must remain at the designated area after the Three-Minute Setup Period.
- Pilots must obtain pilot certification and wear a pilot wristband in order to operate Drones during the match. For details, please refer to the "Technical Assessment" section of the RoboMaster 2026 University Championship Participant Manual.

3.1 Hero

Only Hero robots can launch 42 mm projectiles on the Battlefield.

Table 3-3 Key Features of Hero

Key Features	Description
Initial Zone	Starting Zone
Operating Mode	No restrictions. Up to one remote controller and one custom controller.
Inter-robot Communication	Allowed
Launching Mechanism	One Launching Mechanism (42 mm projectile)
Initial Launching Speed Limit (m/s)	12 m/s; increases to 16.5 m/s when the performance system is set to sniper-focused and enters Deployment Mode.
HP Loss and Penalty for Exceeding Limit	● Exceeding Initial Launching Speed Limit: applicable ● Exceeding Barrel Heat Limit and Cooling: applicable ● Exceeding Chassis Power Consumption Limit: applicable ● Attack Damage or Collision: applicable ● Referee System Disconnection: applicable ● Penalty Damage: applicable For details, see "5.1 HP Loss and Penalty for Exceeding Limit".
HP Recovery and Respawn Mechanism	● HP Recovery Method: Occupy the Resupply Zone or use Remote Exchange for HP. ● Respawn Method: Use Progressive Respawn or exchange for Instant Respawn. For details, see "5.2 HP Recovery and Respawn Mechanism".
Projectile Loading and Reloading	● During the Three-Minute Setup Period, 42 mm and 17 mm projectiles can be pre-loaded. The Initial Projectile Allowance is 0. ● During the Seven-Minute Round, you can either exchange for Projectile Allowance in the Resupply Zone, Base Buff Point, Outpost Buff Point, or perform a remote exchange. For details, see "5.3 Economy System".
Experience and Performance System	Applicable

Key Features	Description
	Note: Parameters such as Chassis Power Limit, Initial HP, Maximum HP, Barrel Heat Limit, and Barrel Heat Cooling Rate are related to levels, chassis types, and launching mechanisms. For details, see “5.4 Experience and Performance Systems”.
Buff Point Occupation	● Base Buff Point ● Central Elevated Ground Buff Point ● Trapezoid-Shaped Elevated Ground Buff Point ● Terrain Crossing Buff Point (Road, Elevated Ground, Launch Ramp, and Tunnel) ● Outpost Buff Point ● Resupply Zone Buff Point For details, see “5.5.3 Battlefield Buff Mechanism”.

3.2 Engineer

Engineers can be equipped with Energy Units to provide a variety of team buffs.

Table 3-4 Key Features of Engineer

Key Features	Description
Initial Zone	Starting Zone
Operating Mode	No restrictions. Up to one remote controller and one custom controller.
Inter-robot Communication	Allowed
Launching Mechanism	Not applicable
Initial Launching Speed Limit (m/s)	Not applicable
HP Loss and Penalty for Exceeding Limit	● Exceeding Initial Launching Speed Limit: not applicable ● Exceeding Barrel Heat Limit and Cooling: not applicable ● Exceeding Chassis Power Consumption Limit: applicable ● Attack Damage or Collision: applicable ● Referee System Disconnection: applicable ● Penalty Damage: applicable

Key Features	Description
	For details, see "5.1 HP Loss and Penalty for Exceeding Limit".
HP Recovery and Respawn Mechanism	● HP Recovery Method: Occupy the Resupply Zone. ● Respawn Method: Use Progressive Respawn or exchange for Instant Respawn. For details, see "5.2 HP Recovery and Respawn Mechanism".
Projectile Loading and Reloading	● During the Three-Minute Setup Period, 42 mm and 17 mm projectiles can be pre-loaded. For details, see "5.3 Economy System".
Experience and Performance System	The Initial HP or Maximum HP is 250, and the Chassis Power Limit is 120 W. Apart from this, the Experience and Performance Systems do not apply.
Buff Point Occupation	● Base Buff Point ● Trapezoid-Shaped Elevated Ground Buff Point ● Terrain Crossing Buff Point (Road, Elevated Ground, Launch Ramp, and Tunnel) ● Outpost Buff Point ● Assembly Zone Buff Point ● Resupply Zone Buff Point ● Resource Zone Buff Point For details, see "5.5.3 Battlefield Buff Mechanism".

3.3 Infantry

An Infantry robot can launch 17 mm projectiles.

Table 3-5 Key Features of Infantry

Key Features	Description
Initial Zone	Starting Zone
Operating Mode	No restrictions. Up to one remote controller and one custom controller.
Inter-robot Communication	Allowed
Launching Mechanism	One Launching Mechanism (17 mm projectile)

Key Features	Description
Initial Launching Speed Limit (m/s)	25
HP Loss and Penalty for Exceeding Limit	● Exceeding Initial Launching Speed Limit: applicable ● Exceeding Barrel Heat Limit and Cooling: applicable ● Exceeding Chassis Power Consumption Limit: applicable ● Attack Damage or Collision: applicable ● Referee System Disconnection: applicable ● Penalty Damage: applicable For details, see "5.1 HP Loss and Penalty for Exceeding Limit".
HP Recovery and Respawn Mechanism	● HP Recovery Method: Occupy the Resupply Zone or use Remote Exchange for HP. ● Respawn Method: Use Progressive Respawn or exchange for Instant Respawn. For details, see "5.2 HP Recovery and Respawn Mechanism".
Projectile Loading and Reloading	● During the Three-Minute Setup Period, 42 mm and 17 mm projectiles can be pre-loaded. The Initial Projectile Allowance is 0. ● During the Seven-Minute Round, you can either exchange for Projectile Allowance in the Resupply Zone, Base Buff Point, Outpost Buff Point, or perform a remote exchange. For details, see "5.3 Economy System".
Experience and Performance System	Applicable Note: Parameters such as Chassis Power Limit, Initial HP, Maximum HP, Barrel Heat Limit, and Barrel Heat Cooling Rate are related to levels and chassis types. For details, see "5.4 Experience and Performance Systems".
Buff Point Occupation	● Base Buff Point ● Central Elevated Ground Buff Point ● Trapezoid-Shaped Elevated Ground Buff Point ● Terrain Crossing Buff Point (Road, Elevated Ground, Launch Ramp, and Tunnel) ● Outpost Buff Point ● Resupply Zone Buff Point ● Fortress Buff Point

Key Features	Description
	For details, see “5.5.3 Battlefield Buff Mechanism”.

3.4 Drone

Drones can initiate air support, have a first-person view (FPV), and fire 17 mm projectiles from the air.

For the purpose of this manual, a Drone is considered alive while providing air support.

Table 3-6 Key Features of Drones

Key Features	Description
Initial Zone	Landing Pad
Operating Mode	No restrictions. Up to two remote controllers and one custom controller.
Inter-robot Communication	Allowed
Launching Mechanism	One Launching Mechanism (17 mm projectile)
Initial Launching Speed Limit (m/s)	25
HP Loss and Penalty for Exceeding Limit	● Exceeding Initial Launching Speed Limit: applicable ● Exceeding Barrel Heat Limit and Cooling: applicable ● Exceeding Chassis Power Consumption Limit: not applicable ● Attack Damage or Collide: not applicable ● Referee System Disconnection: applicable ● Penalty Damage: not applicable For details, see “5.1 HP Loss and Penalty for Exceeding Limit”.
HP Recovery and Respawn Mechanism	Not applicable
Projectile Loading and Reloading	● During the Three-Minute Setup Period, 17 mm projectiles can be preloaded. The Initial Projectile Allowance is 750. ● During the Seven-Minute Round, a Drone is not allowed to exchange for additional Projectile Allowance.

Key Features	Description
	For details, see “5.3 Economy System”.
Experience and Performance System	Applicable Note: Parameters such as Barrel Heat Limit and Barrel Heat Cooling Rate are related to levels. For details, see “5.4 Experience and Performance Systems”.
Buff Point Occupation	Not applicable

3.5 Sentry

Sentry robots can operate in fully automatic or semi-automatic modes and can launch 17 mm projectiles.

Table 3-7 Key Features of Sentry Robots

Key Features	Description
Initial Zone	Starting Zone
Operating Mode	Up to one Remote Controller for commissioning
Inter-robot Communication	A Sentry can send data to all other robots on its own side, but can only receive data from the Radar.
Launching Mechanism	One Launching Mechanism (17 mm projectile)
Initial Launching Speed Limit (m/s)	25
HP Loss and Penalty for Exceeding Limit	● Exceeding Initial Launching Speed Limit: applicable ● Exceeding Barrel Heat Limit and Cooling: applicable ● Exceeding Chassis Power Consumption Limit: applicable ● Attack Damage or Collision: applicable ● Referee System Disconnection: applicable ● Penalty Damage: applicable For details, see “5.1 HP Loss and Penalty for Exceeding Limit”.
HP Recovery and Respawn Mechanism	● HP Recovery Method: Occupy the Resupply Zone or use Remote Exchange for HP. ● Respawn Method: Use Progressive Respawn or exchange for Instant Respawn. For details, see “5.2 HP Recovery and Respawn Mechanism”.

Key Features	Description
Projectile Loading and Reloading	● During the Three-Minute Setup Period, 17 mm and 42 mm projectiles can be pre-loaded. The Initial Projectile Allowance is 300. ● During the Seven-Minute Round, you can exchange for Projectile Allowance in the Resupply Zone, Base Buff Point, and Outpost Buff Point, obtain it directly, or perform a remote exchange. For details, see “5.3 Economy System”.
Experience and Performance System	The Experience or Performance Systems do not apply. A robot’s performance depends on how it is operated, as detailed below: **Automatic Sentry** ● The Initial HP or Maximum HP is 400. ● The Barrel Heat Limit is 260. ● The Barrel Heat Cooling Rate is 30/second. ● The Chassis Power Limit is 100 W. **Semi-automatic Sentry** ● The Initial HP or Maximum HP are 200. ● The Barrel Heat Limit is 100. ● The Barrel Heat Cooling Rate is 10/second. ● The Chassis Power Limit is 60 W. For details, see “5.6.4 Special Mechanism of Sentry”.
Buff Point Occupation	● Base Buff Point ● Central Elevated Ground Buff Point ● Trapezoid-Shaped Elevated Ground Buff Point ● Terrain Crossing Buff Point (Road, Elevated Ground, Launch Ramp, and Tunnel) ● Outpost Buff Point ● Resupply Zone Buff Point ● Fortress Buff Point For details, see “5.5.3 Battlefield Buff Mechanism”.

3.6 Dart System

The Dart System can attack the opponent's Outpost and Base by launching Darts.

Table 3-8 Key Features of the Dart System

Key Features	Description
Initial Zone	Dart Launching Station
Operating Mode	No restrictions. Up to one remote controller.
Inter-robot Communication	Allowed

3.7 Radar

A Radar can acquire Battlefield information autonomously and sent it to robots on its own side or the Referee System player's client via inter-robot communication. A Radar can also intercept the opponent's information or counter Air Support from opponent drones.

Table 3-9 Key Features of a Radar

Key Features	Description
Initial Zone	Radar Foundation
Operating Mode	Autonomous, up to one remote controller for debugging
Inter-robot Communication	A Radar can send data to all robots on its own side, but can only receive data from Sentry.

4. Competition Area

4.1 Battlefield Overview

- Unless otherwise specified, all dimensional tolerances described in this document are within ±5%, Battlefield structure angular tolerances are within ±3°, Battlefield Component angular tolerances are within ±1°, and dimensional errors for structures smaller than 100 mm are within ±5 mm. All dimension parameters in Battlefield Instruction drawings are indicated in mm. To ensure the stability and safety of the Battlefield, adjustments may be made to specific dimensions and workmanship during actual construction. For Battlefield elements or dimensions not explicitly specified in the drawings, please refer to the actual Battlefield.
- The Battlefield has a centrally symmetrical layout. All descriptions and illustrations of Battlefield modules in this document will be based on the Red Team, with the same applying to the Blue Team. The Battlefield has a complex environment that may feature non-uniform magnetic fields, air conditioner winds, and other environmental conditions that are not fully symmetrical.
- The rendering is for illustrative purposes only. For the specific Battlefield dimensions, see the corresponding drawings.
- Refer to the actual battlefield in the Practice Match for any aspects not covered by this section.

The core competition area of the RMUC is called the Battlefield. The Battlefield is 28 m long and 15 m wide, with a wooden inner structure. Its surface material is to be determined. Some structures have metal protective layers. It mainly includes a Base Zone, an Elevated Zone, a Road Zone, and a Flight Zone. On the periphery of the Battlefield is a black steel perimeter wall with a height of 2.4 m from its upper edge to the Battlefield ground.

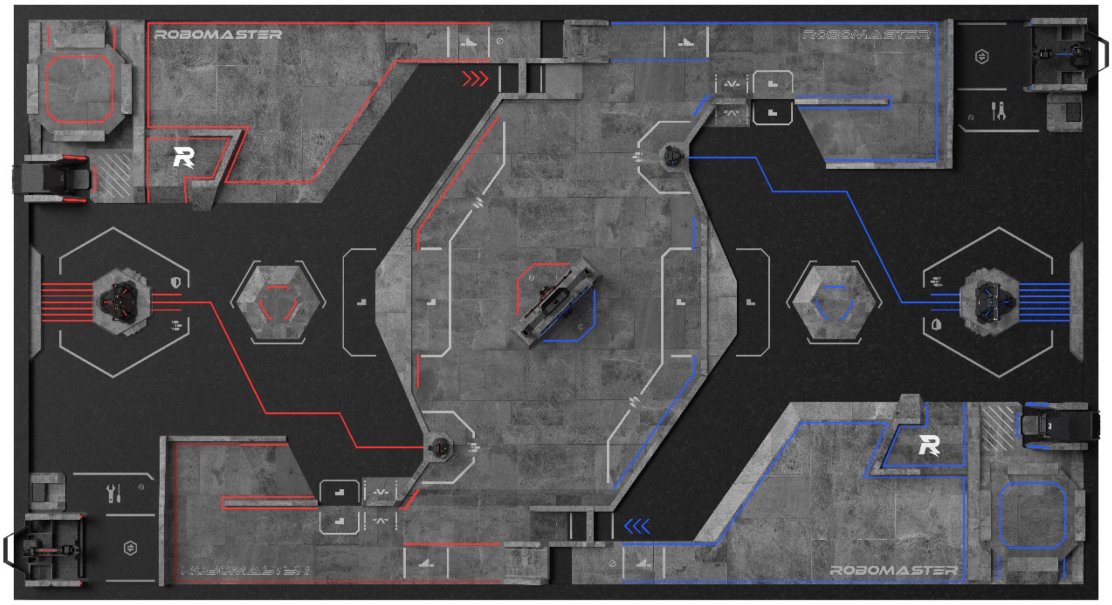


Figure 4-1 Top-view Rendering of the Battlefield

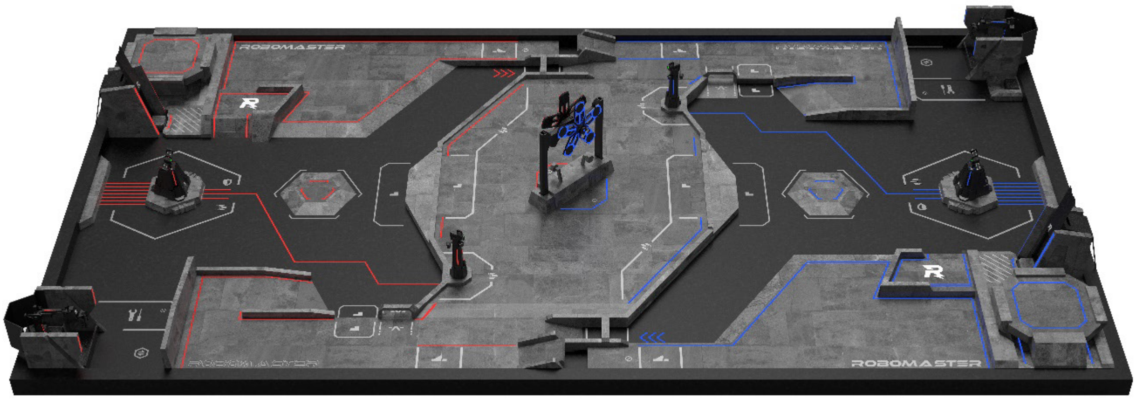
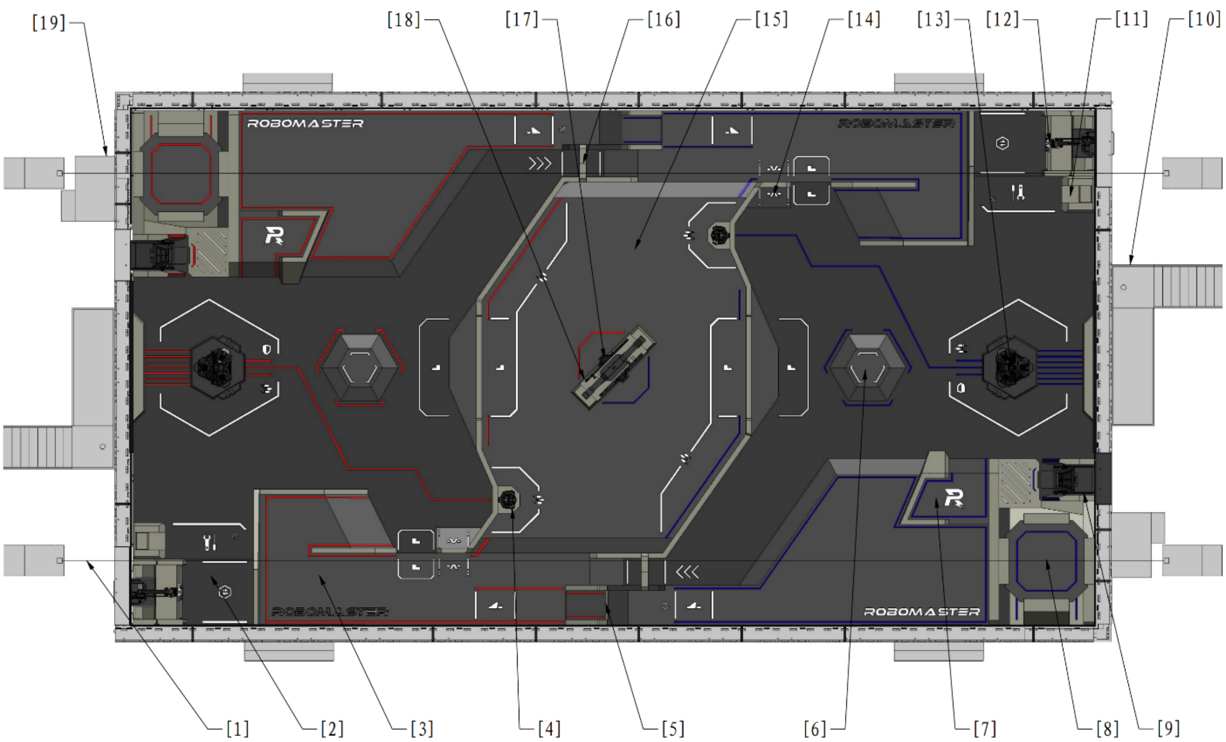


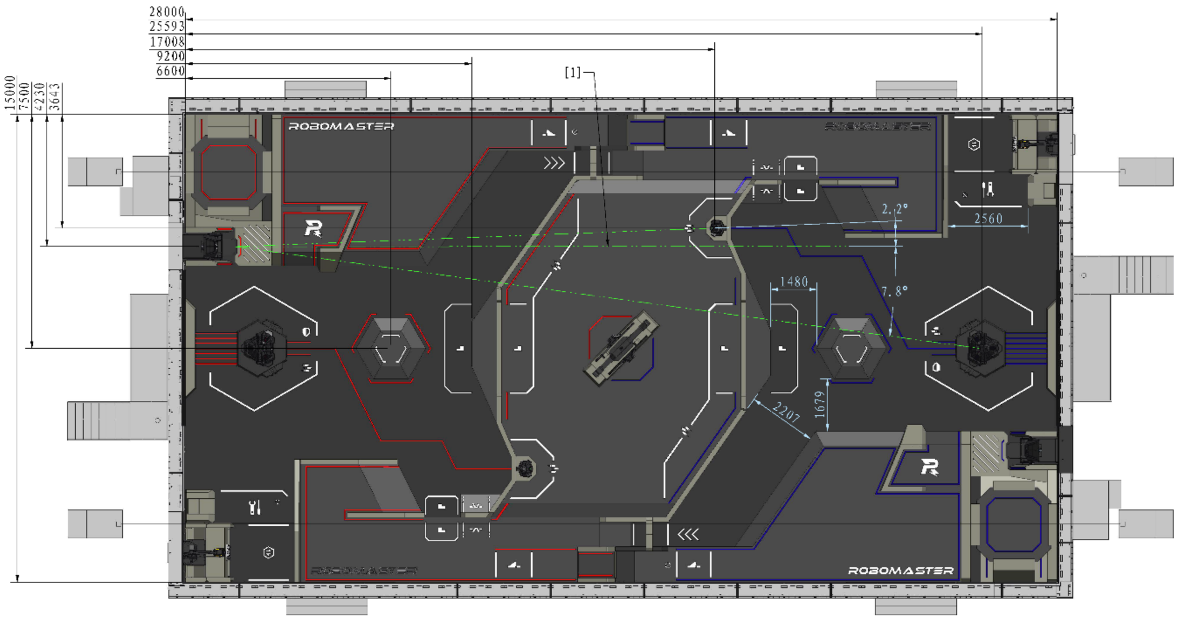
Figure 4-2 Oblique Rendering of the Battlefield

Figure 4-3 Axonometric Rendering of the Battlefield



[1]	Drone Safety Rope	[2]	Resupply Zone	[3]	Road Zone	[4]	Outpost
[5]	Launch Ramp	[6]	Fortress	[7]	Trapezoid-Shaped Elevated Ground	[8]	Landing Pad
[9]	Dart Launching Station	[10]	Radar Foundation	[11]	Wireless Charging Device Zone	[12]	Resource Zone
[13]	Base	[14]	Road Tunnel	[15]	Central Elevated Ground	[16]	Road Tunnel
[17]	Power Rune	[18]	Assembly Zone	[19]	Pilot Operation Zone		

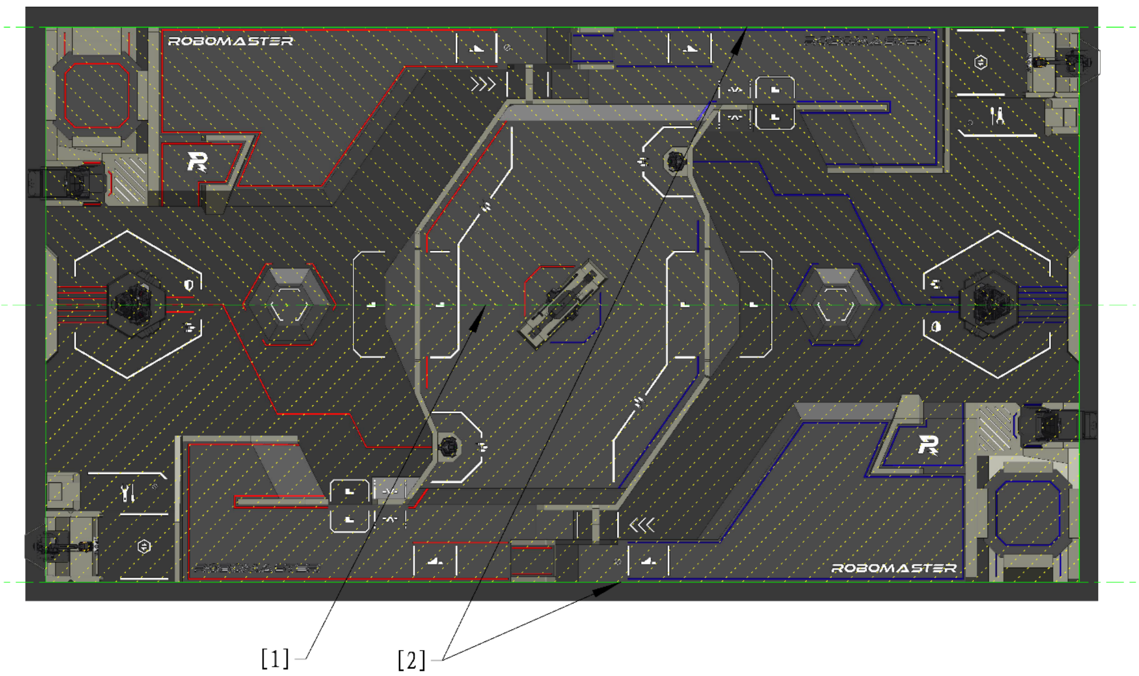
Figure 4-4 Battlefield Module Graph



[1] Parallel with the long side of the Battlefield and the orientation of the Dart Launching Station

Figure 4-5 Battlefield Module Localization Dimension

All site measurements are indicated based on the left and right inclined planes. In practice, the Battlefield floor is not level; it tilts 1° to 2° towards both long sides along the central axis of the Battlefield, as illustrated in the Figure below.

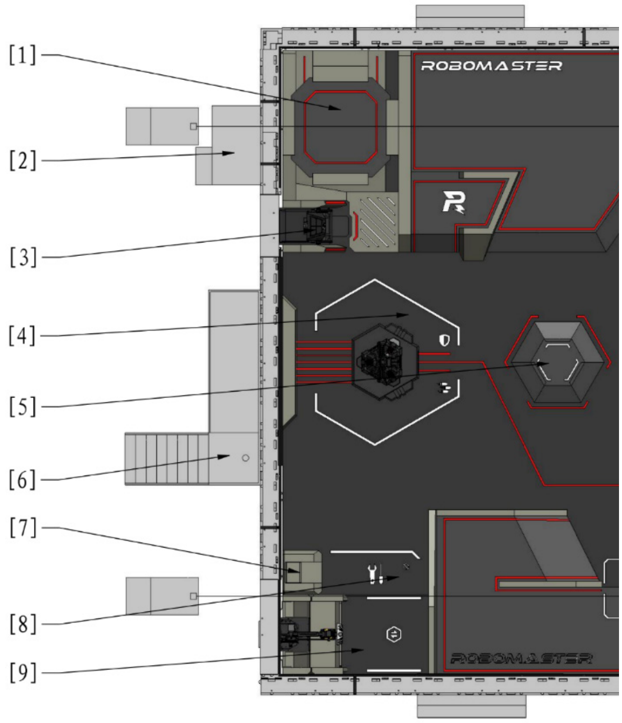


[1] Higher Area [2] Lower Area

Figure 4-6 Battlefield Inclination

4.2 Base Zone

The Base Zone consists of the Starting Zone, Base, Dart Launching Station, Landing Pad, Radar Foundation, Resupply Zone, and Fortress.

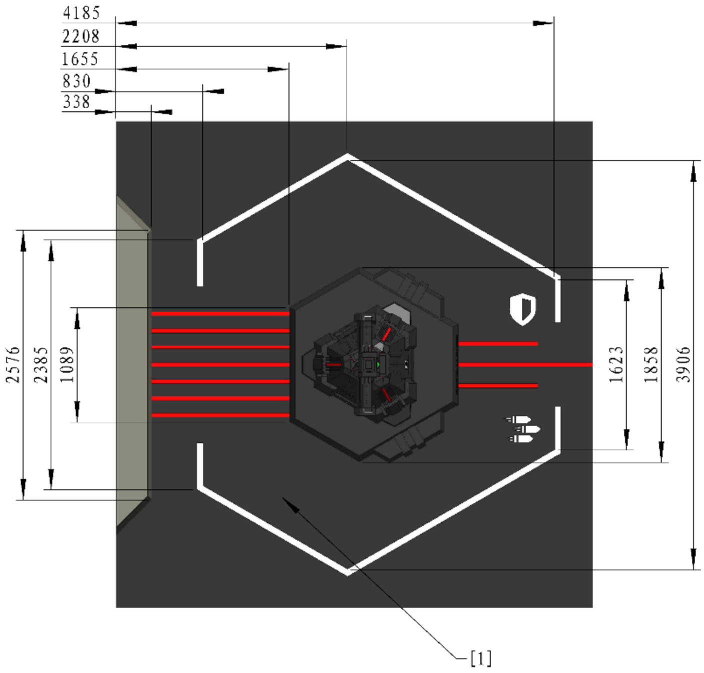


[1]	Landing Pad	[2]	Pilot Operation Zone	[3]	Dart Launching Station	[4]	Starting Zone
[5]	Fortress	[6]	Radar Foundation	[7]	Wireless Charging Device Zone	[8]	Resupply Zone
[9]	Resource Zone						

Figure 4-7 Base Zone

4.2.1 Starting Zone

The Starting Zone is a hexagonal area around the Base, designated as the area for placing Ground Robots before the match begins. The Starting Zone includes a Base Buff Point, as shown below.



[1] Starting Zone

Figure 4-8 Robots Starting Zone

4.2.2 Base

The Base is at the core of offense and defense on both sides, and is placed on the Base Foundation at the center of each team's Starting Zone. The space above the Base Foundation is the Penalty Zone of the Base.

A Base consists of the main body, Armor Modules, Dart Detection Module, Base Protective Armor, etc. The Base Protective Armor has two states: closed or expanded.

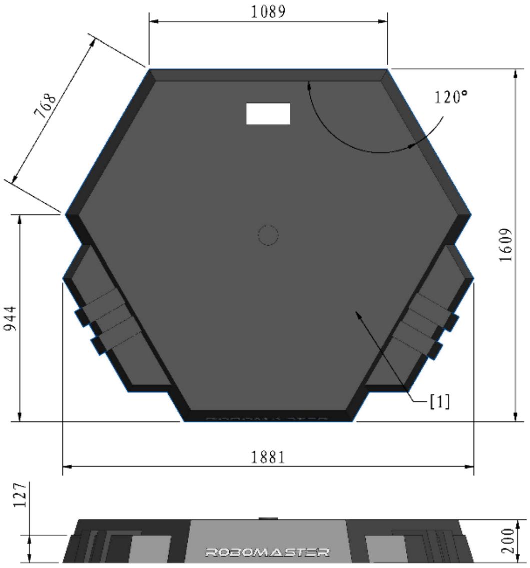
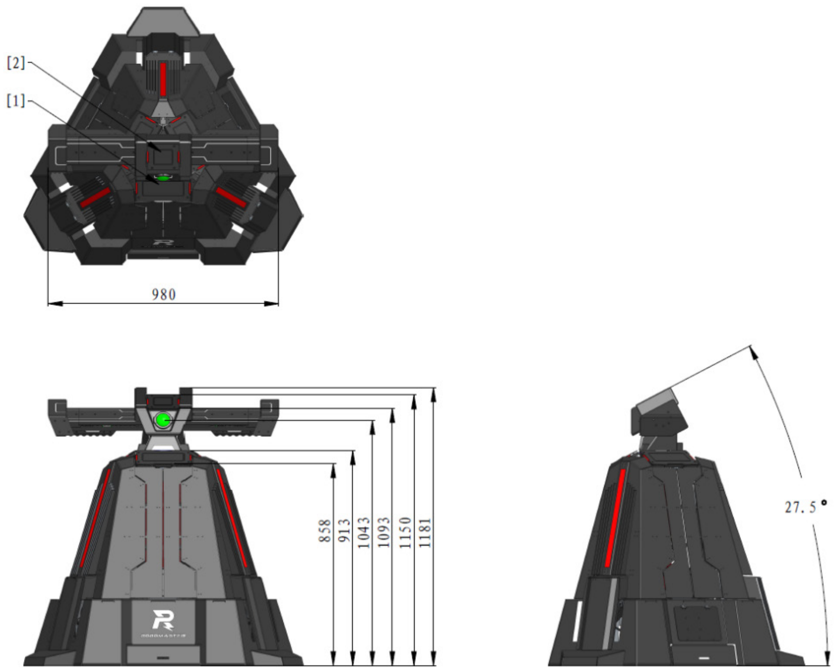


Figure 4-9 Base Foundation



[1] Large Armor Module [2] Dart Detection Module

Figure 4-10 Closed Base Protective Armor

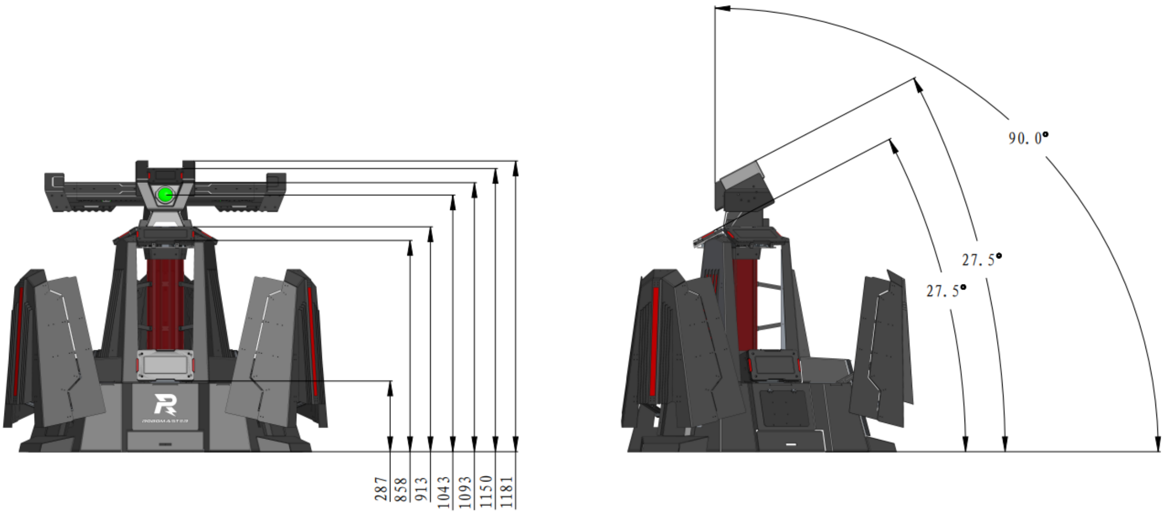
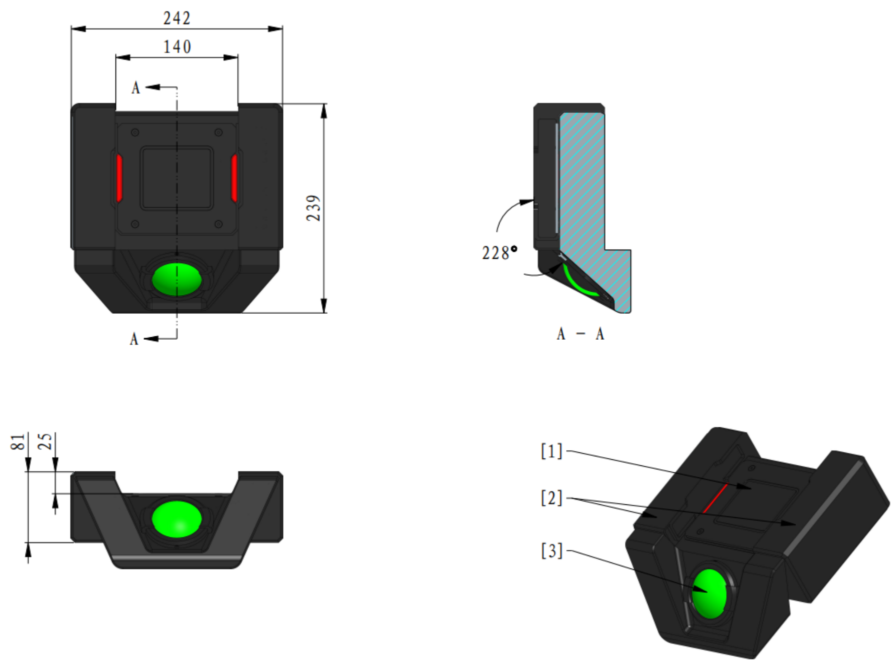


Figure 4-11 Expanded Base Protective Armor

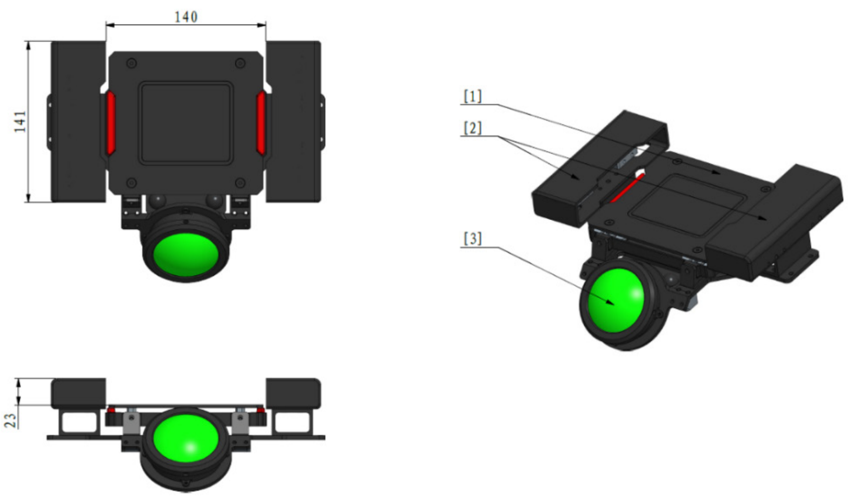
The Dart Detection Module is located on the top of both the Base and Outpost, consisting of a Small Armor Module, Dart detection sensor and Dart guiding light.

The Small Armor Module of the Dart Detection Module can detect attacks from both Darts and 42 mm projectiles. The Dart Detection Sensor can detect the infrared light emitted by a Dart Trigger. When the projection of the Dart Trigger onto the Dart Detection Module falls entirely within the Small Armor Module, and the Dart Detection Module simultaneously detects an infrared beam and a Strike, the system determines the module has been hit by a Dart. If only a Strike is detected, the system considers the module to have been hit by a projectile. The Dart Guiding Light emits green visible light with a wavelength of 520 nm. When approximating a point light source, its luminous intensity is around 10 cd, and the diameter of its light-emitting part is about 55 mm. It is for guiding Darts toward their targets.

If the projection of the Dart Trigger onto the Dart Detection Module does not fall entirely within the Small Armor Module, there is still a certain chance that the Dart Detection Module will successfully recognize it.



Base Dart Detection Module



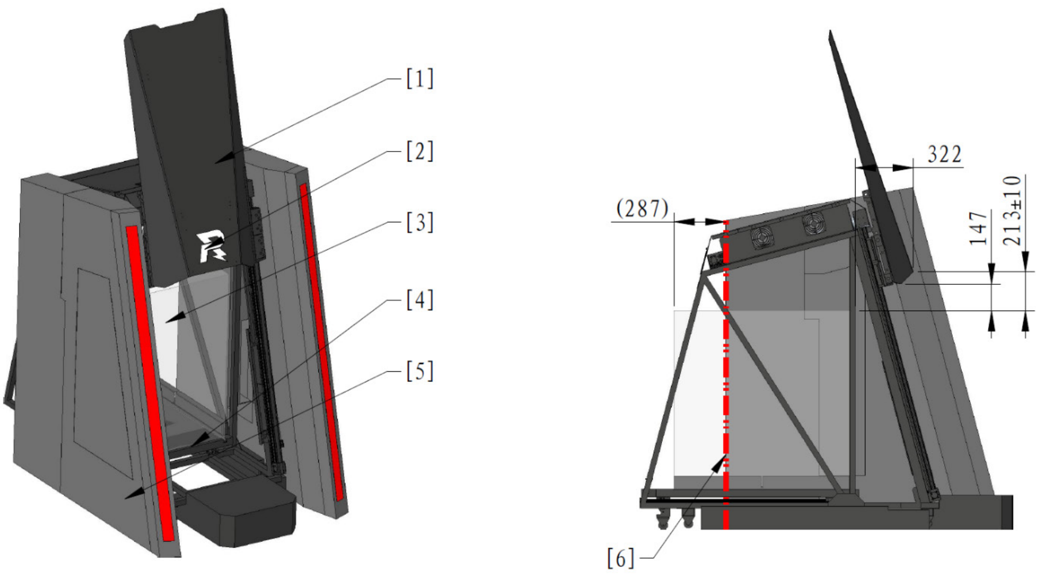
Outpost Dart Detection Module

[1] Small Armor Module [2] Dart Detection Sensor [3] Dart Guiding Light

Figure 4-12 Dart Detection Module

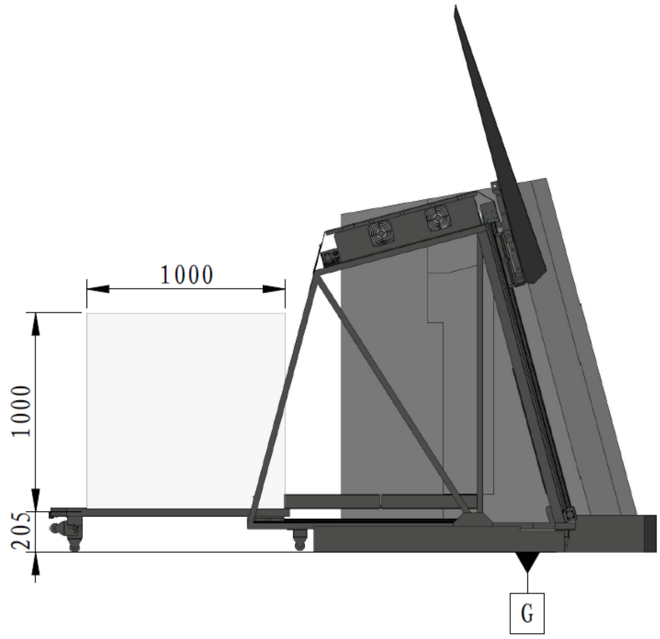
4.2.3 Dart Launching Station

The Dart System can only be placed in the Dart Launching Station, which consists of the main body, gliding platform and gate. A lighting device is installed inside the Dart Launching Station and cannot be turned off. During the Referee System Initialization Period, the gate of the Dart Launching Station opens for about 5 seconds. Throughout this time, the Dart Detection Module remains static at the Fixed Target position at the Base. During the Three-Minute Setup Period and the Fifteen-Second Referee System Initialization Period, the Dart Guiding Lights at the Outpost and Base are illuminated.



[1]	Gate	[2]	Status Indicator	[3]	Space for placing the Dart System
[4]	Gliding Platform	[5]	Main body of Dart Launching Station	[6]	Battlefield edge

Figure 4-13 Dart Launching Station



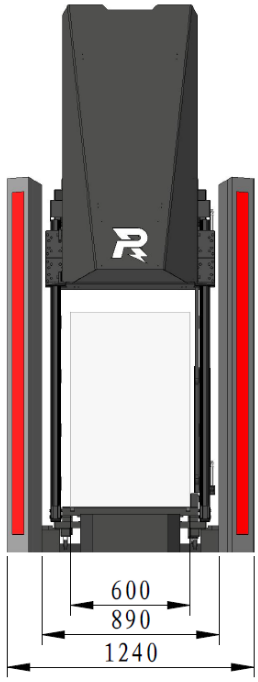
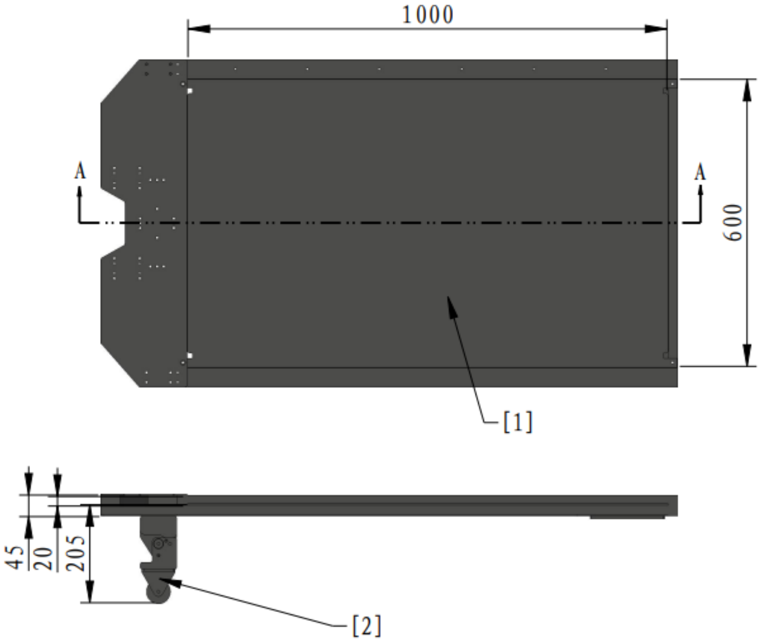


Figure 4-14 Gliding Platform Extension

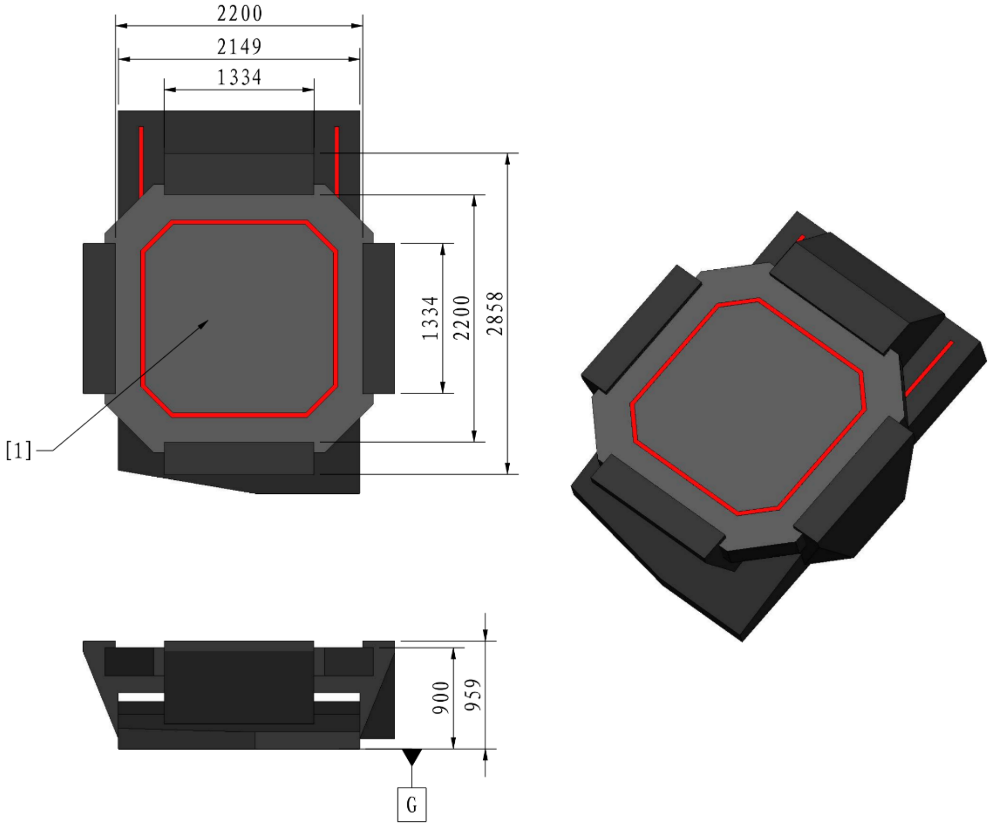


[1] Surface for placing the Dart System [2] Supporting Wheel

Figure 4-15 Gliding Platform Dimensions

4.2.4 Landing Pad

Prior to the start of the match, the Drone must be placed on a Landing Pad platform, its projection must be within the boundaries of the Landing Pad platform, and it must be connected to a Drone Safety Rope in accordance with guidelines. The Landing Pad of one team is the Landing Pad Penalty Zone for the robots of the other team.



[1] Landing Pad Platform

Figure 4-16 Landing Pad

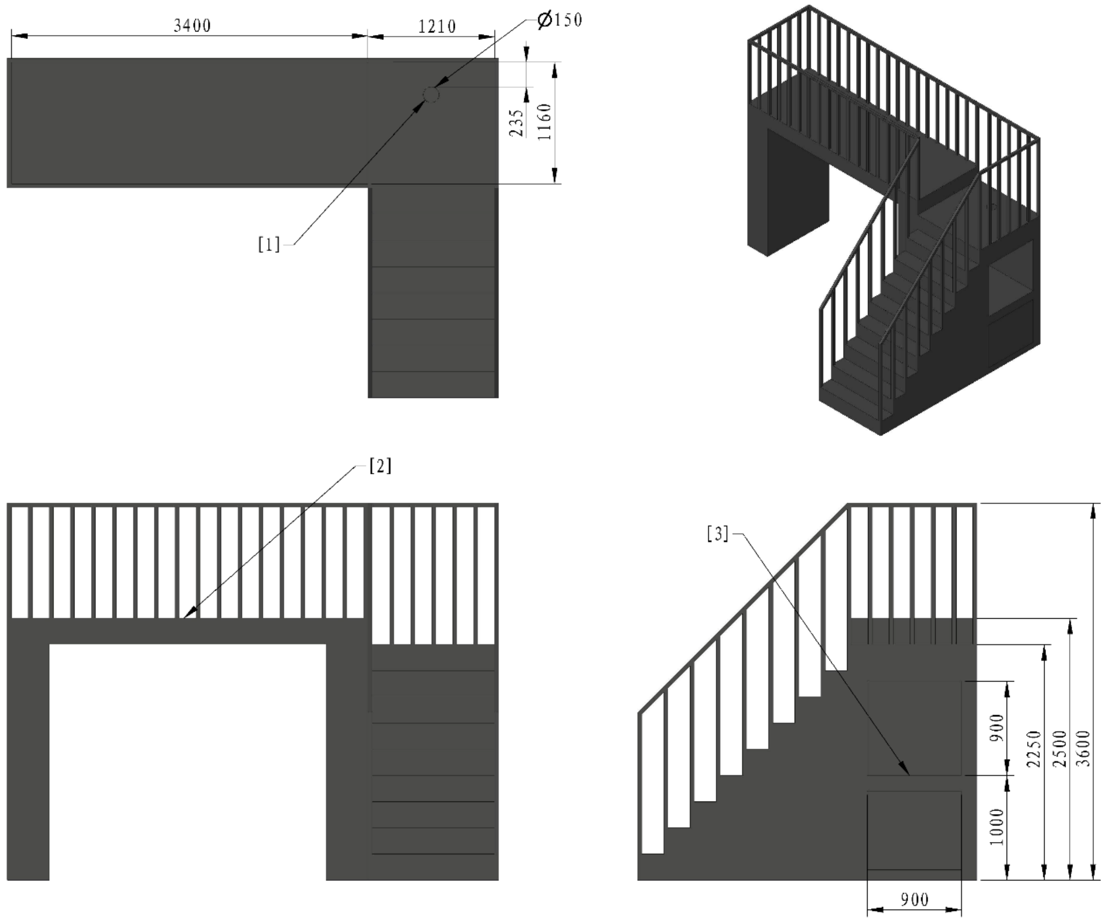
4.2.5 Radar Foundation

The Radar Foundation is used to place the Radar Sensor and laser launch device, featuring a top platform that measures 3.4 m × 1.16 m. The center of the Radar Foundation is aligned with the central axis of the Battlefield. The platform surface is approximately 2.5 m above the Battlefield ground and is surrounded by a 1.1-meter high fence. On the platform are sensor data cable slots, which may be used as needed during the competition.

The Radar Computing Platform is powered by 220 V mains supply. Its platform has the following:

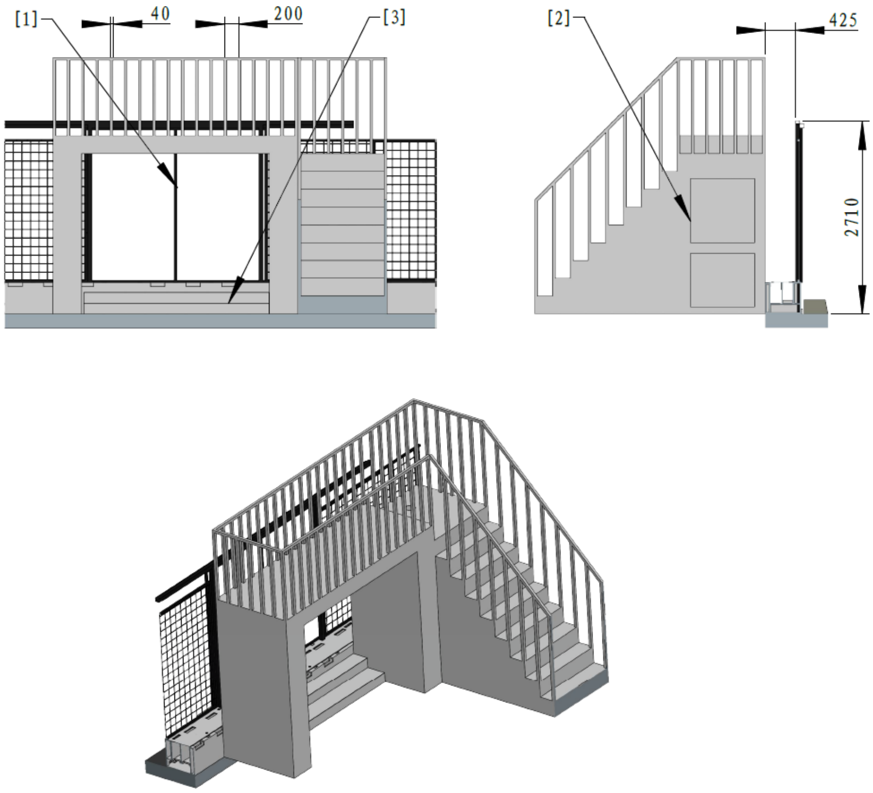
- One official display device that only supports HDMI signal input, with a resolution of 1920×1080. Participating teams may use it to confirm the operational status of the Radar.

- One HDMI cable for connecting the Radar to the official display device.
- One fixed 5-hole socket compliant with China's national standard GB/T 2099.7-2015, with both single-phase two-pin and single-phase three-pin outlets available for simultaneous use. This socked is designated for supplying power to the Radar.



[1]	Sensor data cable slot	[2]	Radar Sensor platform	[3]	Platform for placing the Radar Computing Platform

Figure 4-17 Radar Foundation

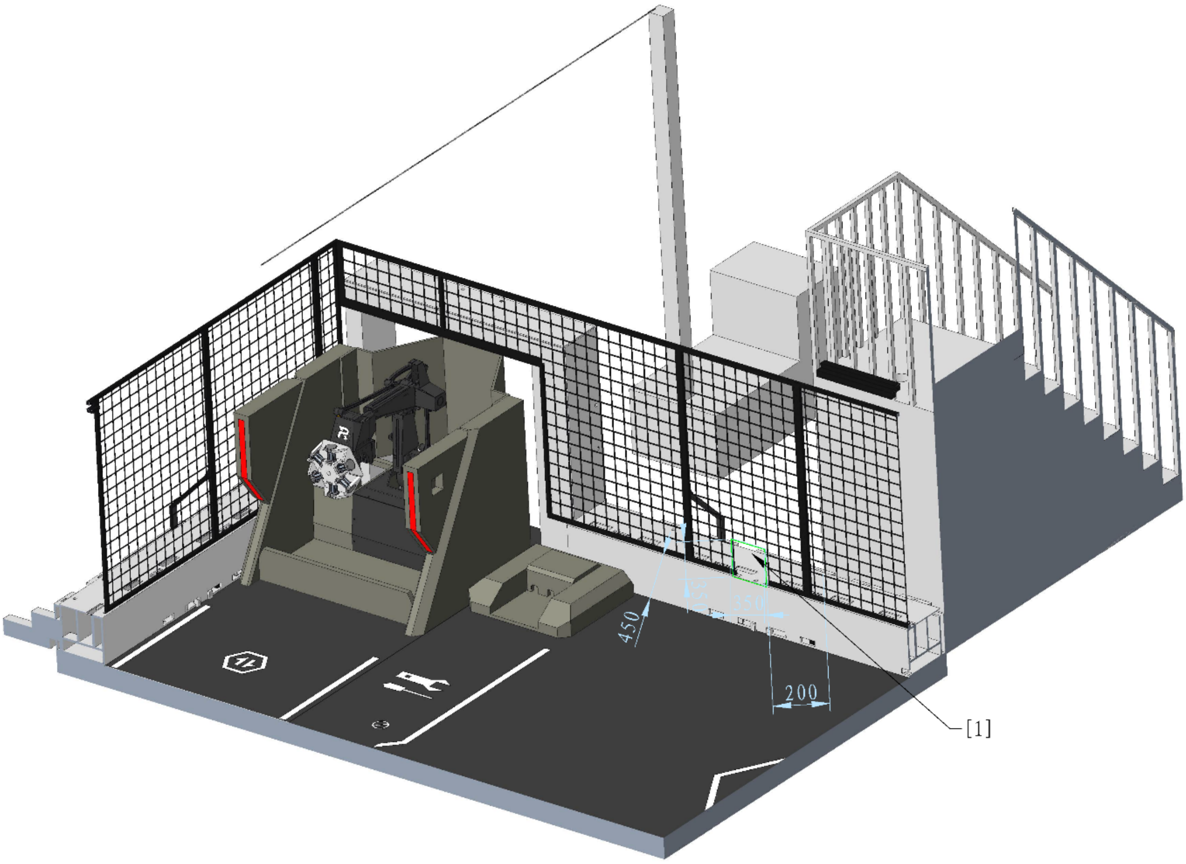


[1] Battlefield Entrance & Exit [2] Radar Foundation [3] Steps

Figure 4-18 Relative Location of the Radar Foundation

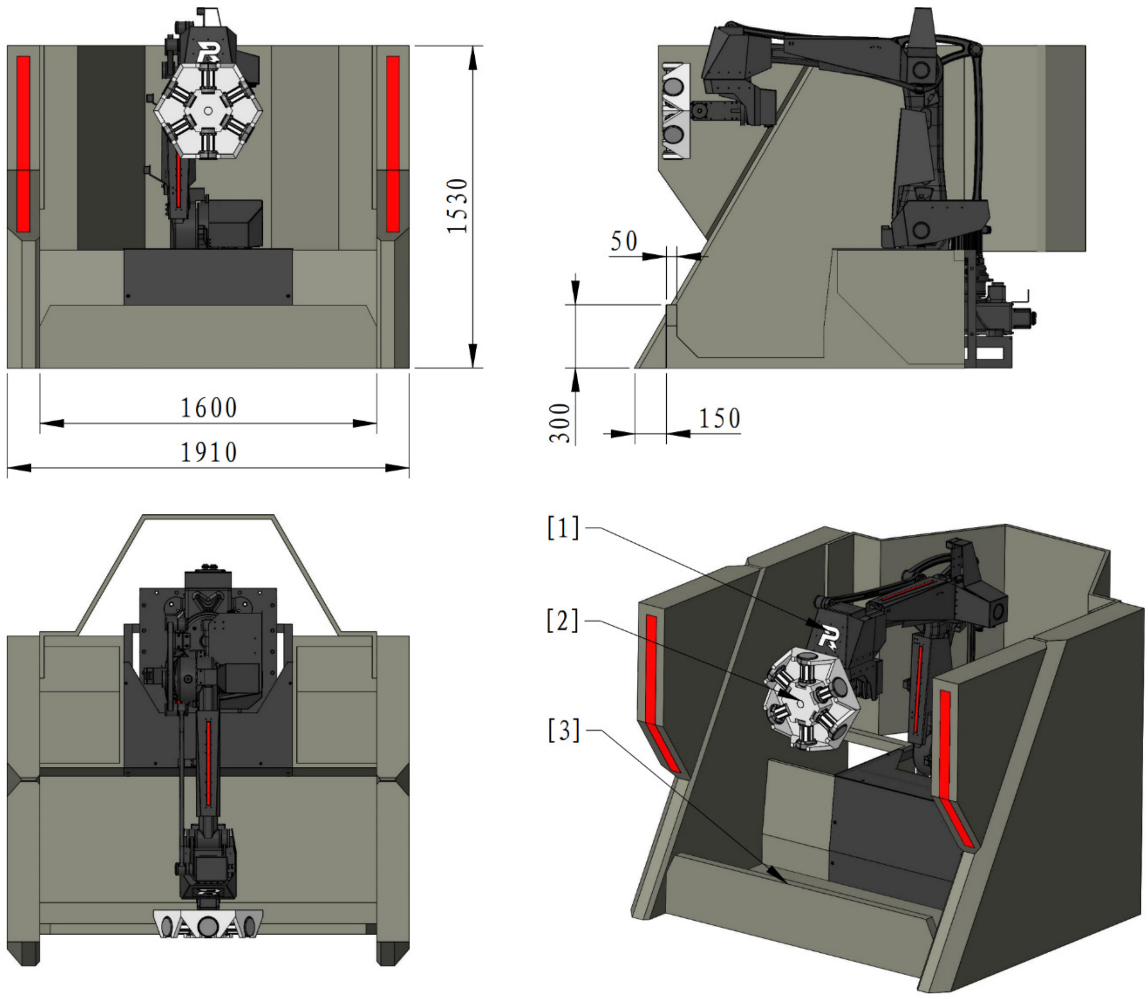
4.2.6 Resupply Zone

The Resupply Zone includes the Resource Zone, Resource Zone Buff Point, Resupply Zone Buff Point, and Wireless Charging Device Zone. It is a key area for robots to perform wireless charging, restore health, replenish their Projectile Allowance, and obtain Energy Units. One team's Resupply Zone is a Penalty Zone for the other team's robots. A Dart delivery window is set in the Perimeter Wall at the edge of the Resupply Zone, allowing robots to pick up Darts from their own side and pass them through the Dart Delivery Window to team members outside the Battlefield for repairs, as shown in the following figure.



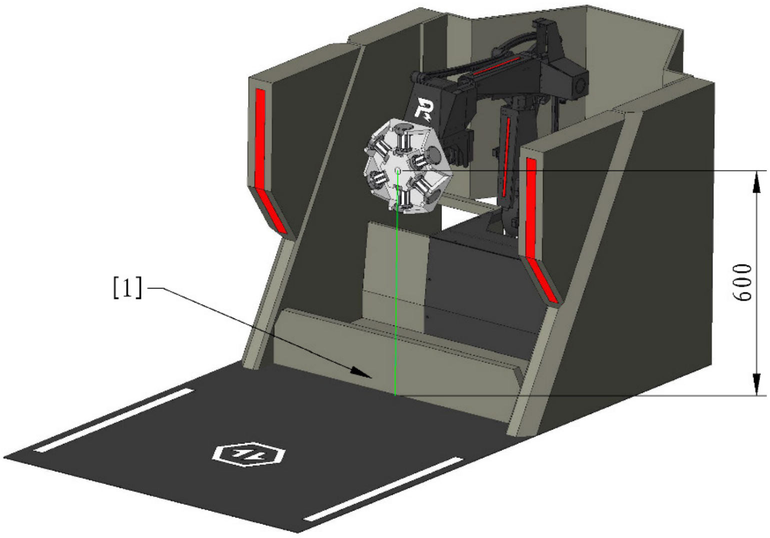
[1] Dart Delivery Window

Figure 4-19 Resupply Zone



[1] Status Indicator [2] Energy Unit Placement Zone [3] Guard

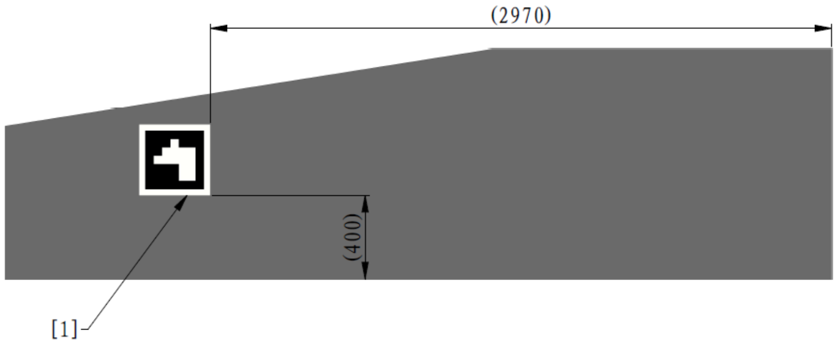
Figure 4-20 Resource Zone



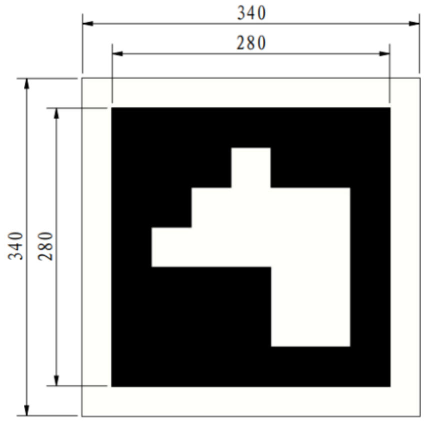
[1] Perimeter Wall

Figure 4-21 Initial Location for Placing Power Units

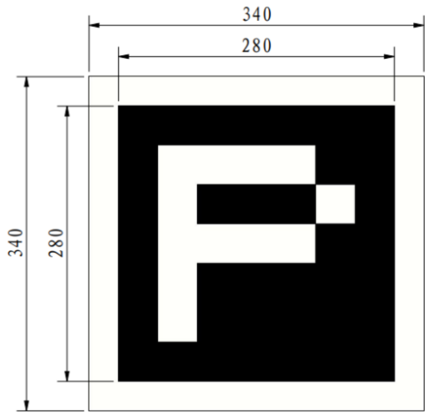
The wall of the Road Zone facing the Resupply Zone contains a metal visual marker, as shown below.



[1] Visual Marker



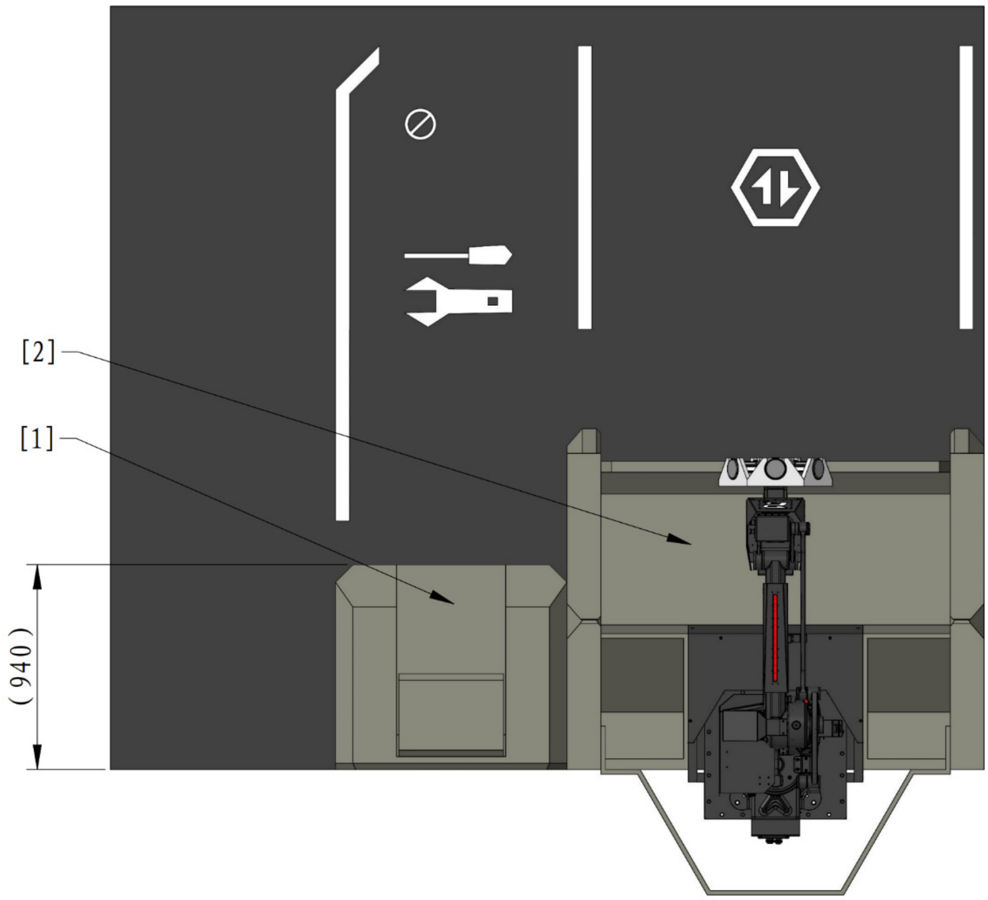
Red Team's Visual Marker



Blue Team's Visual Marker

Figure 4-22 Visual Marker

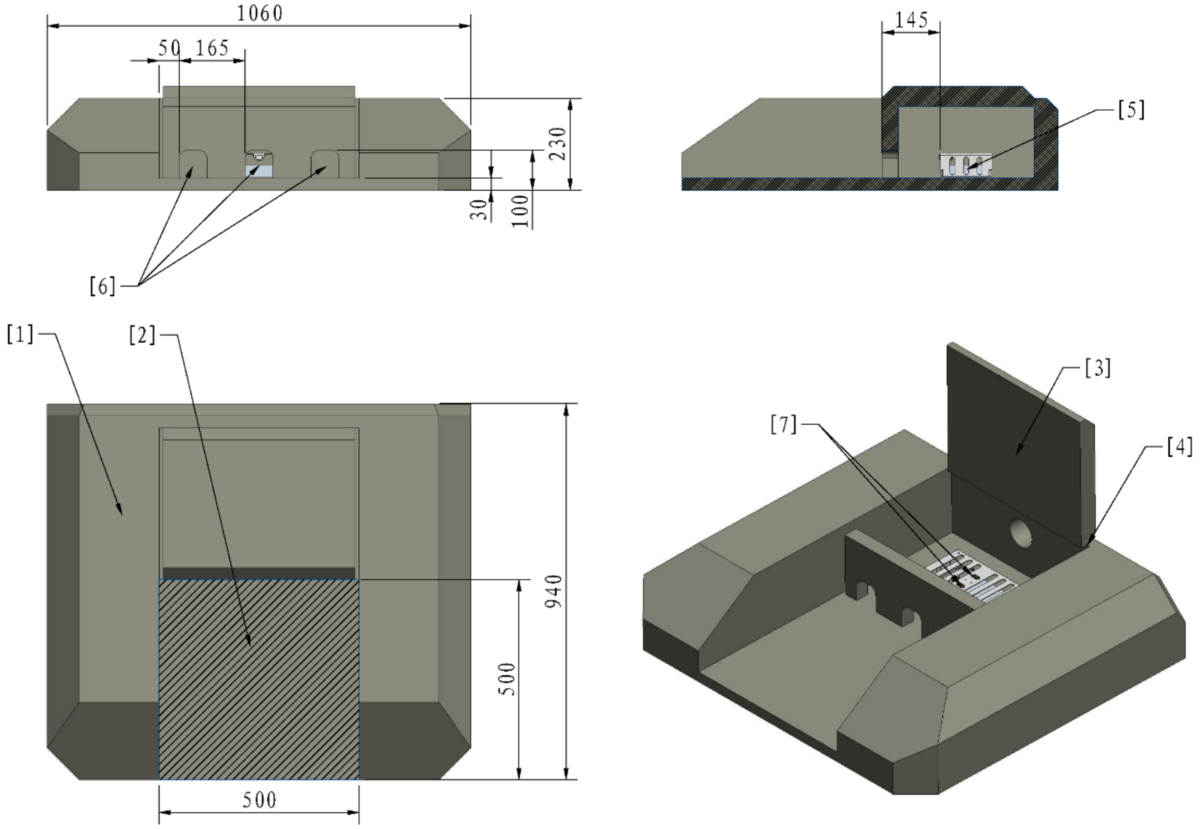
The wireless charging device is placed in the Resupply Zone. Participating teams may install their wireless charging devices (transmitters) in the designated area during the Three-Minute Setup Period, but they must ensure the projection of the devices remains within the area. The specifics are shown in the figure below:



[1] Wireless Charging Device Zone [2] Resource Zone

Figure 4-23 Wireless Charging Device Zone

The RMOC has installed a DC switching power supply with a power supply voltage of 24 V ± 1 V and a maximum output of 200 W in the Wireless Charging Device Zone, using an XT60 receptacle as the power supply interface.

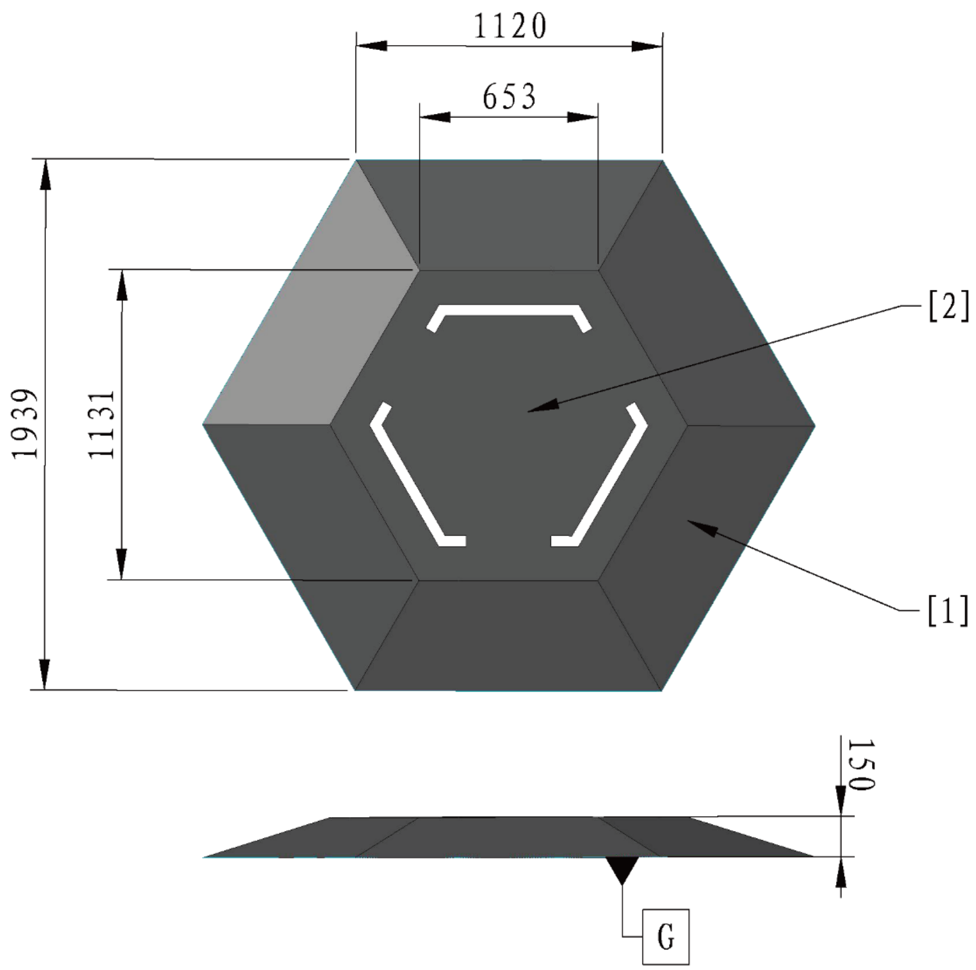


[1]	Perimeter wall	[2]	Wireless Charging Device Zone	[3]	Cable management flip cover
[4]	Rotating shaft of flip cover	[5]	Power supply	[6]	Cable pass-through hole
[7]	XT60 receptacle port				

Figure 4-24 Wireless Charging Device

4.2.7 Fortress

The Fortress is the key defensive area of the Base Zone, and a Buff Point is set within the Fortress.



[1] 20° slope [2] Fortress Buff Point

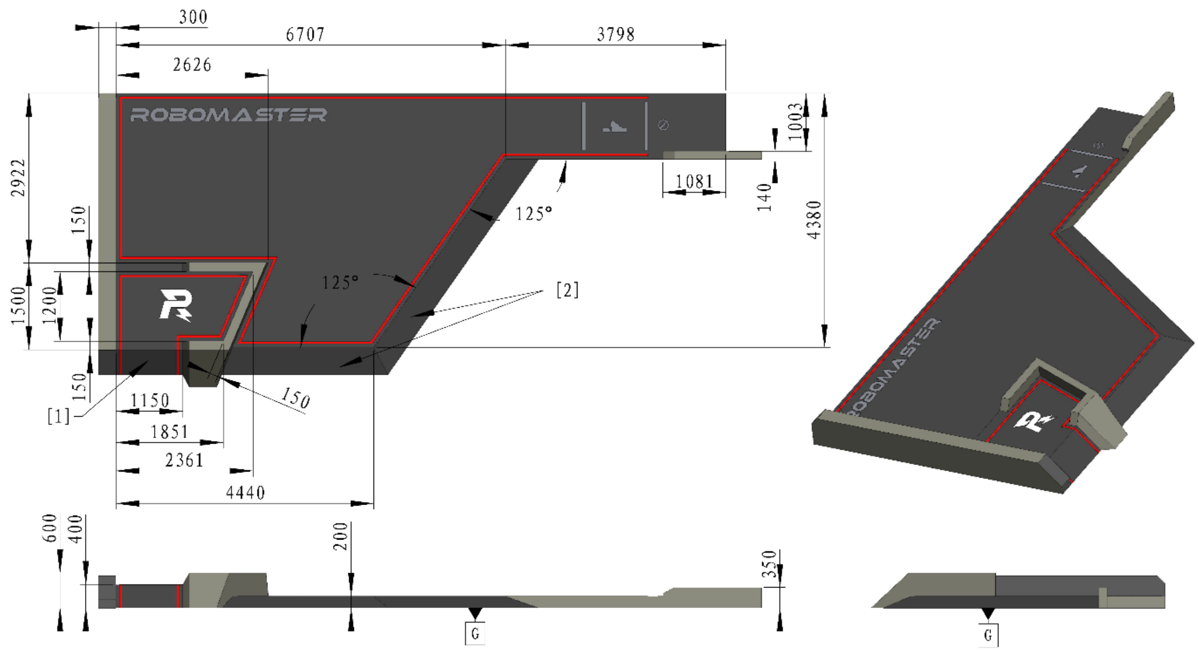
Figure 4-25 The Fortress

4.3 Elevated Zone

The Elevated Zone refers to the elevated areas in the Battlefield that are higher than the Battlefield floor. On each half of the Battlefield, there is one Trapezoid-Shaped Elevated Ground designated for the red and blue teams, respectively. Additionally, there is a Central Elevated Ground located at the center of the Battlefield. These Elevated Grounds divide the Battlefield into different zones and create a three-dimensional space.

4.3.1 Trapezoid-Shaped Elevated Ground

The Trapezoid-Shaped Elevated Ground is located near the Landing Pad, with a height of 200 to 400 mm above the Battlefield surface. A Buff Point is located on the Trapezoid-Shaped Elevated Ground, as shown in the figure below.

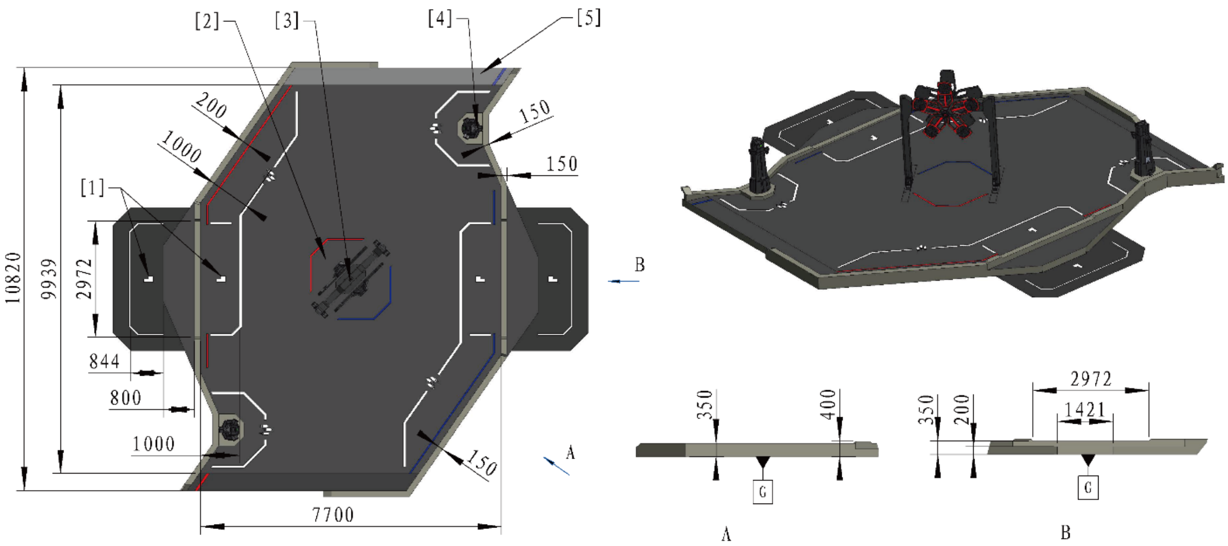


[1] 43° slope [2] 23° Slope

Figure 4-26 Trapezoid-Shaped Elevated Ground

4.3.2 Central Elevated Ground

The Central Elevated Ground is located at the center of the Battlefield, with both ends connected to the Road Zone by slopes. The Central Elevated Ground Buff Points and Terrain Crossing Buff Points (Elevated Ground) are located on the Central Elevated Ground, with no overlap between these Buff Points, as shown below.



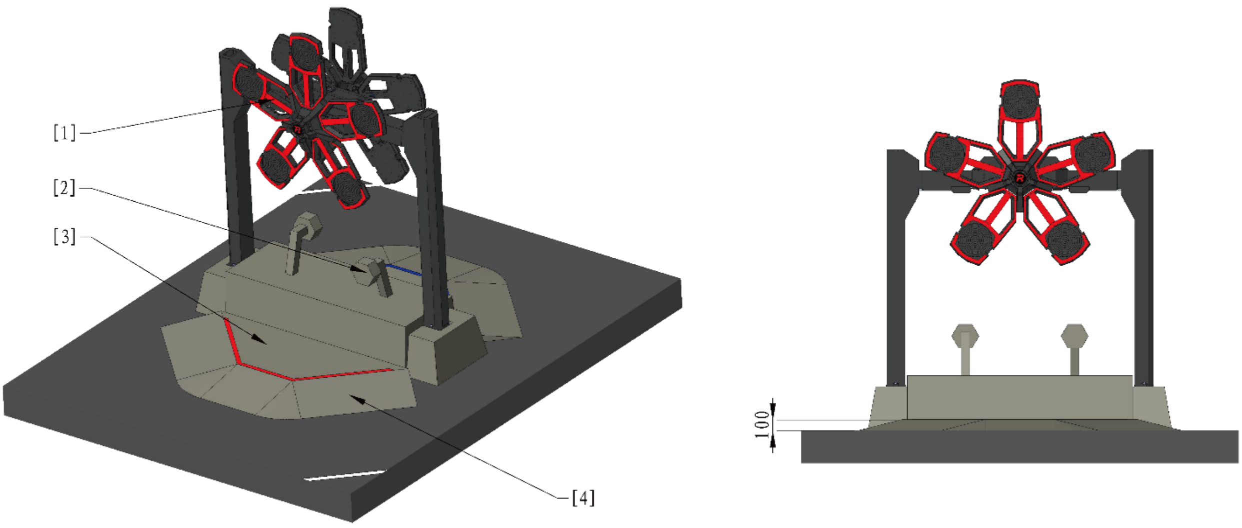
[1]	Terrain Crossing Buff Points (Elevated Ground)	[2]	Assembly Zone	[3]	Power Rune
[4]	Outpost	[5]	10.5° slope		

Figure 4-27 Central Elevated Ground

4.3.2.1 Assembly Zone

The Assembly Zone is located at the center of the Battlefield, directly beneath the Power Rune. The Assembly Zone is divided into red and blue zones, each equipped with a Buff Point. One team's assembly area serves as a Penalty Zone for the other team. A Tech Core is positioned directly below the Power Rune, and Engineer robots can assemble Energy Units onto the Tech Core.

The graphs of the Tech Core and Assembly Zone is for reference only and will be updated later.



[1] Power Rune [2] Tech Core [3] Assembly Zone [4] 10° slope

Figure 4-28 Axonometric View of the Assembly Zone

4.3.2.2 Power Rune

The Power Rune is located directly above the Assembly Zone. The Power Rune is powered by the motor and rotates synchronously according to a specific pattern. The Power Rune is divided into red and blue sides, with one side being the red team's Power Rune and the other side the blue team's Power Rune.

- The Power Rune will have a slight dip in the middle due to its weight. The dip is around 0–50 mm.
- Due to the viewing angle and transmission gap, a team may see parts of the other team's Power Rune.

The Power Rune consists of five evenly distributed light arms. Their locations and dimensions are shown below:

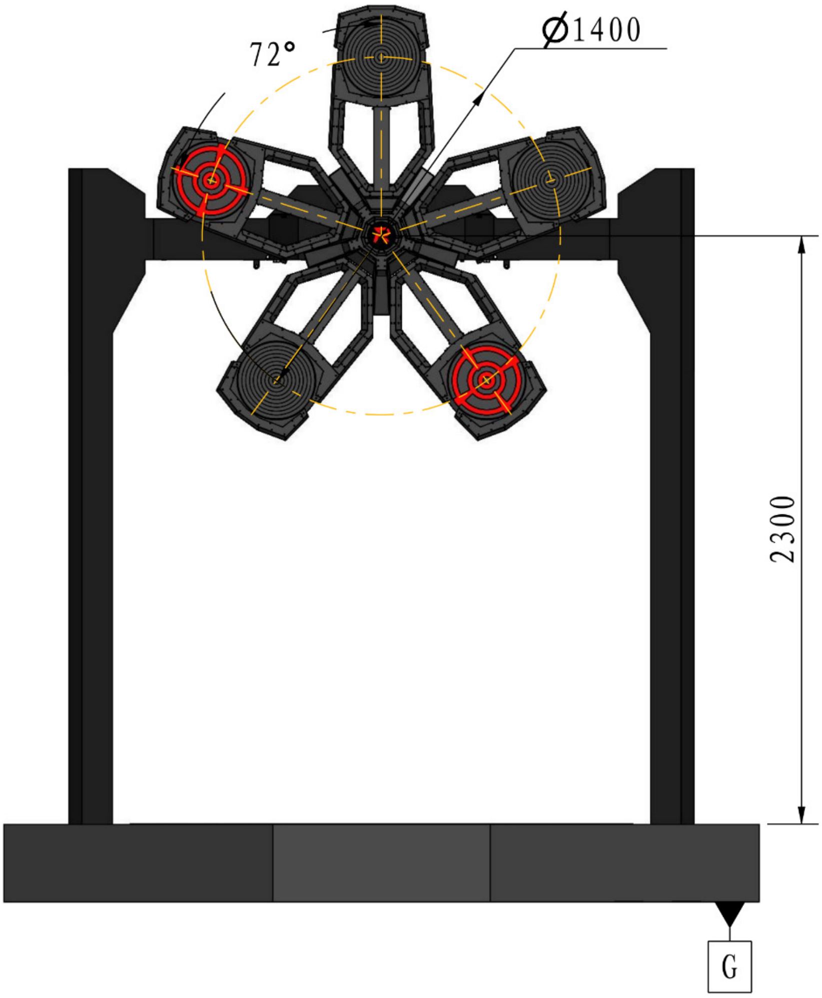


Figure 4-29 Power Rune

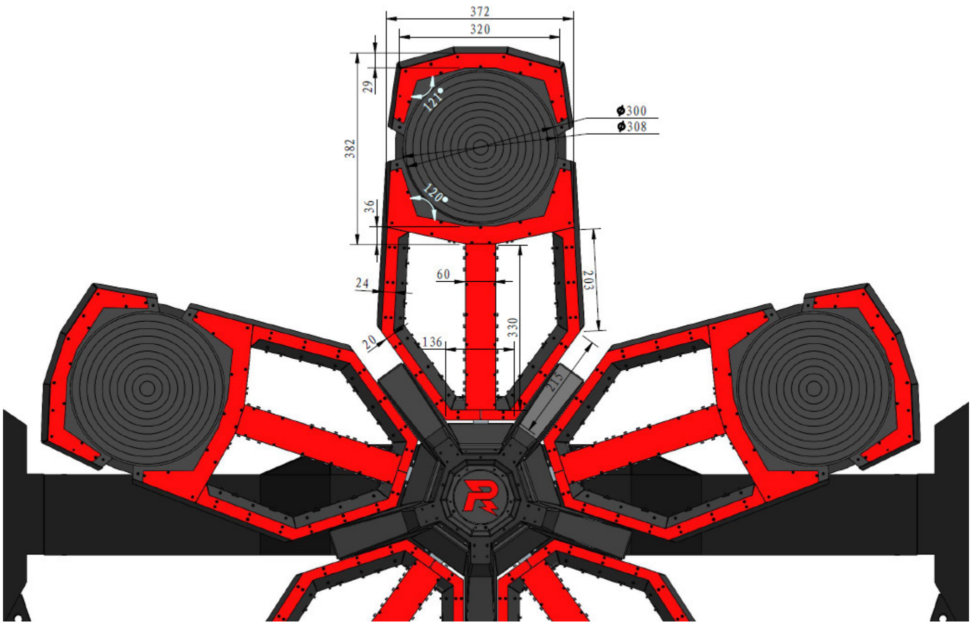


Figure 4-30 Power Rune Light Arm Dimensions

The actual maximum diameter of a light arm's round target is 308 mm, and its effective detection diameter is 300 mm.

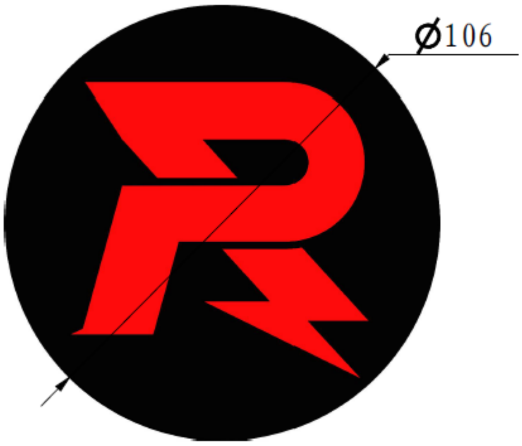
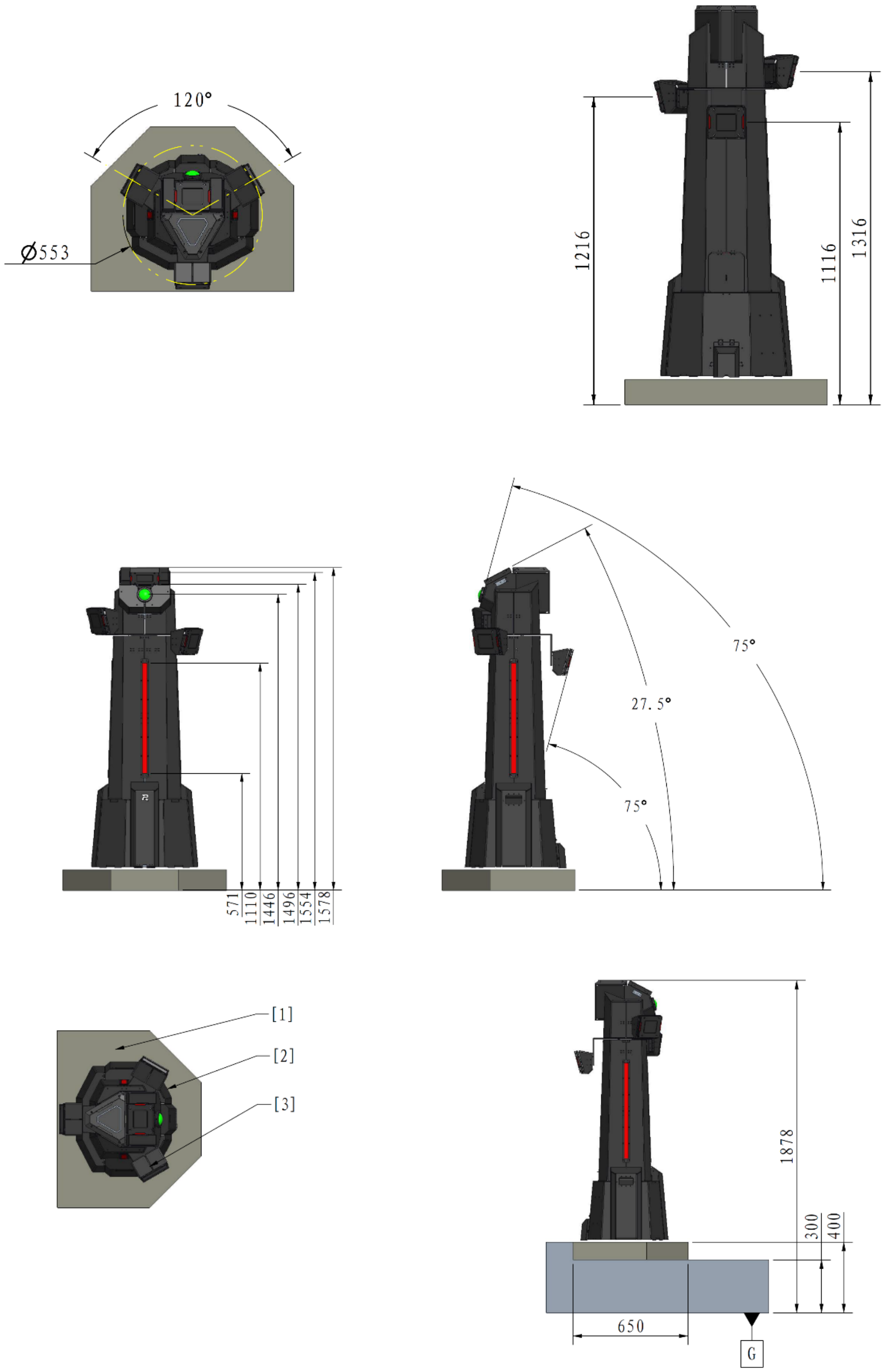


Figure 4-31 Power Rune Central Logo Dimensions

4.3.2.3 Outpost

The Outpost is located on the Outpost Foundation near the Road Launch Ramp and consists of its main body, an Armor Module, and a Dart Detection Module. Please refer to the Dart Detection Module graph in "Figure 4-12 Dart Detection Module". The Outpost Buff Point is located around the Outpost and its specific location is shown below.



[1] Outpost Foundation [2] Outpost [3] Middle Armor

Figure 4-32 Outpost

4.3.2.4 Bumpy Roads

Bumpy Roads are located in the Road Zone, with bumps arranged at regular intervals and in the direction shown in the figure below.

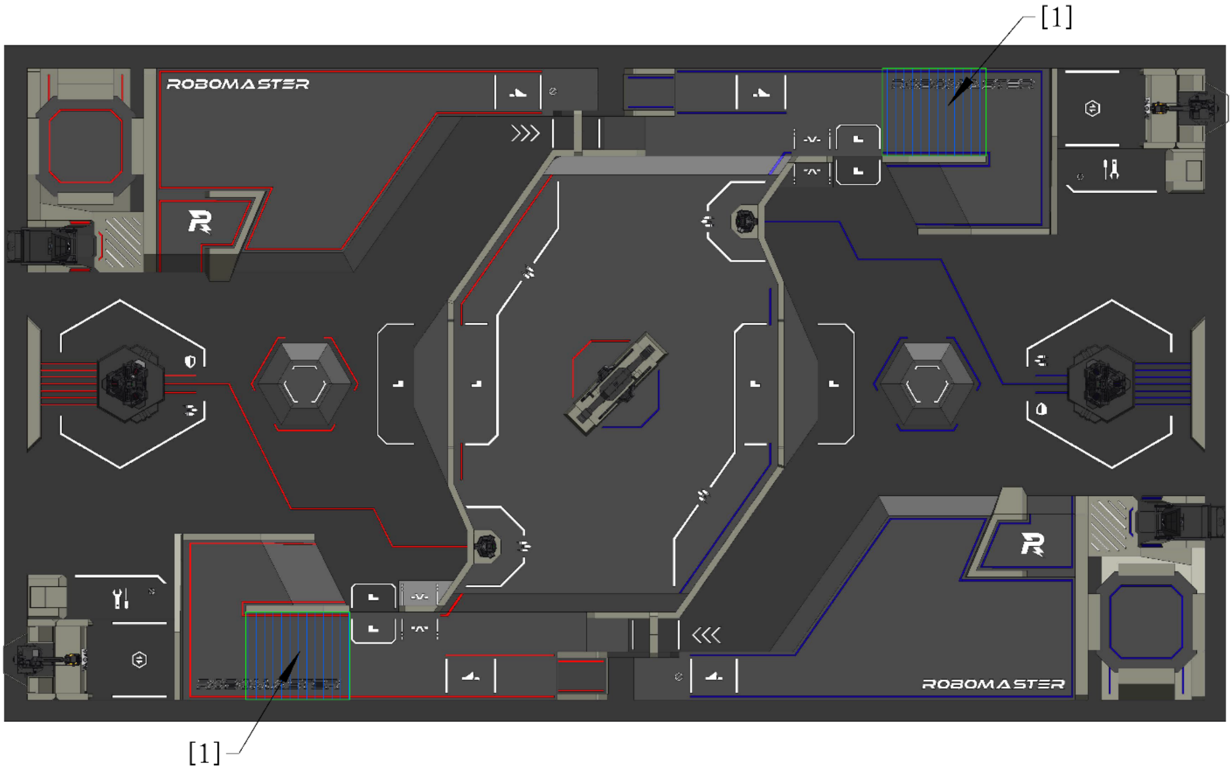
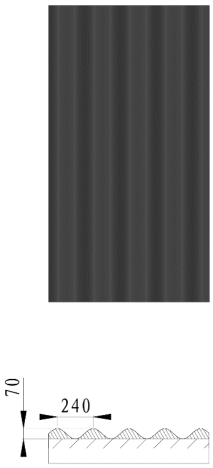


Figure 4-33 Bumpy Roads

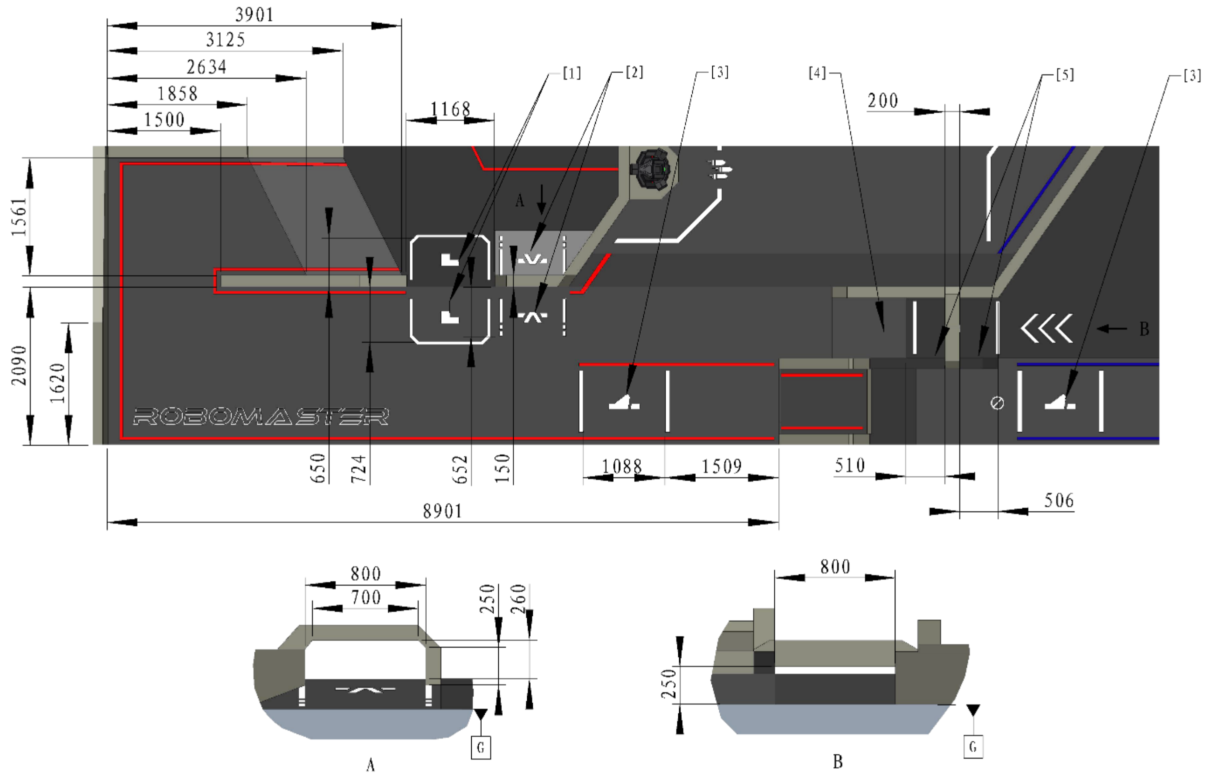


[1] Bump

Figure 4-34 Bump

4.4 Road Zone

The Road Zone includes the Road, the Launch Ramp, and the Tunnel. Each Road has two Terrain Crossing Buff Points (Launch Ramp), which are located respectively on the Roads in front of and behind the Launch Ramp. Part of the road is designated as the Road Penalty Zone, where robots from both sides are prohibited from entering except when using the Launch Ramp.

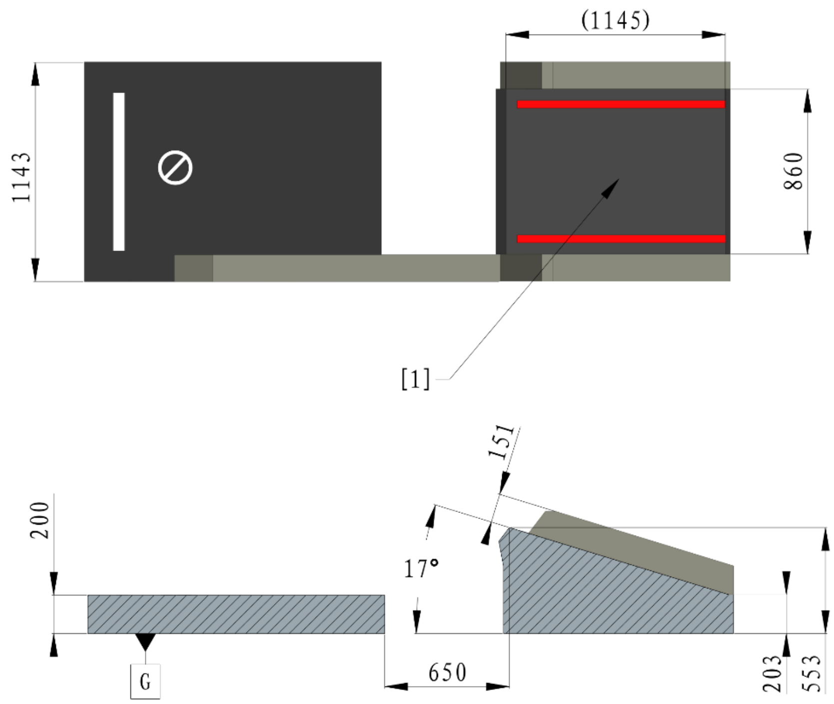


[1]	Terrain Crossing Buff Point (Road)	[2]	Terrain Crossing Buff Point (Tunnel)	[3]	Terrain Crossing Buff Point (Launch Ramp)
[4]	11° slope	[5]	Terrain Crossing Buff Point (Tunnel)		

Figure 4-35 Road Zone

4.4.1 Launch Ramp

The Launch Ramp is located on the Road Zone, with which robots can fly over the ravine and reach the territory of the other team quickly.



[1] 17° slope

Figure 4-36 Launch Ramp

4.5 Flight Zone

The Flight Zone is the flight area for Drones, which includes the Landing Pad and the air space above it, as well as the air space above the road connected to the Trapezoid-Shaped Elevated Ground.

To ensure its safety, a Drone must be attached to a Drone Safety Rope during a match. The Drone Safety Rope is 2.4 m long. The flight distance is restricted by the snap ring of the Drone Safety Rope. The distance of the snap ring from the wide edge of one team's side of the Battlefield is approximately 14 m. When a Drone is at the farthest position, the Pilot should not fly it forward any further.

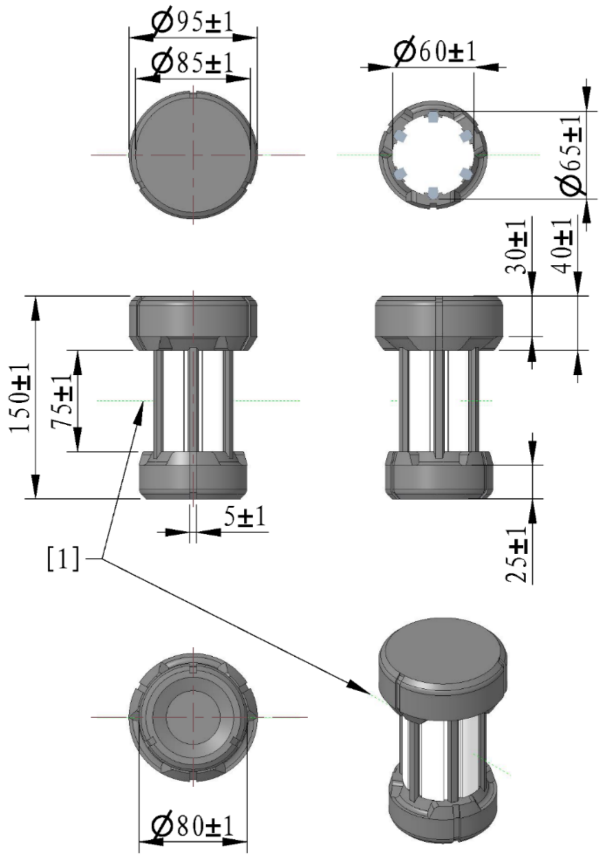
- Drones must be connected to a Drone Withdrawal System with a Drone Safety Rope during competition to ensure safety. A Drone Withdrawal System consists of a wire rope, a withdrawal device, and a retractable tether box. The withdrawal device and the retractable tether box are suspended on the wire rope. A snap ring is set on the wire rope about 14 m away from the edge of a team's side of the field. The retractable tether box cannot pass the snap ring. The retractable tether box weighs approximately 800 g and contains a 2.4 m soft tether with a constant elastic force. The static elastic force of the tether is less than 5 N. The end of the tether is connected to a Drone with a hook.
- When the Drone is flying normally, the withdrawal device that hangs on the wire rope directly above the Pilot Operation Room does not move with the Drone. At the same time, only the retractable tether box moves with the Drone. When the Drone crashes or experiences other situations where the referee determines that it is possible and necessary to recover the Drone, the withdrawal device will be activated and move to the position of the retractable tether box to recover the Drone.

4.6 Miscellaneous

4.6.1 Mobile Component

An Energy Unit is a Mobile Component, which can be grabbed and carried by robots. An Energy Unit is a cylindrical component weighing approximately 400–500 g, with a height of 150 mm, an upper diameter of 95 mm, a lower diameter of 80 mm, and asymmetrical ends.

As the competition progresses, the surface of Mobile Components may suffer mild damage or gather dust. Participating teams must adapt to such condition, but Mobile Components that are severely stained or damaged will not be used in the official competition.



[1] Rotation Axis

Figure 4-37 Energy Unit

4.6.2 Projectile

Robots deal damage by firing projectiles to attack Armor Modules. The parameters and scenarios of use for projectiles in the competition are as follows:

Table 4-1 Projectile Parameters and Scenarios of Use

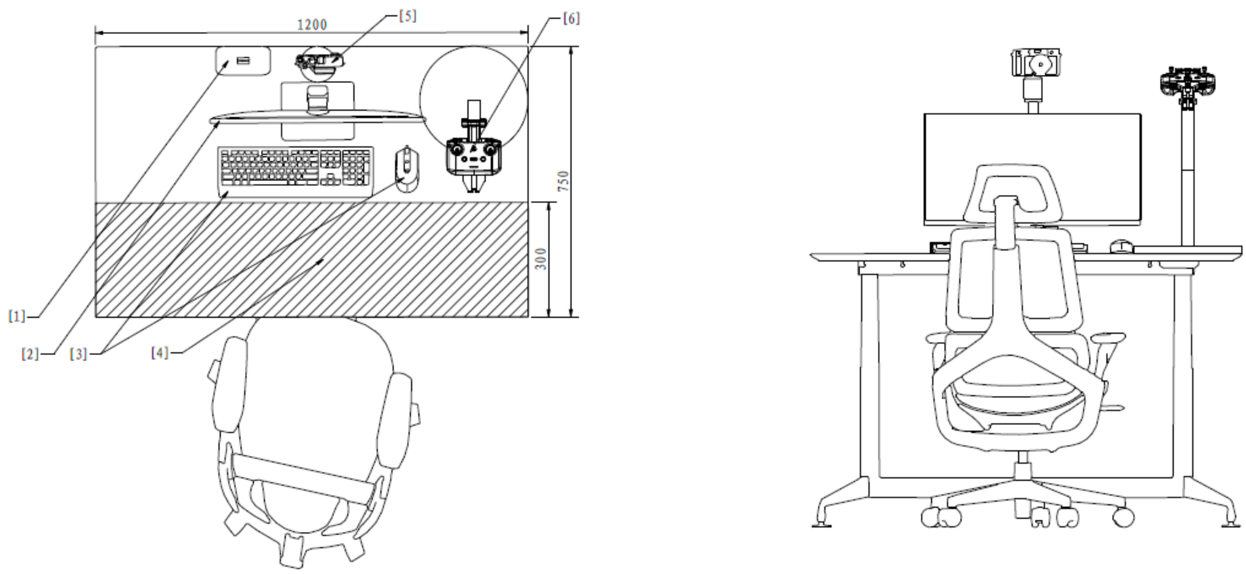
Type	Appearance	Color	Dimensions	Weight	Shore Hardness	Material	Scenarios of Use	Rated Quantity
42 mm Luminous Projectile	Similar to a golf ball	Semi-transparent	42.5 mm ± 0.5 mm	44.5 g ± 0.5 g	90 A ± 5 A	Plastic (TPE)	RMUC 2026 Throughout the entire process	100 projectiles per Round

Type	Appearance	Color	Dimensions	Weight	Shore Hardness	Material	Scenarios of Use	Rated Quantity
17 mm Fluorescent Projectile	Sphere	Yellow -green	16.8 mm ± 0.2 mm	3.2 g ± 0.1 g	90 A ± 5 A	Plastic (TPU)	RMUC 2026 Throughout the entire process	Limited, but no fewer than 4,000 projectiles per Round.

4.6.3 Operation Room

The Operation Room is located near the perimeter of the Battlefield, and serves as the area for the Operators during the competition. The Operation Room consists of the Main Operation Room and the Pilot Operation Room. Each Operator's desk in the Main Operation Room measures 120 cm in length and 75 cm in width, and is equipped with the corresponding number of computers. Each computer comes with a matching monitor, mouse, keyboard, RoboMaster Referee System Video Transmitter Module (Receiver) and its mount, three USB Type-A Receptacle ports, one RS232 Plug port, one 1GbE (RJ45) Ethernet port, a wired headset, and other Official Equipment. The Pilot Operation Room is equipped only with goggles to protect against laser exposure from the opponent's radar.

The Main Operation Room desktop layout is shown below. The Operation Room desktop features a monitor, broadcast camera, RoboMaster Referee System Video Transmitter Module (Receiver) and its stand, keyboard, mouse, and equipment area. The monitor, broadcast camera, and interface are fixed. The stand for the RoboMaster Referee System Video Transmitter Module (Receiver) is mounted on the desk, and the Operator can operate the RoboMaster Referee System Video Transmitter Module (Receiver) within the stand's movable range. The keyboard and mouse can be moved freely, and the Operator may place the Custom Controller and Custom Client in the designated area. The layout is as shown below:



[1]	Interface	[2]	Monitor	[3]	Keyboard and mouse
[4]	Equipment area	[5]	Broadcast camera	[6]	VTM (Receiver) and its stand

Figure 4-38 Layout of the Main Operation Room

If a participating team removes the VTM (Receiver), resulting in video transmission interruption or any other effects, they must bear full responsibility.

5. Competition Mechanism

5.1 HP Loss and Penalty for Exceeding Limit

5.1.1 Attack Damage or Collision

Ground Robots, Base, and Outpost may lose HP when their Armor Modules are attacked or collide with other objects. The mechanism is as follows:

The Dart Detection Module detects attacks from Darts and 42 mm projectiles through the Armor Module and the phototube.

An Armor Module can detect projectile attacks. The shortest detection interval for an Armor Module is 50 ms (when an Armor Module is hit with a 42 mm projectile, the detection interval can be extended to a maximum of 200 ms).

The projectile needs to come into contact with the impact surface of the Armor Module at a certain speed in order to be successfully detected. The speed ranges in which Armor Modules can effectively detect different types of projectiles are as follows:

Table 5-1 Speed Ranges of Different Types of Projectiles for Effective Detection by Armor Modules

Armor Module	17 mm projectile	42 mm projectile
Large Armor Module, Small Armor Module	Higher than 12 m/s	Higher than 10 m/s
Power Rune Armor Module	Higher than 12 m/s	-

- In an actual match, the normal speed of a projectile that touches the Armor Module attack surface is different from its Projectile Speed due to the projectile's speed decay and its incident angle not being normal to the Armor Module attack surface. Damage detection is based on the normal component of the projectile's speed upon contact with the Armor Module attack surface.
- Teams are not allowed to hit a Power Rune with 42 mm projectiles or Darts.
- If a Hero robot has not fired a 42 mm projectile for four consecutive seconds, the opponent's robots, Outpost, and Base will become immune to 42 mm projectile damage until the Hero robot fires a 42 mm projectile again.

- The Referee System will round off the HP loss to the nearest integer when calculating the HP.

When the Armor Module of a Ground Robot, the Base, or an Outpost has a collision, there is a chance that the collision is identified as projectile damage. However, causing damage through collisions (including ramming with robots or throwing objects) is not allowed.

Without buffs, the original damage is as follows:

Table 5-2 HP Loss Mechanism for Attack Damage or Collision

Damage Type / Target	42 mm projectile	17 mm projectile	Dart	Collision
Robot Armor Module	200	20	-	2
Base Armor Module	200	20	-	-
Base Dart Detection Module	200	-	Fixed target: 200 Random fixed target: 300 Random moving target: 625 Terminal moving target: 1000	-
Outpost Middle Armor Module	200	20	-	-
Outpost Dart Detection Module	200	-	750	-

5.1.2 Exceeding the Initial Launching Speed Limit

The Launching Mechanisms of Hero, Infantry, Drone, and Sentry robots will be locked for a certain duration if the robots exceed the initial launching speed limit. The duration does not accumulate.

Locks and unlocks due to exceeding the initial launching speed limit are independent from locks and unlocks caused by other reasons. Assume the initial launching speed limit is V0 (m/s), and the actual speed detected by the Referee System is V1 (m/s). The specific lock durations are shown in the table below.

Table 5-3 Penalty Mechanism for Exceeding the Initial Launching Speed Limit

17 mm Projectile	42 mm Projectile	Lock Duration
$0 < V_1 - V_0 < 5$	$V_0 < V_1 \leq 1.1 \times V_0$	15s
$5 \leq V_1 - V_0 < 10$	$1.1 \times V_0 < V_1 \leq 1.2 \times V_0$	20s

17 mm Projectile	42 mm Projectile	Lock Duration
$10 \leq V_1 - V_0$	**17 mm projectile: $1.2 \times V_0 < V_1$** **42 mm projectile: 18 m/s $< V_1$**	The remainder of the current round

5.1.3 Exceeding Barrel Heat Limit and Cooling

The Launching Mechanisms of Hero, Infantry, Drone, and Sentry robots will be locked when the robots exceed the Barrel Heat Limits. Locks and unlocks due to exceeding the Barrel Heat Limits are independent from locks and unlocks caused by other reasons. The mechanism is as follows:

Set Q_0 as the Barrel Heat Limit and Q_1 as the current Barrel Heat. For each 17 mm projectile detected by the Referee System, the current Barrel Heat Q_1 is increased by 10 (regardless of the 17 mm projectile's initial speed). For each 42 mm projectile detected, the current Barrel Heat Q_1 is increased by 100 (regardless of the 42 mm projectile's initial speed). The Barrel Heat cools at a frequency of 10 Hz. The cooling value per detection cycle is equal to the cooling rate/10.

When $Q_2 > Q_1 > Q_0$, the FPV visibility of the robot operator's computer will be reduced, and the Launching Mechanism of the robot will be locked. These restrictions will remain in place until $Q_1 = 0$.

When $Q_1 > Q_2$, the Launching Mechanism of the robot will be locked for the remainder of the current round.

For the 17 mm Launching Mechanism, $Q_2 = Q_0 + 100$.

For the 42 mm Launching Mechanism, $Q_2 = Q_0 + 200$.

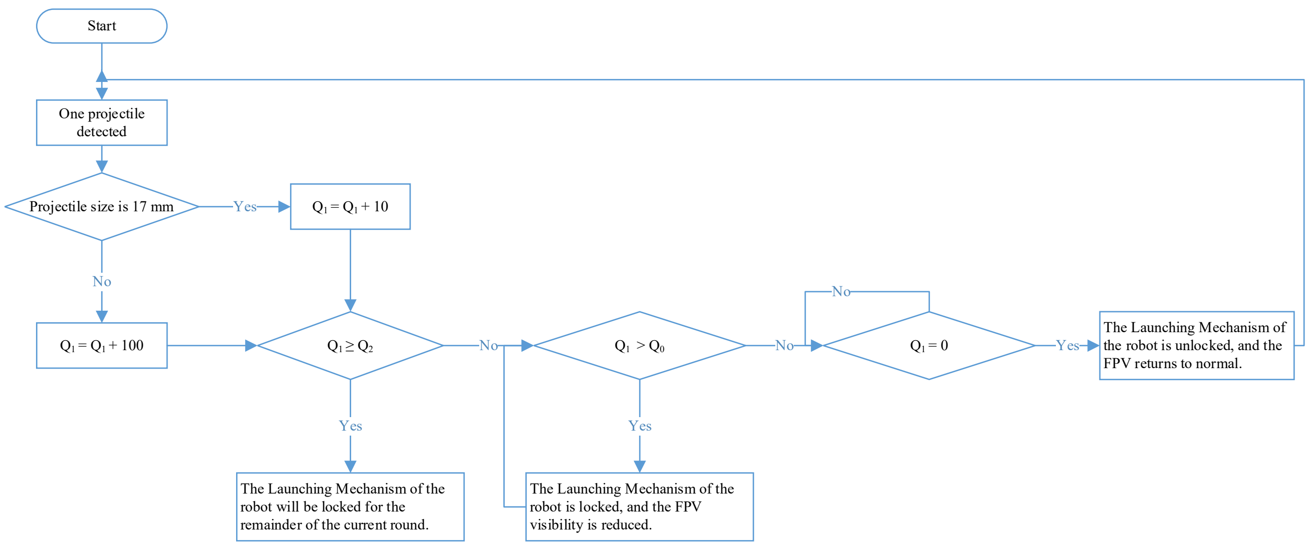


Figure 5-1 Penalty Mechanism for Exceeding the Barrel Heat Limit

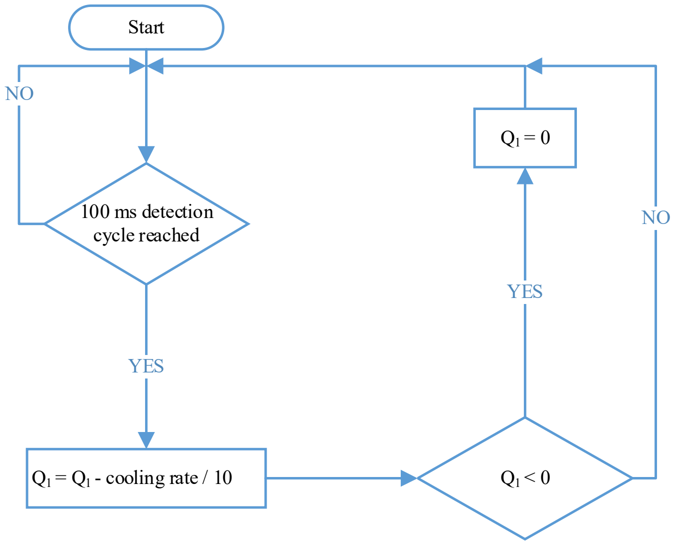


Figure 5-2 Barrel Heat Cooling Mechanism

5.1.4 Exceeding Chassis Power Consumption Limit

The chassis power of robots will be continuously monitored by the Referee System, and the robot chassis needs to run within the Chassis Power Limit. When a Ground Robot exceeds the Chassis Power Consumption Limit, Buffer Energy will be deducted from that robot. Upon the exhaustion of buffer energy, if the chassis power is still above the limit, the chassis will be powered off for 5 seconds. The Referee System calculates chassis power consumption at a frequency of 10 Hz.

Let Z represent Buffer Energy, Q represent the Buffer Energy Limit, P_r represent instantaneous Chassis output power, and P_l represent the power limit. Q = 60 J

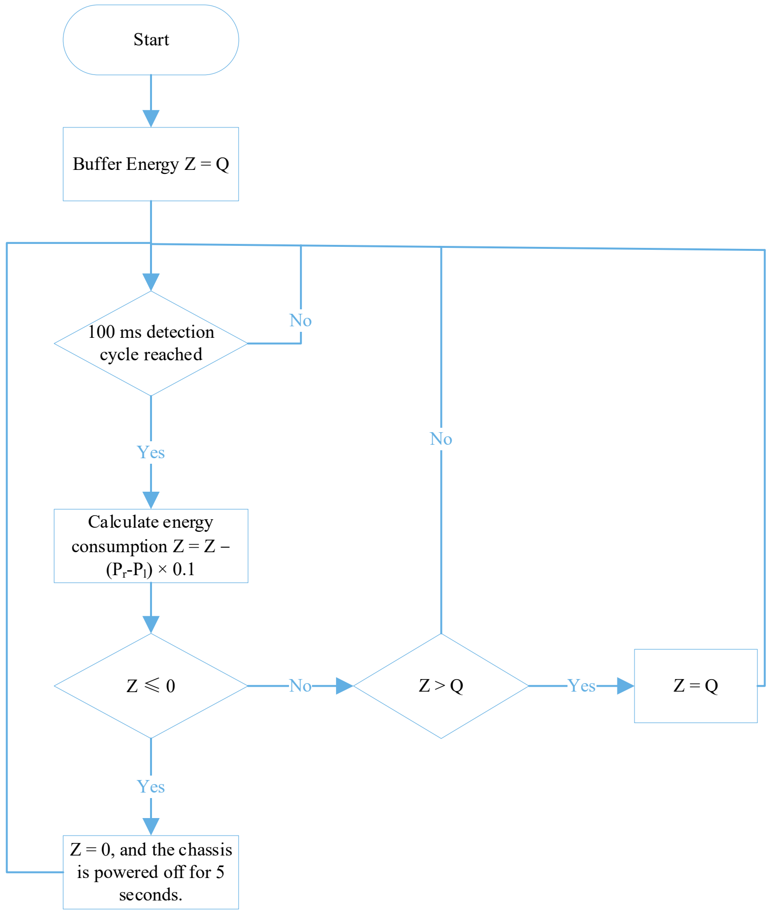


Figure 5-3 Robot Chassis Power Detection and Penalty for Exceeding Limit

5.1.5 Referee System Disconnection

Participating teams must mount the corresponding Referee System Module on their robots in accordance with the requirements of the RoboMaster 2026 University Series Robot Specification Manual, and ensure stable connection between each module of the Referee System and the server throughout the competition. The Referee System server detects the connectivity of each module at a frequency of 2 Hz.

5.1.5.1 Abnormal Disconnection

If a robot's Main Control Module becomes disconnected and enters an "Abnormal Disconnection" status during a competition, it can reconnect to the competition. Experience points and level will continue to be calculated during the offline period.

Table 5-4 Penalty Mechanism for Robots in Abnormal Disconnection Status

Robot Type	Consequences of Abnormal Disconnection
Ground Robot	● The Launching Mechanism (if any), Gimbal, and Chassis are powered off, and 5% of the Maximum HP is dealt for each second elapsed until it drops to zero.

Robot Type	Consequences of Abnormal Disconnection
	● The RFID Interaction Module is disabled. ● The robot no longer detects any damage caused by collision and projectile hits and any HP Loss. ● The respawn process stops. ● The inter-robot communication disconnects.
Drone	● The Launching Mechanism is powered off and air support cannot be requested. ● The system does not detect illumination from the laser launch device. ● Image transmission is disconnected. ● The inter-robot communication disconnects.
Radar	● The inter-robot communication disconnects. ● The laser launching device and its actuator are powered off.
Dart	● If it is disconnected when the Dart Launching Station gate closes, the Dart Launching Station gate cannot be opened. ● If it is disconnected when the Dart Launching Station gate is open, the Dart Launching Station gate will immediately close and cannot be opened.

5.1.5.2 Armor Module or Supercapacitor Management Module Disconnection

If a Ground Robot's design or structural issues cause the disconnection of its Armor Module or Supercapacitor Management Module, the robot will lose HP.

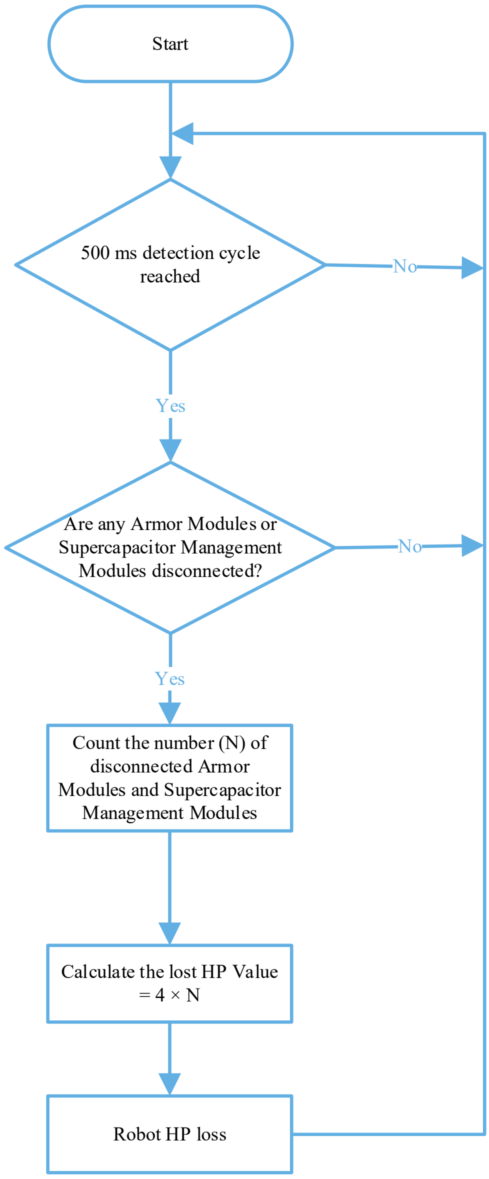


Figure 5-4 HP Loss Mechanism for Armor Module and Supercapacitor Management Module Disconnection

5.1.5.3 Speed Monitor Module Disconnection

If a Speed Monitor Module (17 mm or 42 mm projectile) mounted on a Hero, Infantry, Sentry, or Drone robot goes offline, the robot's Launching Mechanism (17 mm or 42 mm projectile) will be immediately locked. The locking and unlocking caused by the Speed Monitor Module disconnection is independent from any locking or unlocking caused by other issues.

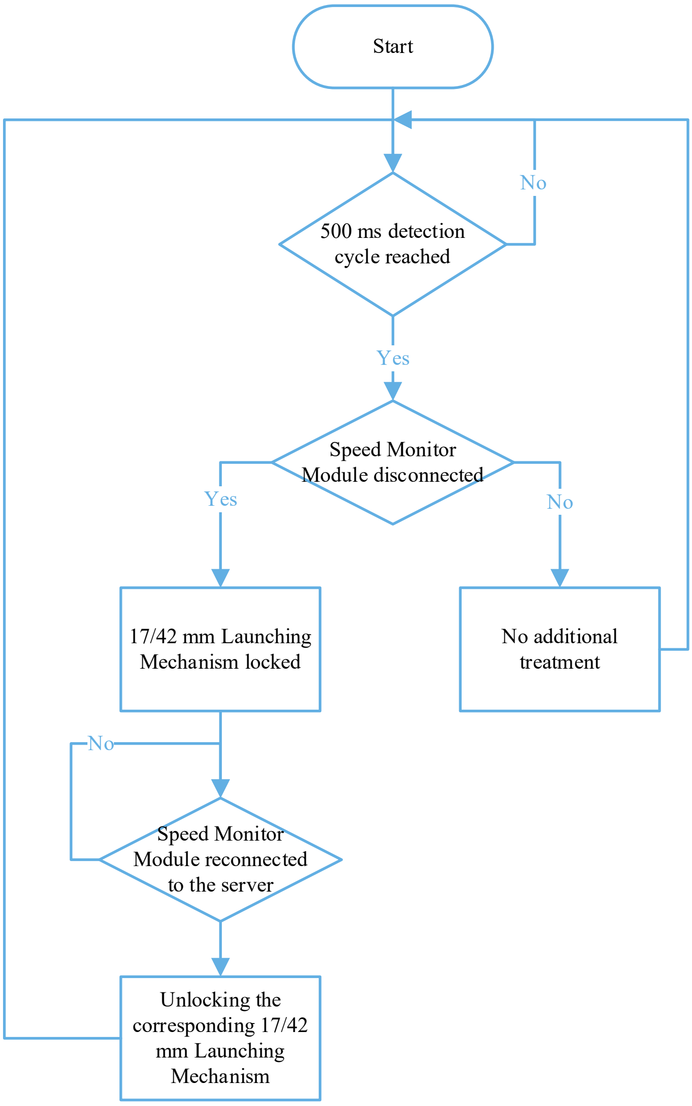


Figure 5-5 Power-off Mechanism for Speed Monitor Module Disconnection

5.1.5.4 UWB Positioning Module Disconnection

If the UWB Positioning Module installed on a Hero, Engineer, Infantry, Drone, or Sentry robot is disconnected, the opponent Radar will always consider its identify mark as “accurate”, while the Radar on the robot’s own side will always consider its identify mark as “inaccurate”.

5.1.6 Penalty Damage

When a Ground Robot receives a Yellow Card, Red Card, or other Penalty, it may lose HP. For details, see “7.1 Penalty System”.

5.2 HP Recovery and Respawn Mechanism

Except when in an Abnormal Disconnection state or after being ejected, Ground Robots can recover their HP and respawn.

5.2.1 HP Recovery Mechanism

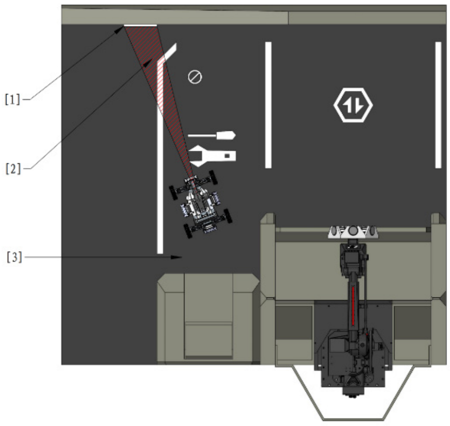
HP Recovery in the Resupply Zone

A Ground Robot that occupies the Resupply Zone Buff Point of its team will receive an HP Recovery Buff, which is 10% of its Maximum HP per second. After four minutes into the match, when robots are out of combat and occupy their own Resupply Zone Buff Point, they will receive an HP Recovery Buff equal to 25% of their maximum HP per second; if the robots are not out of combat, the 25% per second HP Recovery Buff will immediately expire.

Remote Exchange for HP

When a Hero, Infantry, or Sentry robot is out of combat, they have the option to remotely exchange Gold Coins for HP. Six seconds after a remote HP exchange is confirmed, the robot will receive an additional 60% of its current HP, without exceeding the Maximum HP.

- If the robot is defeated within six seconds after a remote HP exchange is confirmed, the exchange will lapse and the Gold Coins will not be returned.
- When a robot's RFID Interaction Module detects the RFID Interaction Module Card of the Resupply Zone, or its Video Transmitter Module (VTM) recognizes a Visual Marker of the Resupply Zone when the robot is located in a Resupply Zone defined by the rule, the robot will be deemed to be in the Resupply Zone. An alive robot in its team's Resupply Zone is deemed to have occupied the Resupply Zone.
- The VTM is only able to recognize a Visual Marker when the Visual Marker is in the image center of the VTM and the VTM has no obvious roll angle.



[1]	Visual Marker	[2]	Image Transmission Range of Robots	[3]	Recognizable Zone

Figure 5-6 Relative Location of the Robot's VTM and Visual Marker

5.2.2 Respawn Mechanism

When a robot is defeated, its Barrel Heat and Buffer Energy will be reset to 0. A defeated robot can respawn by completing the respawn timer or exchanging Gold Coins for instant respawn before the timer is completed.

- When seeking to complete the respawn timer, the length of the respawn timer is related to the remaining time of the competition when the robot is defeated and the robot's cumulative number of instant respawn exchanges. Its formula is as follows:

$$\text{Required respawn timer} = 10 + \frac{420 - \text{time remaining}}{10} + 20 \times \text{cumulative instant respawn exchanges}$$

The time remaining is measured in seconds. The number is rounded off to the nearest integer.

Example: If a robot is defeated with 120 seconds remaining in the match, and it has already used Instant Respawn twice before, the length of its respawn timer for this time will be as follows: $10 + \frac{420-120}{10} + 20 \times 2 = 80$.

The respawn timer of a robot starts immediately after it is defeated. From the moment the robot is defeated, the respawn timer increases by 1 per second; if the robot is in the Resupply Zone Buff Point or the Base HP of its side is below 2,000, the respawn timer increases by 4 per second.

When seeking to complete the respawn timer:

- The respawned robot retains its level and Experience Points from before its defeat and remains invincible for 30 seconds.

- ➢ Its HP is restored to 10% of its Maximum HP.
- ➢ The robot enters the "Weakened" state.
  - The Launching Mechanism is locked.
  - The robot cannot occupy any Buff Points or rebuild Outposts.

Thereafter, when its RFID Interaction Module detects the RFID Interaction Module Card of an occupiable Outpost, Base Buff Point, or Resupply Zone Buff Point, its "Weakened" and invincible states will be lifted.

- When using Gold Coins for an instant respawn:
  - ➢ The respawned robot retains its level and Experience Points from before its defeat and remains invincible for three seconds.
  - ➢ Its HP is restored to 100% of the Maximum HP.
  - ➢ Its Launching Mechanism is immediately unlocked.
  - ➢ Its Chassis Power Limit is doubled (up to 200 W), for 4 seconds.

5.3 Economy System

5.3.1 Overview of the Economy System

During the Three-Minute Setup Period, Ground Robots can pre-load 42 mm and 17 mm projectiles while Drone can pre-load 17 mm projectiles only.

During the competition, both teams will earn Gold Coins regularly. They can also receive additional Gold Coins by equipping Energy Units. Gold Coins can be exchanged for air support, 17 mm and 42 mm Projectile Allowance, remote HP recovery, and instant respawn.

At the start of a round, each team receives a certain quantity of initial Gold Coins. As the competition progresses, both sides passively gain Gold Coins, as detailed in the table below.

Table 5-5 Passive Gold Coin Gains

Game Countdown	Red Team	Blue Team
06:59	400 (initial)	400 (initial)
05:59	50	50
04:59	50	50
03:59	50	50

Game Countdown	Red Team	Blue Team
02:59	50	50
01:59	50	50
00:59	150	150

The ratings obtained by a team for its "Project Documents" and "Technical Proposal" during the Final Robot Assessment will impact the team's initial Gold Coin quantity for each round during the Regional Competition. The correspondence between the impact and the ratings is as follows:

Table 5-6 Impact of Project Document Rating

Project Document Rating	Degree of Impact
S	+50
A	+25
B	0
C	-25
D	-50

Table 5-7 Impact Level of Technical Proposal Rating

Technical Proposal Score	Degree of Impact
S	+150
A	+75
B	0
C	-25
D	-50

Table 5-8 Exchange Rules

Exchange Item	Exchange Ratio	Exchange Limit
17 mm Projectile Allowance	● Remote Exchange: 150 Gold Coins / 100 rounds ● Non-Remote Exchange: 10 Gold Coins / 10 rounds	1,000 rounds per team

Exchange Item	Exchange Ratio	Exchange Limit
42 mm Projectile Allowance	● Remote Exchange: 150 Gold Coins / 10 rounds ● Non-Remote Exchange: 10 Gold Coins / 1 round	100 rounds per team
Air Support Time	1 Gold Coin / 1 second	No restrictions
HP (Remote Exchange)	$50 + \mathrm{ROUNDUP}(\frac{420-\mathrm{time\ \ remaining}}{60} \times 20)$Gold Coin per use	Unlimited
Instant Respawn	$\mathrm{ROUNDUP}\left(\frac{420-\mathrm{time\ \ remaining}}{60}\right) \times 80 + \mathrm{robot\ \ level} \times 20$ Gold Coin / 1 unit	Unlimited

- Time is measured in seconds.
- ROUNDUP means to round up to the nearest integer.
- The exchange limit of the 17 mm Projectile Allowance refers to the maximum Projectile Allowance obtained through the exchange of Gold Coins.

5.3.2 Projectile Allowance Mechanism

For every round of projectiles fired by a robot, the Projectile Allowance corresponding to the type of projectiles fired is reduced by one round. When the Launching Mechanism is locked, its power stays off; when it is unlocked and the Projectile Allowance for the corresponding projectile type is greater than zero, its power remains on, otherwise it is powered off.

In the following special circumstances, the Armor Modules of the opponent's robots, Outpost, and Base will become immune to 42 mm projectile damage until the Hero robot is alive and has a Projectile Allowance greater than 0:

1. Three seconds after the Hero robot becomes defeated or abnormally disconnected
2. When the Hero robot launches a 42 mm projectile despite having a Projectile Allowance of 0 (it launches more projectiles than its allowance)
3. After the Hero robot launches the third 42 mm projectile after it is defeated

Each robot's Initial Projectile Allowance and its mechanism are provided below:

Table 5-9 Projectile Allowance for Robots

Robot	Initial Projectile Allowance	Projectile Allowance Mechanism
Hero	0	Projectile Allowance can be obtained via exchange in the Resupply Zone, at the Base Buff Point, and at Outpost Buff Point, or via remote exchange.
Infantry	0	
Sentry	300	Projectile Allowance can be obtained via exchange in the Resupply Zone, at the Base Buff Point, and at the Outpost Buff Point, via remote exchange, or from the Resupply Zone.
Drone	750	Extra Projectile Allowance is not allowed.

Resupply Zone, Base Buff Point, or Outpost Buff Point Exchange

When a Hero, Infantry, or Sentry robot occupies a Resupply Zone, Base Buff Point, or Outpost Buff Point, the Operator can Exchange Projectile Allowance via the Referee System Player's Client. Sentry robots can also independently exchange Projectile Allowance via the Referee System Serial Port. The specific projectile types and minimum quantities that can be exchanged for are shown in the table below.

Table 5-10 Minimum Exchange Units of Projectile Allowance for Different Types of Projectiles in the Resupply Zone, at the Base Buff Point, or at the Outpost Buff Point

17 mm Projectile Allowance	42 mm Projectile Allowance
10 rounds	1 round

Remote Exchange

If a robot is out of combat, the operator can remotely exchange for Projectile Allowance through the Referee System player's client. A Sentry can remotely exchange for Projectile Allowance through the Referee System Serial Port on its own. The specific projectile types and minimum quantities that can be exchanged for are shown in the table below.

Table 5-11 Minimum Exchange Units for Different Types of Projectiles during Remote Exchange

17 mm Projectile Allowance	42 mm Projectile Allowance
100 rounds	10 rounds

When a remote exchange is completed, the Projectile Allowance will be effective after six seconds.

Exchange in the Resupply Zone

After the competition starts, every minute, the Sentry can obtain 100 rounds of Projectile Allowance by occupying the Resupply Zone Buff Point on its own side. The allowance not obtained through this method can be accumulated.

Example: If a Sentry has never occupied the Resupply Zone Buff Point on its own side before and occupies it with 30 seconds remaining in the competition, the Sentry's Projectile Allowance increases by 600.

5.3.3 Tech Core – Energy Unit Mechanism

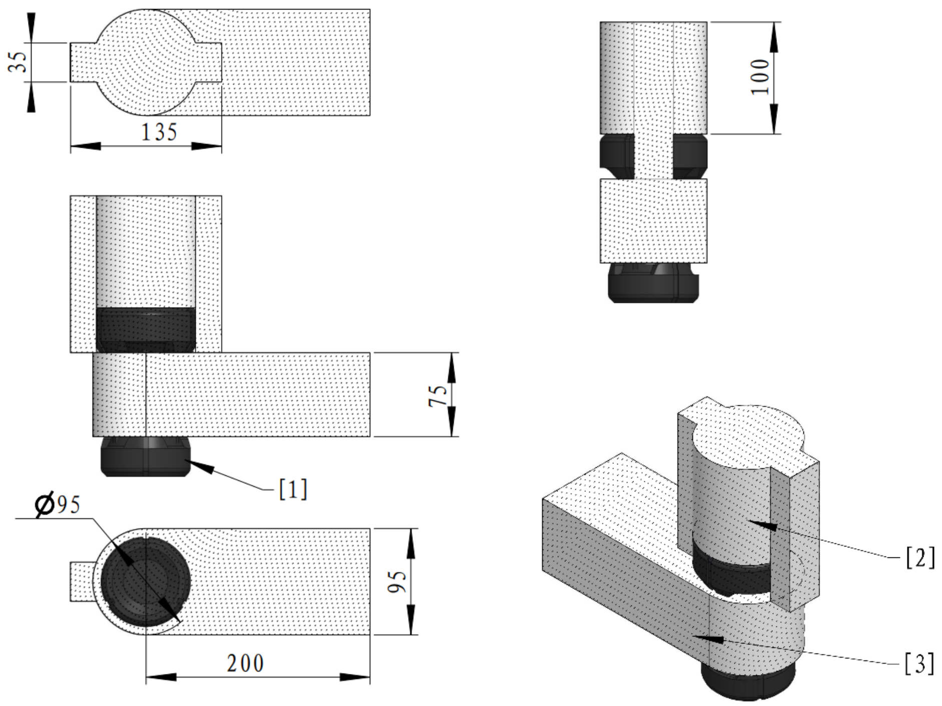
During the competition, the Engineer installs its carried energy units into the Tech Core in exchange for Gold Coins and team buffs.

For each Tech Core, establish a right-handed Cartesian coordinate system named Assembly Coordinate System OXYZ, where the origin O is set at the midpoint of the intersection line between the front of the base and the Battlefield surface. The negative X-axis is defined by the direction of the base front normal pointing toward the Tech Core, and the positive Z-axis points vertically upward. A right-hand orthogonal coordinate system EX'Y'Z' is established using the geometric center of the front surface of the Tech Core's entry as the origin E. The X', Y', and Z' axes align parallel to, and share the same positive directions as, the X, Y, and Z axes. The unit normal vector of the front surface of the Tech Core's entry is denoted as $\vec{e}$. A spherical coordinate system (r, θ, φ) is established based on the coordinate system of the Tech Core. Here, θ is the angle between the projection of the unit normal vector $\vec{e}$ on the X'Y' plane and the positive X' axis, where θ ∈ [-90, 90]; φ is the angle between $\vec{e}$ and the positive Z' axis, where φ ∈ [0, 90]. Additionally, α is defined as the rotation angle of the Tech Core around the direction of $\vec{e}$. A counterclockwise rotation is considered positive for α, where α ∈ [-45, 45].

The pose of the Tech Core is defined as the combination of the coordinates (x, y, z) of the center point E on the front surface of the Tech Core's entry in the OXYZ coordinate system, the orientation coordinates (θ, φ) of the unit normal vector $\vec{e}$ in the spherical coordinate system of the Tech Core, and the rotation angle α of the Tech Core around the direction of $\vec{e}$, forming (x, y, z, θ, φ, α).

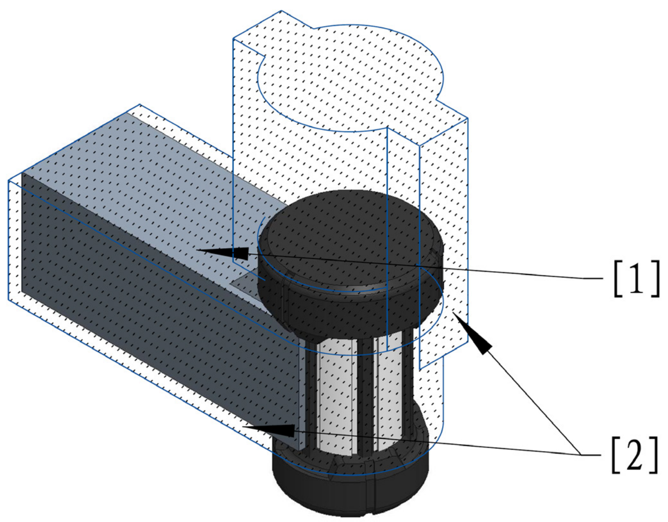
There is space inside the Tech Core, allowing an Energy Unit to be inserted up to a depth of 100 mm and moved horizontally up to 100 mm relative to the insertion surface. After placing the Energy Unit and performing horizontal movement, there is still additional space, allowing the Energy Unit to rotate up to 90° around the rotating shaft. For more information about the Energy Unit rotation axis, see "Figure 4-37 Energy Unit".

The detailed shape and dimensions of the Tech Core will be updated later. However, when the Energy Unit is inside the Tech Core, there is an area relative to the Energy Unit for each difficulty level in which the Engineer can move freely without any structural interference with the Tech Core, as shown in the following figure:

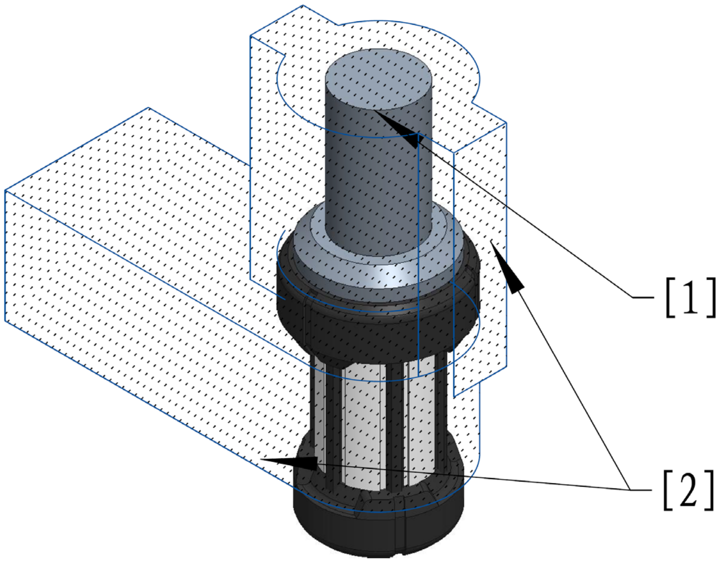


[1] Energy Unit [2] Interaction Area for Level 1 Difficulty [3] Interaction Area for Level 2, 3, and 4 Difficulty

Figure 5-7 Energy Units - Interaction Areas within the Tech Core



[1] Gripper [2] Interaction Area



[1] Suction Cup [2] Interaction Area

Figure 5-8 Interaction Areas and Typical Working Conditions

For example, when assembling an Engineer at the Level 1 difficulty, its terminal executive mechanisms can secure the Energy Unit from both its top and side Interaction Areas. However, when assembling an Engineer at the Level 3 difficulty, the Energy Unit can only be secured from its side Interaction Area.

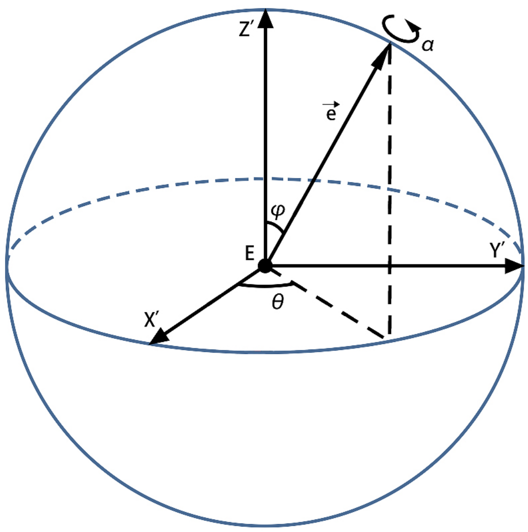


Figure 5-9 Spherical Coordinate System

The initial pose of the Tech Core is: x = -200, y = 200, z = 800, θ = 0, φ = 90, α = 0 (length unit: mm; angle unit: °).

The pose range of the Tech Core must meet the following conditions:

- No part of the Tech Core may extend beyond the plane of the front face of the base.
- The pose of the Tech Core is related to the difficulty level. Their relationship is shown below:

Difficulty	x	y	z	θ	φ	α	Panning Requirements	Spinning Requirements
Level 1	[-100,0]	[-100,100]	800	0	90	0	No panning requirements	No spinning requirements
Level 2	[-100,0]	[-100,100]	[700,900]	[-90,90]	[0,90]	[-45,45]	100 mm movement after insertion	
Level 3	[-100,0]	[-100,100]	[700,900]	[-90,90]	[0,90]	[-45,45]	100 mm movement after insertion	After the translation is complete, use the specified axis of rotation, as detailed in “Figure 4-37 Energy Unit“, and rotate it to the designated angle within the permissible 90° range, with 5° precision.
Level 4	[-100,0]	[-100,100]	[700,900]	[-90,90]	[0,90]	[-45,45]	100 mm movement after insertion	

In each round, the red and blue teams’ Tech Cores have the same pose changes at the same difficulty level. The pose value range and panning and spinning requirements are the same for Difficulty Level 3 and Difficulty Level 4. However, under Difficulty Level 4, a team’s Tech Core will be temporarily moved to the other team’s location. At this point, the team’s Tech Core coordinates x’y’z’ relate to the other team’s Tech Core coordinates xyz as follows: x’ = x, y’ = y - 400, z’ = z, with the same angle pose. The assembling side must complete the assembly for both teams’ Tech Cores. After selecting Difficulty Level 4, the Engineer must complete assembly within 45 seconds after the Tech Core is in place. A failed assembly at Difficulty Level 4 will prevent that side from attempting Difficulty Level 4 again within the next 90 seconds.

Priority for Difficulty Level 4 Assembly

- If neither side is assembling, and the other side selects Difficulty Level 4, the Tech Core will immediately move to the designated location.
- If one side is attempting assembly at a difficulty level other than Level 4, and the other side selects Level 4,

the side currently assembling will have a 15-second buffer to tap OK or withdraw from the assembly. Afterward, the Tech Core will immediately move to the designated location. In this case, if the side attempting Difficulty Level 4 fails to assemble, they will not be able to attempt Difficulty Level 4 again in this round, and the Gold Coin earnings every 10 seconds will be reduced by 25.

If the Engineer does not leave the Tech Core during the buffer period, it may cause mechanical interference or damage.

- If one side is attempting assembly at Difficulty Level 4, the other side cannot choose Difficulty Level 4.

Engineers can select the difficulty level independently, but must complete the assembly for the previous difficulty level before moving on to the next one. The available difficulty levels are subject to certain conditions, as shown in the table below:

Match Time	Available Difficulty Levels
At the start of the competition	Both sides can only select Difficulty Level 1.
One minute after the competition begins	Both sides can select Difficulty Level 1 or 2.
Two minutes after the competition begins	Both sides can select Difficulty Level 1, 2, or 3.
Three minutes after the competition begins	Both sides can select Difficulty Level 1, 2, 3, or 4.

After a successful assembly, teams will receive different rewards depending on whether it is their first time completing the assembly. Details are shown in the following table:

Level Completed	First-time completion reward	Non-first completion reward
Level 1	Afterwards, the side earns 50 Gold Coins every 10 seconds.	Afterwards, the side will receive an additional 5 Gold Coins every 10 seconds.
Level 2	• Unlock Robots Level Cap: Before completing the assembly task at this difficulty level, your robots level cap is Level 5. Upon completion, the level cap increases to Level 7. • Afterwards, the side will receive an additional 25 Gold Coins every 10 seconds.	Afterwards, the side will receive an additional 10 Gold Coins every 10 seconds.
Level 3	• Ground Robots, Outpost, and Base on the side gain a 25%	Afterwards, the side gains an

	Defense Buff, lasting until the End of Round. ● The level cap for robots on this side is increased to 10. ● Afterwards, the side will receive an additional 25 Gold Coins every 10 seconds.	additional 15 Gold Coins every 10 seconds.
Level 4	● Ground Robot, Outpost, and Base on the side receive a 50% Defense Buff, lasting until the End of Round. ● Base HP increases by 2,000, and any excess HP will be converted into an equivalent amount of Virtual Shield. When the Base takes Damage afterward, the Virtual Shield will be depleted first, followed by the Base HP. ● Afterwards, the side gains an additional 50 Gold Coins every 10 seconds.	Can only be completed once.

Once robots reach their level cap, they will no longer gain experience.

Example: If a team successfully completes five assembly tasks in one round, at Difficulty Levels 1, 2, 3, 2, and 1 respectively, they will receive a total of 50+25+25+10+5=115 Gold Coins every 10 seconds.

Exchange Process:

To assemble the Energy Unit at Difficulty Levels 1, 2, and 3, the following steps must be completed:

1. After the Engineer occupies the Assembly Zone Buff Point, the Operator selects the assembly difficulty on the Referee System player's client.
2. Once the Tech Core moves to the designated pose (ready for assembly), the corresponding light effect will appear. The Engineer then inserts the Energy Unit it carries into the Tech Core in the correct orientation and performs the required panning or spinning operations as specified.
3. The Operator presses the corresponding button to confirm the assembly. If the Tech Core detects that the Energy Unit has been fully inserted and assembled correctly according to the panning and spinning requirements, the Energy Unit will be reclaimed and the respective rewards will be granted to that side.

To assemble an Energy Unit at Difficulty Level 4, the following steps must be completed:

1. After the Engineer occupies the Assembly Zone Buff Point, the Operator selects the assembly difficulty on the Referee System player's client.

2. Once the Tech Core moves to the designated pose (ready for assembly), the corresponding light effect will appear. The Engineer then inserts the Energy Unit it carries into the Tech Core in the correct orientation and performs the required panning or spinning operations as specified.

3. Within three seconds after successfully assembling the first Energy Unit, the Engineer must successfully assemble an additional Energy Unit at the opponent's Tech Core.

4. The Operator presses the corresponding button to confirm the assembly. If the Tech Core detects that both Energy Units have been fully inserted and assembled correctly according to the panning and spinning requirements, and the time requirement is met, the Energy Units will be recovered and the corresponding rewards will be granted to that side; otherwise, the assembly will be deemed unsuccessful.

In addition, the Operator should pay attention to the following during the assembly process:

- After the Operator selects the assembly difficulty, if an Energy Unit is present inside the Tech Core, the assembly difficulty level cannot be changed. In addition, before the Energy Unit is successfully assembled, the pose of the Tech Core will remain unchanged for the same assembly difficulty level.
- When the Engineer is installing the Energy Unit, its Operator can use the "Eject Energy Unit" function. At this time, the Tech Core will return to its initial location and eject any Energy Unit stored inside the Tech Core.
- After the Operator confirms the assembly, they must not continue to affect the Energy Unit; otherwise, the assembly may fail.
- The Exchange Station will not detect obstacles in its path. If the Engineer's mechanisms suffer damage as they come into contact or collide with the Ore receptacle during the operation of the Exchange Station, it may cause damage to the Engineer, and any damage resulting from this will be the responsibility of the participating team.

5.4 Experience and Performance Systems

5.4.1 Experience System

At the start of the competition, Hero, Infantry, and Drone robots all begin at Level 1, and can level up by gaining Experience Points. The Experience System does not apply to other robots.

During the competition, a robot earns experience points in various ways, as shown below:

Behavior Type	Experience Points Gained
Launching projectiles	• Infantry and Drone: Gain 1 Experience Point for each projectile launched. • Hero: Gains 10 Experience Points for each projectile launched.

<table>
<tr><th>Behavior Type</th><th>Experience Points Gained</th></tr>
<tr><td>Dealing Attack Damage</td><td>

Dealing attack damage to robots: The attacking side gains four Experience Points for every one point of damage.
Dealing attack damage to an Outpost Armor Module: The attacking side gains two Experience Points for every one point of damage.
Dealing Attack Damage to the Base Armor Module: The attacking side gains one Experience Point for every two points of damage.

When a team’s robots, Base or the Outpost receives attack damage, but either that the Referee System cannot detect the specific source of the damage, the source of the damage cannot gain Experience Points or that the source of damage is from a penalized robot/robot on the same team: if the damage is from projectiles, the Experience Points will be equally divided among all of the other team’s alive robots that can deal projectile damage of that type and gain experience; if the damage is not from projectiles, the Experience Points will be equally divided among Hero, Infantry, and Drone robots of the other team that are alive at the moment. The average is rounded up and must be accurate to one decimal place.

Example: An Infantry robot of the blue team received 120 points of damage from a 17 mm projectile, but the Referee System is unable to identify the source of damage. At this time, the red team has two alive Infantry robots, one Drone providing air support, and one alive Hero robot. Each Infantry robot and Drone receives the calculated Experience Points, while the Hero does not receive any Experience Points.$\frac{4\times120}{3}=160$
</td></tr>
<tr><td>Defeating robots</td><td>
In the following calculations, both Engineer and Sentry robots are considered to be Level 1.

If a robot is destroyed and the destroyer is eligible to gain experience, the number of Experience Points gained is as follows:

When the destroyed robot’s level is higher than or equal to that of the destroyer, the Experience Points are calculated as follows:

The experience points gained by destroyer = 50 × the level of the destroyed robot × (1 + 0.2 × the difference between the level of the destroyed robot and that of the destroyer)
When the destroyed robot’s level is lower than that of the destroyer, the level

</td></tr>
</table>

Behavior Type	Experience Points Gained
	difference is considered to be 0. The experience points are calculated as follows: The Experience Points gained by destroyer = 50 × the level of the destroyed robot ● If the Referee System does not identify a destroyer or if the destroyer is ineligible for experience: The destroyer's level is considered to be equivalent to the level corresponding to the average Experience Points of the opponent's alive Hero, Infantry, and Drone robots. After calculation according to the formula above, the Experience Points are divided equally among the alive Hero, Infantry, and Drone robots of the opposing team. The average value of Experience Points is rounded down to the nearest whole number. If a robot is defeated or goes disconnected abnormally, or the Referee System is unable to detect a destroyer for reasons other than suffering a hit on its Armor Module, it will be deemed that no destroyer has been found. **When a Level 2 Infantry destroys a Level 6 Infantry of the opponent, the Experience Points gained by the destroyer is 50 × 6 × (1 + (6 - 2) × 0.2) = 540.** **Example 2: One Level 9 Infantry from the red team is defeated, and no destroyer is detected. At this point, the blue team has two Infantry robots with total Experience Points of 1250 and 4000, and one alive Drone with total Experience Points of 3000. The average Experience Points is 2750, corresponding to robots at Level 6. Therefore, each of the blue team's two Infantry robots and one Drone receives:** $\frac{50\times9\times(1+(9-6)\times0.2)}{3}=240$
Activating the Small Power Rune	For details, see "5.5.2 Power Rune Mechanism".
Activating the Large Power Rune	For details, see "5.5.2 Power Rune Mechanism".
Dealing damage in Deployment Mode	A robot receives 100 Experience Points for each instance of damage it deals in Deployment Mode. For details, see "5.6.1 Special Mechanism of Hero".
Gaining Terrain Crossing Buff	Robots receive 300 Experience Points the first time they acquire the Terrain Crossing Buff (Road), Terrain Crossing Buff (Elevated Ground), Terrain Crossing Buff (Launch Ramp), and Terrain Crossing Buff (Tunnel).

Behavior Type	Experience Points Gained
Hitting the Outpost/Base using a Dart	For details, see “5.5.1 Base and Outpost Mechanisms”.

Table 5-12 Levels and Experience of Hero, Infantry, and Drone Robots

Level	Total Experience Required to Level Up
1	0
2	550
3	1100
4	1650
5	2200
6	2750
7	3300
8	3850
9	4400
10	5000

5.4.2 Performance System

After the Three-Minute Setup Period begins, the Operator of the Hero robot may choose the Performance System type, while the Operator of the Infantry robot may select the Chassis and Launching Mechanism types for their robots. For Drones, the Launching Mechanism type cannot be selected, and the Performance System does not apply to other robots.

Once the Seven-Minute Round has begun, the selected chassis and launching mechanism types cannot be changed during the entire round. If the robot, chassis or launching mechanism types are not selected, after the start of the round, the robot type will be automatically set to “sniper-focused”, the chassis type will be automatically set to “HP-focused”, and the launching mechanism type will be automatically set to “Cooling-focused”. After a robot levels up, any increase in Maximum HP is immediately added to the robots’ current HP.

Table 5-13 Hero's Attributes

Launching Mechanism Type	Level	Maximum HP	Chassis Power Limit (W)	Barrel Heat Limit	Barrel Heat Cooling Rate (per second)
Melee-focused	1	200	70	140	12
	2	225	75	150	14
	3	250	80	160	16
	4	275	85	170	18
	5	300	90	180	20
	6	325	95	190	22
	7	350	100	200	24
	8	375	105	210	26
	9	400	110	220	28
	10	450	120	240	30
Sniper-focused	1	150	50	100	20
	2	165	55	102	23
	3	180	60	104	26
	4	195	65	106	29
	5	210	70	108	32
	6	225	75	110	35
	7	240	80	115	38
	8	255	85	120	41
	9	270	90	125	44
	10	300	100	130	50

Table 5-14 Infantry's Chassis Attributes

Chassis Type	Level	Maximum HP	Chassis Power Limit (W)
Power-focused	1	150	60
	2	175	65
	3	200	70
	4	225	75
	5	250	80
	6	275	85
	7	300	90
	8	325	95
	9	350	100
	10	400	100
HP-focused	1	200	45
	2	225	50
	3	250	55
	4	275	60
	5	300	65
	6	325	70
	7	350	75
	8	375	80
	9	400	90
	10	400	100

Table 5-15 Infantry's 17 mm Launching Mechanism Attributes

Launching Mechanism Type	Level	Barrel Heat Limit	Barrel Heat Cooling Rate (per second)
Burst-focused	1	170	5
	2	180	7
	3	190	9
	4	200	11
	5	210	12
	6	220	13
	7	230	14
	8	240	16
	9	250	18
	10	260	20
Cooling-focused	1	40	12
	2	48	14
	3	56	16
	4	64	18
	5	72	20
	6	80	22
	7	88	24
	8	96	26
	9	114	28
	10	120	30

Table 5-16 Drone's 17 mm Launching Mechanism Attributes

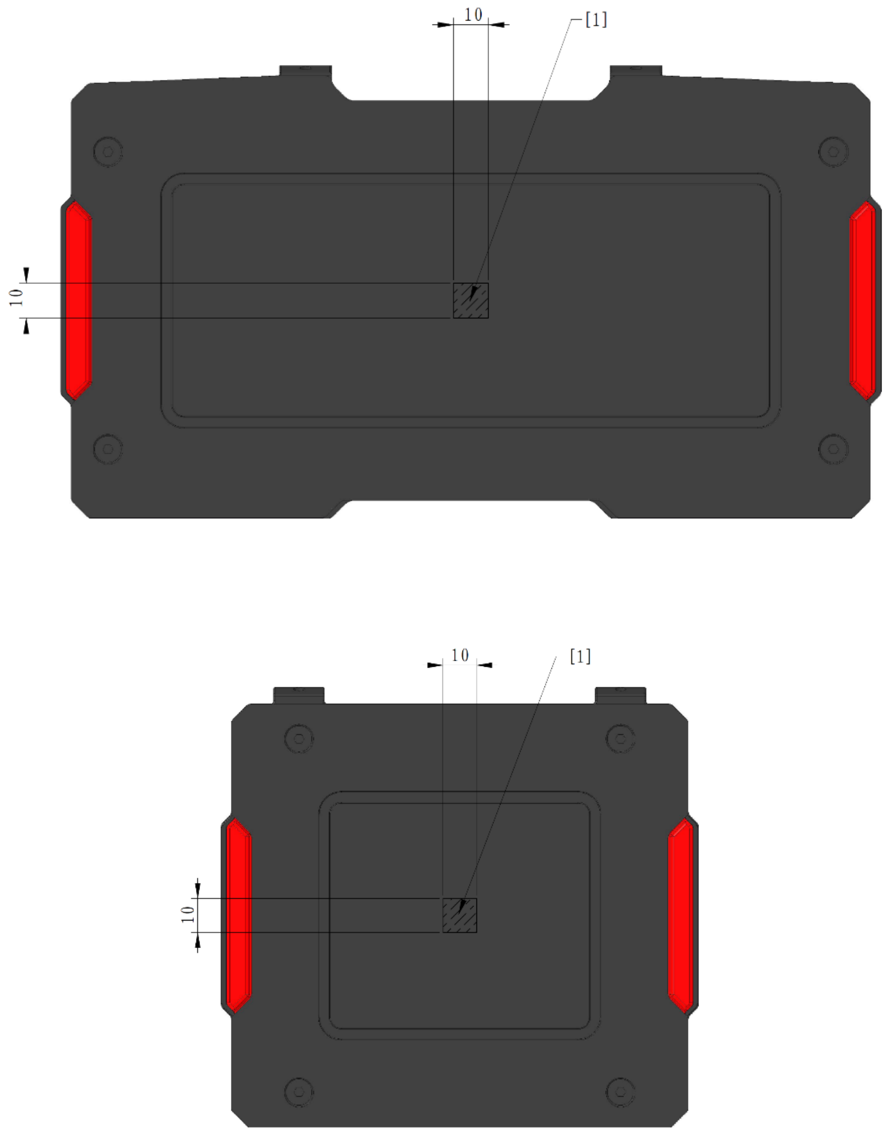
Level	Barrel Heat Limit	Barrel Heat Cooling Rate (per second)
1	**100**	**20**
2	**110**	**30**
3	**120**	**40**
4	**130**	**50**
5	**140**	**60**
6	**150**	**70**
7	**160**	**80**
8	**170**	**90**
9	**180**	**100**
10	**200**	**120**

5.5 Battlefield-related Mechanism

5.5.1 Base and Outpost Mechanisms

Base HP is 5,000, and Outpost HP is 1,500.

The center of the Armor Module of the Base and Outpost (excluding the Dart Detection Module) features a 10 mm × 10 mm rectangular area, as shown in the figure below. If a projectile hits this area, the attack receives a 150% buff.



[1] 10 mm × 10 mm rectangular area

Figure 5-10 The Rectangular Area

When a team's Outpost is alive, the Base is invincible. When its Base HP declines to 2,000 or below, the Base Protective Armor will be enabled.

After a team's Outpost is destroyed, the other team can occupy the Buff Point associated with the destroyed Outpost. For details, see 5.5.3.6Outpost Buff Mechanism.

Each time a team loses 1,000 Base HP, they gain one opportunity to rebuild the Outpost. The opportunities are accumulative. Hero, Infantry, and Sentry robots can rebuild a destroyed Outpost by continuously scanning their Outpost Buff Point RFID Interaction Module Card for 10 seconds. Engineer robots can do so by scanning for 5 seconds. Occupation durations for different robots are calculated independently. This process restores the Outpost to an alive state with 750 HP. The Outpost cannot be rebuilt after 5 minutes into the match.

The central armor of the Outpost can rotate. For its initial position, see Figure 4-32 Outpost. After the match begins, the central armor starts rotating, reaching a speed of 0.8π rad/s within 5 seconds, then maintains a constant speed in a random direction. In each round, the Outposts of both the red and blue teams will rotate in the same, fixed direction.

A team's Outpost armor will stop rotating if the following conditions are met:

- The team's Outpost is destroyed for the first time.
- The other team's Base Protective Armor is expanded.
- The competition has been in progress for three minutes.

The Outpost armor will not rotate again in the current round after it has stopped rotating.

When the Outpost is alive, the process for the central armor to stop rotating is as follows: the central armor decelerates immediately and returns to its initial position within 10 seconds.

- When the Outpost middle armors of both teams stop rotating simultaneously, the time when they restore to their initial positions may differ slightly.
- When an Outpost is alive, the Dart guiding light on the Outpost is solid on, and the Dart guiding light on the Base is off. When the Outpost is destroyed, the Dart guiding light on the Outpost is off, and the Dart guiding light on the Base is on.

5.5.2 Power Rune Mechanism

Robots can activate Power Runes by launching projectiles. Both teams can strike the Power Rune at the same time. Once the red team's Power Rune is activated, it provides buff exclusively to the red team; the same applies to the blue team.

The Power Rune continuously rotates during the match, but its rotation characteristics change depending on its status:

The Power Runes of both teams rotate on the same axis, i.e., the red team's Power Rune rotates in the clockwise direction while the blue team's Power Rune rotates in the counterclockwise direction (as per the rotation direction when facing the respective team's Power Rune). Before the start of a round, the Power Runes rotate in a random direction. During the round, the Power Rune will maintain the same direction throughout.

- The rotating speed of a Small Power Rune is set at 1/3 π rad/s.
- The rotating speed of a Large Power Rune that is not activating is set at 1/3 π rad/s.
- The rotating speed of the Large Power Rune in the activating state varies periodically according to a

trigonometric function. The Target Function for Speed is: $spd = a \times \sin(\omega \times t) + b$, where spd is measured in rad/s, t is measured in s, the value range for a is 0.780 to 1.045, for ω is 1.884 to 2.000, and b always satisfies $b = 2.090 - a$. Each time when the Large Power Rune becomes available, all parameters will be reset, where t shall be 0, and a and ω will be any value within their respective ranges. The actual rotating speed of the Power Rune has a time deviation within 500 ms from the target function for speed.

After a team activates its Power Rune, all alive robots on the team will receive a certain buff while the Power Rune is activated. After the buff effect of the Power Rune ends, the Power Rune will be unavailable. Both teams can have Power Rune buffs at the same time.

According to the time, the Power Rune is divided into two phases: Small Power Rune and Large Power Rune. From the start of the match until the first three minutes, the Small Power Rune stage applies; after three minutes from the start of the match, the Large Power Rune stage applies.

- **Small Power Rune:** The status of each team's Small Power Rune is independent from each other. At the start of the match and at 1 minute 30 seconds, each team receives one opportunity to activate their Small Power Rune, and these opportunities can be accumulated. The Infantry Operator can trigger an inactive Small Power Rune to the Activating state through the Referee System player's client, or the Sentry can trigger it via the Referee System Serial Port. If the Small Power Rune remains unactivated 20 seconds after entering the Activating state, it will revert to the inactive state. After a team's robot activates its Small Power Rune, all robots, Outpost, and the Base of the team will receive a 25% Defense Buff that lasts 45 seconds. During the buff period of the Small Power Rune, all Hero, Infantry, and Drone robots experience a 100% boost in earned experience. The whole team can gain a maximum of 1,200 extra Experience Points in one buff period of the Small Power Rune.
- **Large Power Rune:** The status of each team's Large Power Rune is independent of the other. At 3 minutes, 4 minutes 15 seconds, and 5 minutes 30 seconds after the match begins, each team will receive one opportunity to activate the Large Power Rune, and these opportunities can be accumulated. The Infantry Operator can use the Referee System player's client to trigger a Large Power Rune that is inactive and put it into the Activating state. The Sentry can do so by using the Referee System Serial Port. If the Large Power Rune remains unactivated for 20 seconds after entering the Activating state, it will revert to the unactivated state. Each Armor Module of the Large Power Rune is divided into rings 1-10. After the Large Power Rune is activated by one team, the system will grant varying durations and effects of attack, Defense Buff, and heat cooling buffs to all robots on that team, based on the average ring struck and the number of activated light arms. The team's Outpost and Base will also receive corresponding Defense Buffs. For details, see "Table 5-17 Average Ring and Corresponding Buffs". Meanwhile, when the Large Power Rune is activated, 750 Experience Points will be equally assigned to the team's Hero, Infantry, and Drone robots that are alive.

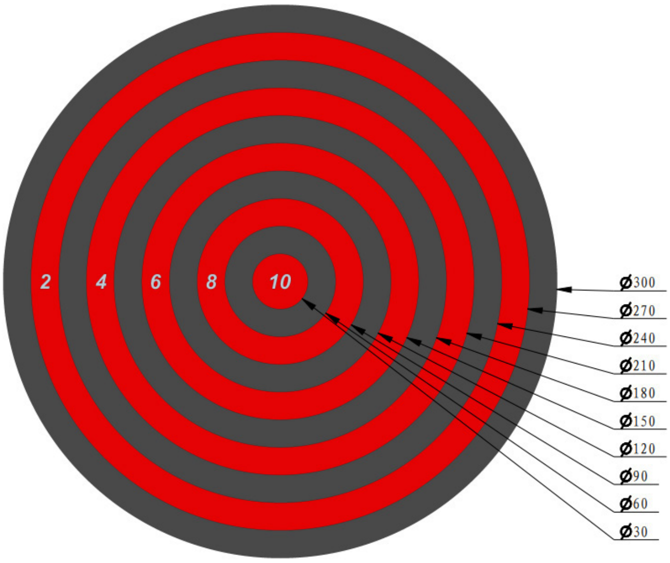


Figure 5-11 Power Rune Strike Area

The Power Rune Armor Module detects impact positions at an accuracy of 1 mm in the radial direction.

Table 5-17 Average Ring and Corresponding Buffs

Average Ring Range	Attack Buff	Defense Buff	Heat Cooling Buff
[1, 3]	150%	25%	None
(3, 7]	150%	25%	2 times
(7, 8]	200%	25%	2 times
(8, 9]	200%	25%	3 times
(9, 10]	300%	50%	5 times

Table 5-18 Number of Activated Light Arms and Corresponding Buff Duration

Number of Activated Light Arms	Buff Duration (seconds)
5	30
6	35
7	40
8	45

Number of Activated Light Arms	Buff Duration (seconds)
9	50
10	60

5.5.2.1 Power Rune Status

The Power Rune has four possible statuses: Inactive, Activating, Activated, and Activation Failed.

1. Inactive

The Power Rune enters the Inactive status, as shown below:

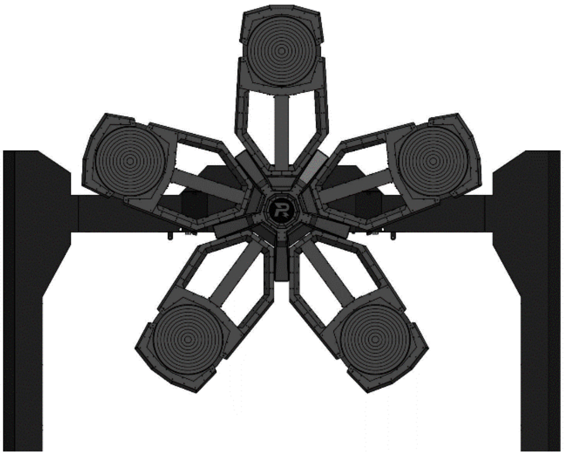


Figure 5-12 The Power Rune in the Inactive Status

2. Activating

Small Power Rune

When the Small Power Rune enters the Activating state, the Power Rune will randomly illuminate one of the five Armor Modules. The illuminated Armor Module displays a special light effect, and the corresponding Light Arm's central shaft shows a Flowing Arrow Light effect. At this time, if a projectile hits any illuminated Armor Module within 2.5 seconds, the corresponding Light Arm will be fully lit with the appropriate Light Effect. Meanwhile, the Power Rune will randomly light up another Armor Module from the remaining four, and the process repeats.

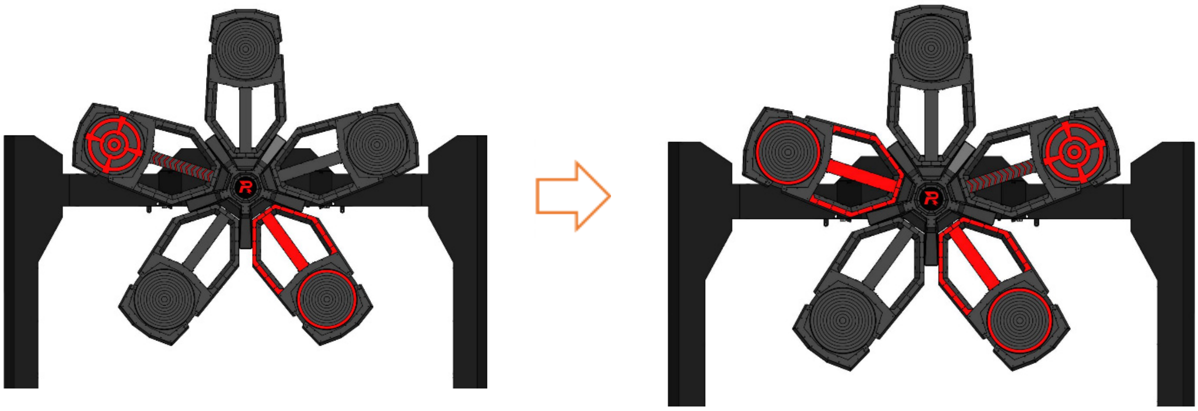


Figure 5-13 Small Power Rune in the Activating State

Large Power Rune

When the Large Power Rune enters the Activating state, two out of the five armor modules on the Power Rune will be randomly illuminated. Each lit armor module will display a special light effect, and the central shaft of the corresponding light arm will feature a Flowing Arrow Light effect. At this point, if a projectile hits any illuminated Armor Module within 2.5 seconds, the corresponding Light Arm will change its Light Effect. Within the following 1 second, robots can hit another illuminated Armor Module. Regardless of success, the Power Rune will then randomly light up any two of the five Armor Modules again, and the process repeats.

The light effects are as shown below:

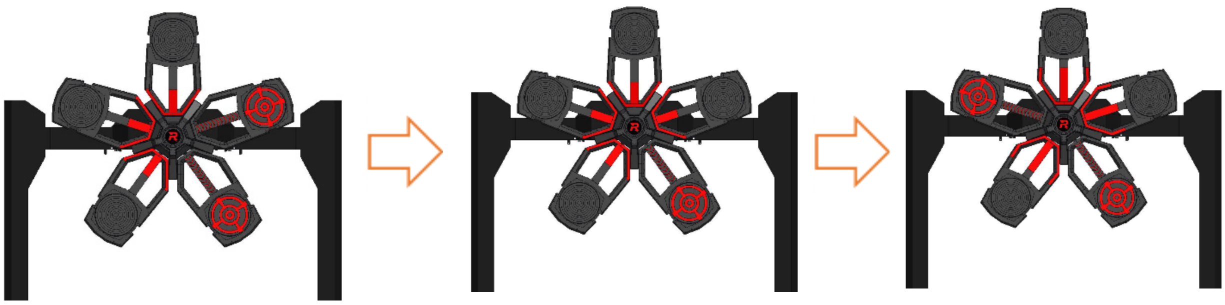


Figure 5-14 Large Power Rune in the Activating State

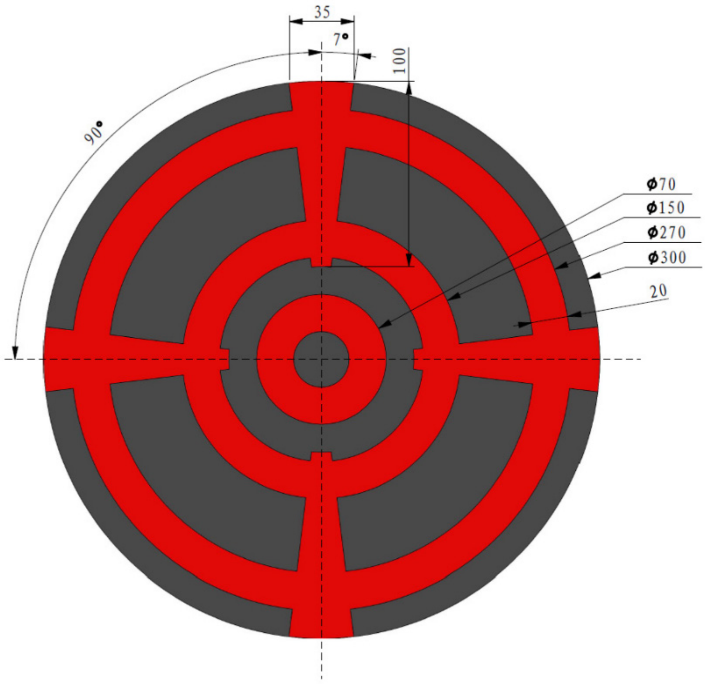


Figure 5-15 Power Rune in the Available State

3. Activated

If all five light arms are activated, the Power Rune is activated as shown below:

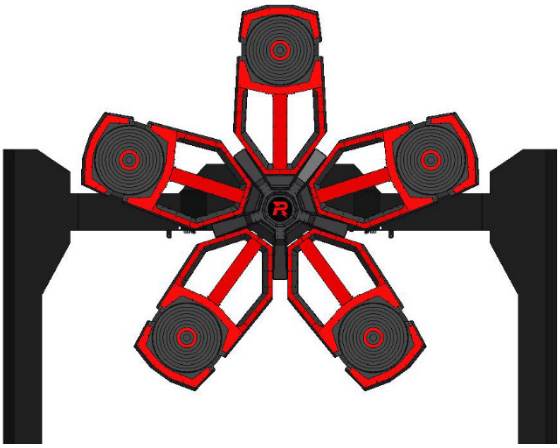


Figure 5-16 The Power Rune in the Activated State

4. Activation Failed

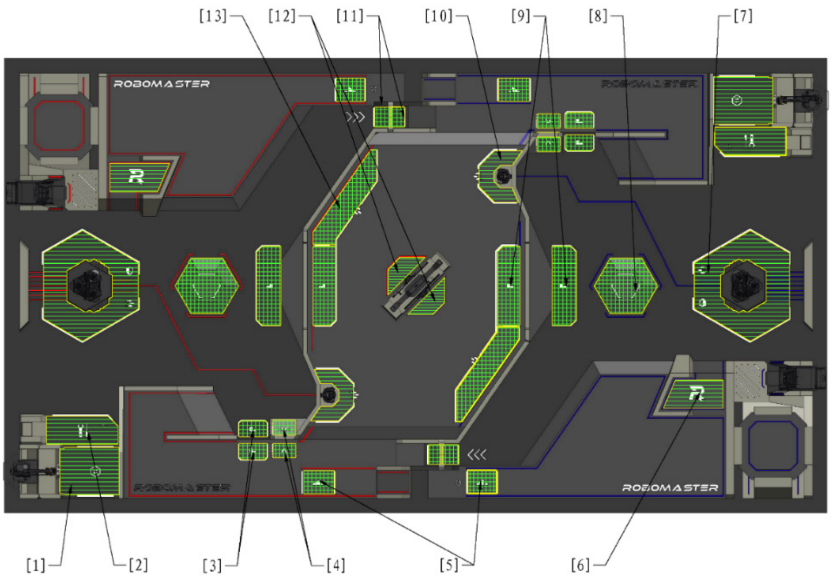
If any of the following conditions occur during activation, the activation will fail and the Power Rune will be reset to the Activating state again:

- Failure to hit a lit Armor Module within 2.5 seconds
- Hit an Armor Module that is not randomly lit

5.5.3 Battlefield Buff Mechanism

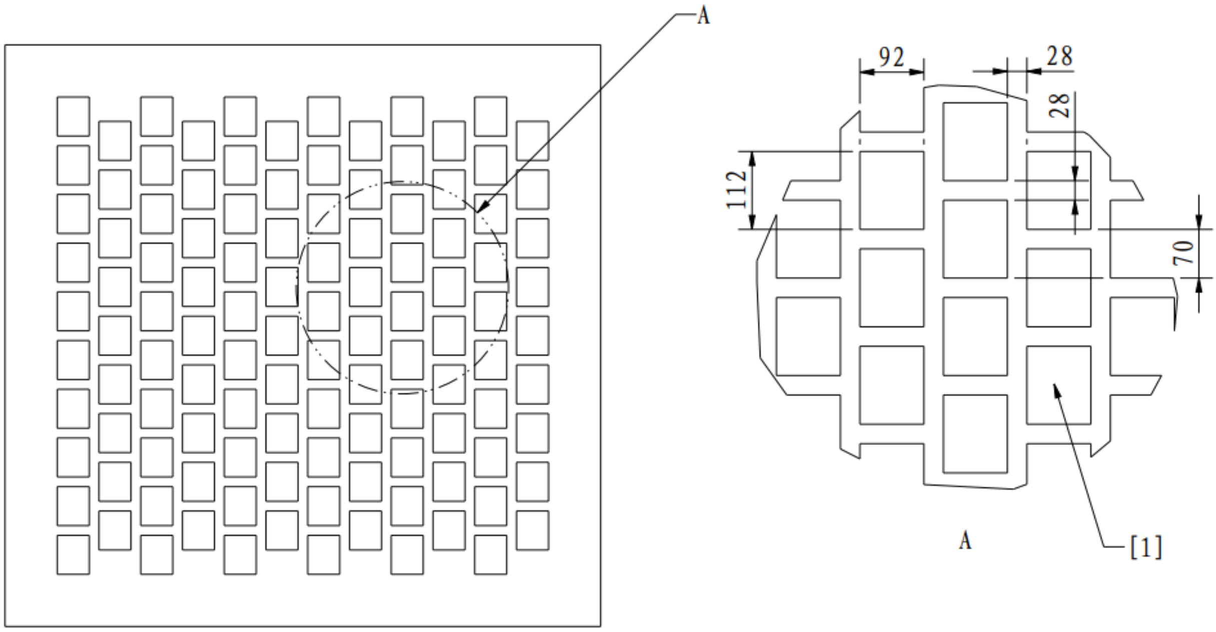
5.5.3.1 Battlefield Buff Points

The Battlefield Buff Points are as shown below: All Buff Points in the Battlefield are laid with RFID Interaction Module Cards, and there may be Deadbands for the RFID Interaction Module Cards. Teams should make adjustments and adapt accordingly. Detection by the RFID Interaction Module may experience a delay, and the expiration of robots' occupy status is also subject to a 2-second delay. If the robot that occupies the buff point is defeated, the buff gained will expire.



[1]	Resource Zone Buff Point	[2]	Resupply Zone Buff Point	[3]	Terrain Crossing Buff Point (Road)	[4]	Terrain Crossing Buff Point (Tunnel)
[5]	Terrain Crossing Buff Point (Launch Ramp)	[6]	Trapezoid-Shaped Elevated Ground Buff Point	[7]	Base Buff Point	[8]	Fortress Buff Point
[9]	Terrain Crossing Buff Points (Elevated Ground)	[10]	Outpost Buff Point	[11]	Terrain Crossing Buff Point (Tunnel)	[12]	Assembly Zone Buff Point
[13]	Central Elevated Ground Buff Point						

Figure 5-17 Distribution on Battlefield Buff Points



[1] Locations where RFID Interaction Module Cards are lodged

Figure 5-18 RFID Interaction Module Card Layout

Table 5-19 Buff Types

Type	Notes
Attack Buff	Increased damage caused by projectile attacks
Defense Buff	Changes the damage suffered from a projectile attack or collision. Negative Defense Buff is also known as “Vulnerability”. Defense Buff is not applicable to HP loss caused by violation penalties, Referee System disconnection, and Darts.
Barrel Heat Cooling Buff	Increases the Barrel Heat Cooling Rate.
Buffer Energy Buff	The robot receives extra buffer energy for chassis power.
HP Recovery Buff	The robot recovers a certain amount of HP every second until it reaches the maximum HP.

Attack Buff changes the projectile damage dealt by robots to: Original Damage × Attack Buff.

Defense Buff changes the Attack Damage received by robots, the Base, and Outposts to: Original Damage × (1 - Defense Buff).

For more information on original damage, see “Table 5-2 HP Loss Mechanism for Attack Damage or Collision”.

Example 1: Say the red team’s robot has a 200% Attack Buff and the blue team’s robot has a 25% Defense Buff, and the progress marked by the radar of the opponent exceeds 100. If the red team’s robot launches a round of 42 mm projectiles and hits the blue team’s robot, the damage suffered by the latter should be: 200 × 200% × (1-25%+15%) = 360.

Example 2: Say the red team’s robot has a 200% Attack Buff and the progress of the blue team’s Hero marked by the radar of the opponent exceeds 100. If the red team’s robot launches a round of 42 mm projectiles and hits the blue team’s Hero, the damage suffered by the latter should be: 200 × 200% × (1 + 15%) = 460.

When a robot receives more than one buff of the same type, the maximum buff effect should be applied, including Attack Buff, Defense Buff, Vulnerability, HP Recovery, and Barrel Heat Cooling Buff.

Example 1: The red team’s Infantry obtains a 25% Defense Buff by activating the Large Power Rune. If the robot then secures the Base Buff Point on its own side, the Defense Buff rises to 50%. However, if a subsequent 15% Vulnerability is applied, the net Defense Buff is 35%.

Example 2: The blue team’s Infantry obtains a 25% Defense Buff by activating the Large Power Rune. If the robot then occupies the Opponent’s Fortress Buff Point, it will have a 75% Vulnerability. If it subsequently receives another 15% Vulnerability, it still retains a 75% Vulnerability.

If the final damage or HP recovery after buff calculation is a decimal number, it will be rounded off.

5.5.3.2 Base Buff Points Mechanism

A team’s Base Buff Point can be occupied by the team’s Ground Robots. Multiple robots from one team can occupy the Base Buff Point simultaneously.

During the seven-minute round, robots that occupy the Base Buff Point on their own side will receive a 50% Defense Buff, exchange for Projectile Allowance, and remove the “Weakened” state. For more information on the “Weakened” state, see “5.2.2 Respawn Mechanism”.

5.5.3.3 Central Elevated Ground Buff Mechanism

Hero, Infantry, and Sentry robots can all occupy the Central Elevated Ground Buff Point. Unconnected Central Elevated Ground Buff Points are occupied separately. Multiple robots from one team can occupy the Central Elevated Ground Buff Point simultaneously. If a robot of one side occupies a Central Elevated Ground Buff Point, no robots of the other side can occupy it at the same time.

Robots that occupy the Central Elevated Ground Buff Point receive a 25% Defense Buff.

5.5.3.4 Trapezoid-Shaped Elevated Ground Buff Point Mechanism

All Ground Robots can occupy the Trapezoid-Shaped Elevated Ground Buff Point on their own side. Multiple robots from one team can occupy the Trapezoid-Shaped Elevated Ground Buff Point simultaneously.

If a Trapezoid-Shaped Elevated Ground Buff Point is occupied by a robot from the same team, the robot occupying the Trapezoid-Shaped Elevated Ground Buff Point will receive a 50% Defense Buff.

5.5.3.5 Terrain Crossing Buff Point Mechanism

There are two Terrain Crossing Buff Points (Elevated Ground) on each side of the central elevated ground. Each team's Road Zone contains two Terrain Crossing Buff Points (Road), two Terrain Crossing Buff Points (Launch Ramp), and four Terrain Crossing Buff Points (Tunnel).

All Ground Robots can occupy the Terrain Crossing Buff Points. Multiple robots from one team and robots from both teams, can occupy a Terrain Crossing Buff Point simultaneously. To activate a Terrain Crossing Buff, a robot must detect the RFID Interaction Module Cards for two Terrain Crossing Buff Points of the same type on one side within X seconds. For Road and Elevated Ground Terrain Crossing Buffs, the robot must detect the lower Buff Point first and then the higher one. The X values are as follows: 3 seconds for Terrain Crossing Buff Points (Road) and (Tunnel), 5 seconds for Terrain Crossing Buff Points (Elevated Ground), and 10 seconds for Terrain Crossing Buff Points (Launch Ramp). The specific buffs are as follows:

Terrain Crossing Buff (Launch Ramp)

- 25% Defense Buff for 30 seconds

Terrain Crossing Buff (Elevated Ground)

- 50% Defense Buff for 30 seconds

Terrain Crossing Buff (Road)

- 25% Defense Buff for 5 seconds
- A robot that receives a Terrain Crossing Buff (Road) cannot receive the same buff again within 15 seconds.

Terrain Crossing Buff (Tunnel)

- Double Barrel Heat Cooling Buff for 120 seconds

When a robot already has any Terrain Crossing Buff that provides a Defense Buff and obtains another Terrain Crossing Buff that provides a Defense Buff, it will gain a 50% Defense Buff. The duration of the Defense Buff will be refreshed to the longer duration of the two Terrain Crossing Buffs obtained.

Example: If a robot consecutively obtains the Terrain Crossing Buff (Launch Ramp) and Terrain Crossing Buff (Road), the duration of its buffs will be 30 seconds.

5.5.3.6 Outpost Buff Mechanism

When a team's Outpost is alive, the team's Hero, Engineer, Infantry, and Sentry robots can all occupy its Outpost Buff Point. Within the first 5 minutes of the match, if the other team's Outpost is destroyed and the team's Outpost is still alive, the other team's Hero, Engineer, Infantry, and Sentry robots can occupy their own Outpost Buff Point. Multiple robots from one team can occupy the Outpost Buff Point simultaneously.

Robots that occupy the Outpost Buff Point can exchange for Projectile Allowance (see 5.3.2Projectile Allowance Mechanism), gain a 25% Defense Buff, and remove the "Weakened" state along with its corresponding Invincible state. For more information on the "Weakened" state, see "5.2.2Respawn Mechanism".

5.5.3.7 Assembly Zone Buff Point Mechanism

Only Engineer robots of a team can occupy the team's Assembly Zone Buff Point.

The Engineer robot that occupies the Assembly Zone Buff Point will become invincible for up to 180 seconds in total.

5.5.3.8 Resupply Zone Buff Point Mechanism

All Ground Robots can occupy their own side's Resupply Zone Buff Point. Multiple robots from one team can occupy the Resupply Zone Buff Point simultaneously.

A robot that occupies its team's Resupply Zone Buff Point can speed up the respawn timer or receive an HP Recovery Buff. See 5.2HP Recovery and Respawn Mechanism for detailed implementation methods and figures.

5.5.3.9 Resource Zone Buff Point Mechanism

Only Engineer robots of a team can occupy the team's Resource Zone Buff Point.

After occupying the Resource Zone Buff Point, the Engineer robot will become invincible.

5.5.3.10 Fortress Buff Point Mechanism

Only Infantry or Sentry robots from a team can occupy the team's Fortress Buff Point. A team's Fortress Buff Point can be occupied by robots from both teams, but can only be occupied by the team's own robots at any given time. A team's Fortress Buff Point can be occupied by multiple opponent robots at the same time.

A team's Fortress Buff Point remains unavailable until its Outpost is destroyed. A team's Fortress Buff Point becomes available when its Outpost is destroyed.

Occupation by Own-Side Robots

Define Δ as the difference between the maximum and current HP of the own-side Base.

Robots on the own side that occupy the Fortress Buff Point receive a 50% Defense Buff, as well as a Barrel Heat Cooling Buff equal to w, where w is related to Δ and follows the formula: $w = \frac{\Delta}{40}$ (rounded down). The upper limit for w is 75. The Barrel Heat Cooling Buff from the Fortress Buff Point is applied at the highest rate among all available Barrel Heat Cooling Buffs.

Example:

A robot with a Barrel Heat Cooling Rate of 80/second has received a 150 Barrel Heat Cooling Buff from the Fortress Buff Point and a 5-time Barrel Heat Cooling Buff at the same time. Since 80 * 5 > 80 + 150, the actual Barrel Heat Cooling Rate of the robot is 400/second.

After a robot occupies the Fortress, the Fortress Buff Point will provide the robot with a “reserved Projectile Allowance,” which is independent from the robot’s own Projectile Allowance, with an upper limit of N. When a robot occupying the Buff Point of the Fortress launches projectiles, they will prioritize consuming the “reserved Projectile Allowance”. Once the reserved Projectile Allowance is depleted, the robot’s own Projectile Allowance will be used.

$$N = 100 + 2 \times \frac{\Delta}{15} \text{ (rounded down), with N up to 500.}$$

Occupation by Opponent Robots

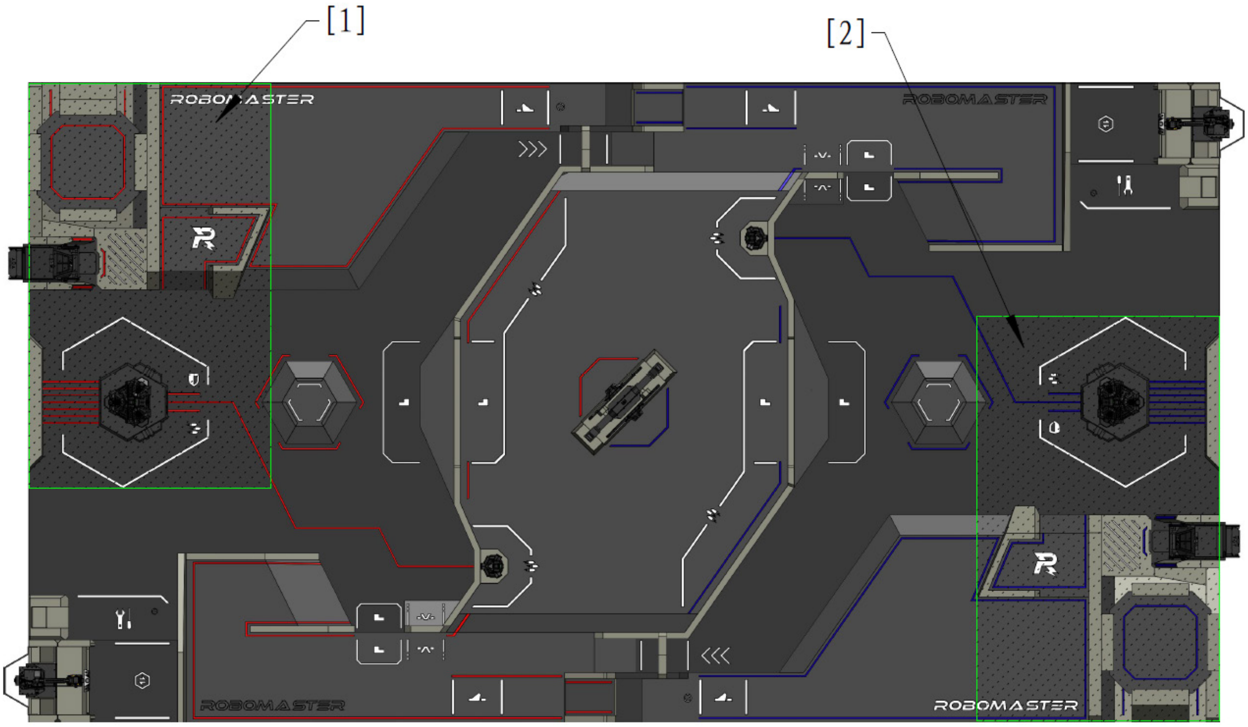
After the match has been underway for 3 minutes and the opponent’s Outpost has been destroyed, Infantry and Sentry robots can occupy the opponent’s Buff Point.

Robots occupying the opponent’s Buff Point become 100% more susceptible to damage. When a single robot accumulates 20 seconds of uninterrupted occupation, the opponent’s Base Protective Armor will deploy. The timer for occupying the opponent’s Buff Point can be retained for 3 seconds after the robot is destroyed or the Occupy status expires, and the timer will pause during this retention period.

5.6 Special Mechanism of Robots

5.6.1 Special Mechanism of Hero

When a Hero selects the “Long-Rang Attack-Focused” type and is located within the deployable area of its team (see Figure 5-19 Own-Side Deployable Zone), it can choose to enter Deployment Mode. The Hero enters Deployment Mode two seconds after confirmation. In this mode, the Hero’s chassis powers off, video transmission is lost, a 25% defense buff is gained, the maximum initial launching speed increases to 16.5 m/s. When it launches 42 mm projectiles to attack the Base, a 150% attack buff is granted.



[1] Red Team's Deployable Area [2] Blue Team's Deployable Zone

Figure 5-19 Own-Side Deployable Zone

- Criteria for determining whether the Hero is within the deployable area of its team:

  ① The Hero's RFID Interaction Module does not detect an RFID Interaction Module Card of the opponent team's deployable zone.

  ② The Hero's RFID Interaction Module detects an RFID Interaction Module Card in the deployable area of its team, or the coordinates from the UWB Positioning Module are within the deployable area of its team.

  The Hero is considered to be within its team's deployable area when Conditions ① and ② are all met.

- For information on how a Hero enters and exits the Deployment Mode, see the Referee System Player's Client Interface Instructions.

5.6.2 Special Mechanism of Engineer

During the first three minutes of the competition, the Engineer has a 50% Defense Buff.

5.6.3 Special Mechanisms of Drone

At the start of the competition, Drone has 30 seconds of air support time, with an additional 20 seconds added every minute thereafter.

The Gimbal Operator can call for or pause air support via the Referee System player's client. During air support, Drones will receive the FPV of the Battlefield and can launch projectiles while not in contact with the Landing Pad. During the Seven-Minute Round, Drones cannot obtain projectiles from any source. During air support, the robot's Launching Mechanism is unlocked; otherwise, it is locked.

If the Aerial Gimbal Operator does not pause Air Support after the Air Support time runs out, each additional second of Air Support will consume one Gold Coin. For details, see Table 5-8 Exchange Rules.

In each round of the competition, the maximum allowance for a Drone is 750 rounds of projectiles.

A laser launching device can be installed on the Radar system for sighting and illuminating the Laser Detection Module on the Drone. The laser launching device and actuator on a team's Radar system are powered on only when the other team's Drone initiates Air Support. If the Laser Detection Module installed on the Drone is continuously illuminated for 10 seconds, its Launching Mechanism will be locked and will unlock after 45 seconds. In each round, the above mechanism can be triggered up to three times.

5.6.4 Special Mechanism of Sentry

After the Three-Minute Setup Period begins and before the 15-Second Referee System Initialization Period, the Gimbal Operator may choose either automatic control or semi-automatic control mode for the corresponding Sentry. If the control mode is not selected within the specified time, it will default to automatic control at the start of the seven-minute round. The Sentry can exchange for Projectile Allowance, remotely exchange for projectiles and HP, confirm respawn, exchange for instant respawn, and switch modes, by communicating directly with the Referee System server.

In addition, the Gimbal Operator can use relevant commands from the Referee System player's client to intervene in the actions of the Sentry. The specific commands include:

- Forward the Mini-map marker
- Purchase Projectile Allowance in the Resupply Zone, at the Base Buff Point, or at the Outpost Buff Point.
- Confirm respawn
- Perform remote exchange for projectiles
- Perform remote exchange for HP
- Perform remote exchange for instant respawn
- Switch modes

For semi-automatic Sentry models, the above operations do not require any Gold Coin; for fully automatic Sentry models, the Gimbal Operator must spend 50 Gold Coins for each operation performed.

The Sentry features a "pose" system that enhances certain abilities while reducing others. Sentry starts each round in the Mobility mode by default. Switching modes has a 5-second cooldown. If a Sentry remains in the same mode for more than 3 minutes during a round, the effect of that mode will decrease, as detailed in the table below:

Mode	Default Effect	Decreased Effect
Offense	Gain triple heat cooling buff, have Chassis Power halved, and become 25% more vulnerable.	Gain 2× heat cooling buff, have Chassis Power halved, and become 25% more vulnerable.
Defense	Gain a 50% Defense Buff, have Chassis Power halved, and have the heat cooling rate reduced to one-third of the original.	Gain a 25% Defense Buff, have Chassis Power halved, and have the heat cooling rate reduced to one-third of the original.
Mobility	Have the Chassis Power Limit increased to 1.5 times the original, become 25% more vulnerable, and have the heat cooling rate decreased to one-	Have the Chassis Power Limit increased to 1.2 times the original, become 25% more vulnerable, and have the heat cooling rate

	third of the original.	decreased to one-third of the original.

All of the calculations above are rounded to the nearest integer.

5.6.5 Special Mechanism of Dart System

In each round, the Dart System can load up to four Darts, the gate of a Dart Launching Station can be opened twice, once after 30 seconds of the competition have elapsed, and the other after 4 minutes. The Gimbal Operator can choose when to open the gate, and any unused opportunities are cumulative.

Opening and Closing of the Dart Launching Station Gate

During the match, the Gimbal Operator will use the keyboard and mouse on the Referee System player's client to open and close the gate of the Dart Launching Station. The Gimbal Operator is not allowed to launch Darts when the gate is opening or closing. The Referee System player's client will display the status of the gate.

It takes around seven seconds for the gate to open completely.

When the gate of the Dart Launching Station is fully open, the Gimbal Operator can then launch Darts by controlling the Dart System within a 30-second period. Meanwhile, the Dart Detection Module on the Outpost or Base of the other team will update the detection window, which lasts for 40 seconds. The launched Dart needs to hit the Dart Detection Module within the detection window period, or the attack will be invalid.

When the gate closes for the first time, the Dart Launching Station will enter a 15-second cooling period. The gate can only open for the second time after the end of the cool-down period. During the cool-down period, the Dart Launching Station's gate cannot open.

Selecting a Dart Target

After an Outpost is destroyed, the Gimbal Operator can choose the Dart's target as a "Fixed Target", "Random Fixed Target", "Random Moving Target", or "Terminal Moving Target" via the Referee System player's client before the Dart Launching Station's gate opens.

- If "Fixed Target" is selected, the Dart Detection Module will remain in its initial position.
- If "Random Fixed Target" is selected, the Base Dart Detection Module will shift to another random location before the gate fully opens, and move back to the initial location when the Dart Detection Window ends. In the detection window, if a Dart hits the target, the Dart Detection Module will shift to a new location.

- If “Random Moving Target” is selected, the Base Dart Detection Module will begin moving as the gate starts to open. After a launched dart is detected, the module shifts to a new location distinct from where it was at the time of the launch within 600 ms and then remains stationary at the new location. In the detection window, after a Dart hits the target or 10 seconds after the first dart launch is detected, the Dart Detection Module resumes moving until the next launch is detected.
- If “Terminal Moving Target” is selected, when the gate of the Dart Launching Station opens, the Dart Detection Module at the Base will begin to move. After 1.2 seconds from detecting the launch of any Dart, it will start to move to a new location different from where it was at the moment the Dart was launched, completing the move and coming to a stop within 600 ms. In the detection window, after a Dart hits the target or 10 seconds after the first dart launch is detected, the Dart Detection Module resumes moving until the next launch is detected.

The Dart Detection Module and Dart Guiding Light maintain their relative locations while moving, i.e., moving within the same plane of their original locations. Their movement extends parallel to the field’s width, ranging from –280 mm to 280 mm from their starting point.

To ensure detection accuracy, there is a delay in the determination of damage caused by 42 mm projectiles hitting the Dart Detection Module in the detection window.

Gains for Successful Dart Hits

The following are the additional gains for a successful dart hit, apart from the Damage dealt to the target itself:

- When a Dart hits the other team’s Outpost, the Fixed Target or the Random Fixed Target of the other team’s Base for the first, second, third, and fourth time, the operating interface of all the team’s operators, the location of robots on the mini-map, and the custom UI will be blacked out for 10 seconds (first time), 5 seconds (second time), 3 seconds (third time), and 2 seconds (fourth time), respectively. When a Dart hits the other team’s Base, the alive Hero, Infantry, and Drone of the attacking side share 200 (Fixed Target) or 600 (Random Fixed Target) Experience Points.
- When a Dart hits a Random Fixed Target at the Opponent’s Base, any Terrain Traverse or Power Rune Buff Effect currently held by the other team will temporarily expire. Timing continues as normal, but no Buff Effect is granted during this period. The expiration duration matches the blackout duration mentioned above.
- When a Dart hits a Random Moving Target or Terminal Moving Target at the other team’s Base, the operating interface of all the team’s Operators, the location of robots on the mini-map, and custom UI will be blacked out for 10 seconds. The alive Hero, Infantry, and Drone of the attacking side will each share 2,500 Experience Points, and any Power Rune Buff Effect and terrain crossing bonus currently held by the other team will expire.
- If the Dart Detection Module of the Base is hit after “Random Moving Target” or “Terminal Moving Target” is selected, all alive Ground Robots of the other team will immediately lose 10% (for Random Moving

Target) or 25% (for Terminal Moving Target) of their maximum HP respectively. This HP loss is counted in the total damage for the other team, but this HP loss itself and any resulting Defeated Robots will not generate any experience. If the target is hit continuously, the duration when the operating interface is blacked out will increase accordingly.

- If four Darts all hit the other team's Base when "Fixed Target" or "Random Fixed Target" is selected, or if any of the Darts hits the other team's Base when "Random Moving Target" or "Final Moving Target" is selected, the Base Protective Armor of the attacked Base will be expanded immediately.
- When the Dart Guiding Light on the Base or Outpost is illuminated, its buff points will expire temporarily for 30 seconds if the Base or Outpost is hit by a Dart; if it is hit successively, the expiry period needs to be refreshed. After each dart hit, the detection window for the corresponding target will close for 2 seconds.

- During periods when the operator interface, the location of robots on the mini-map, or the custom UI are blocked, the Operator can choose to control the robots in a semi-automatic control mode. Specifically, the Referee System player's client will display a large map interface similar to what the Gimbal Operator sees. The Operator can send command instructions to control the robots by clicking on the large map interface, with at least 3 seconds required between consecutive commands.
- The custom client can still normally receive video transmission and other data.

5.6.6 Special Mechanism of Radar

The functions of the Radar include radar detection, radar countermeasures, and radar information wave analysis, as detailed below:

Radar Detection

A team's Radar can detect the locations of the other team's Ground Robots and send the coordinates of the robots to the Referee System server.

- If the straight-line distance error between this coordinate and the actual planar coordinate detected by the corresponding UWB Positioning Module on the robot is less than 0.8 m, it will be marked as "accurate."
- If the error is greater than or equal to 0.8 m but less than 1.6 m, it is considered "semi-accurate."
- If it is greater than or equal to 1.6 m, it will be marked as "inaccurate".

Each Ground Robot and Drone has a Tracking Progress value, ranging from 0 to 150. The accuracy of a marking and the previous marking as well as the Radar's coordinates transmission interval will affect the Tracking Progress.

The specific rules are shown below. Assuming the impact of the previously received coordinates data on a robot's Tracking Progress is "x", and the initial value of x is "0".

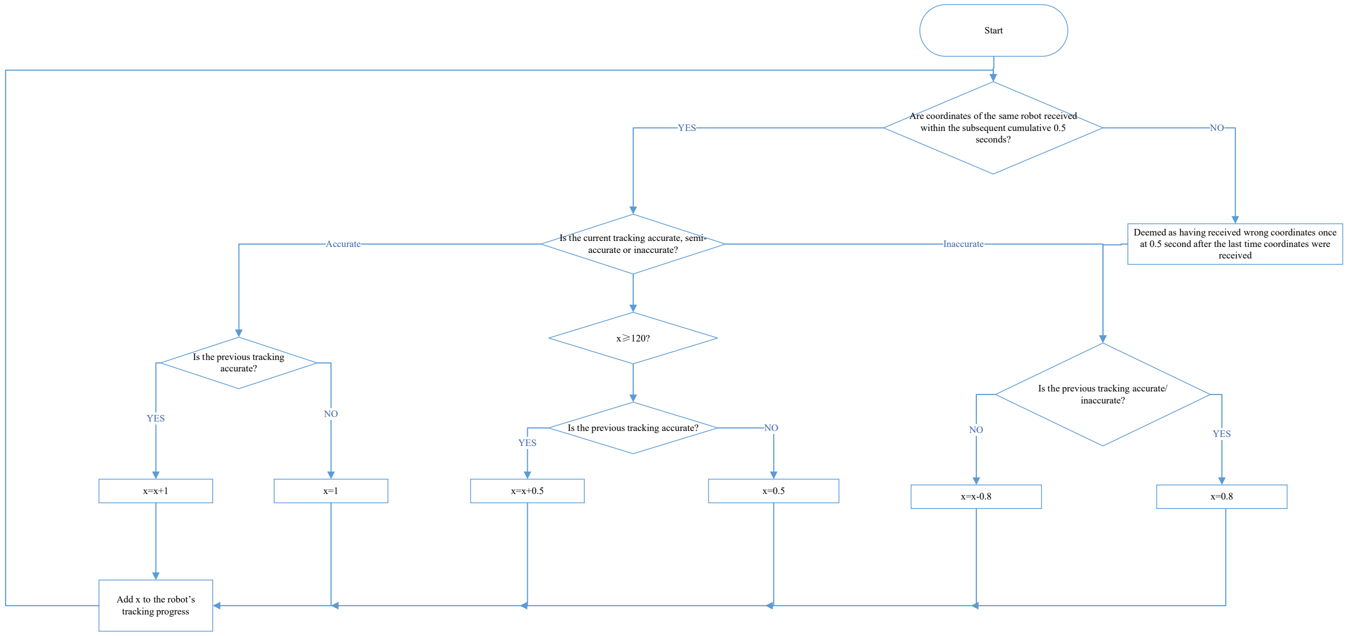


Figure 5-20 Radar Mechanism Logic Diagram

Example: If a team's Radar sends the coordinates of the other team's Engineer at a frequency of 5 Hz and an accuracy of 100% continuously to the Referee System server, the Tracking Progress of the Engineer within 0 to 1 second at an interval of 0.2 seconds will be 1, 3, 6, 10, and 15 respectively. At the 3rd second, the robot's Tracking Progress will reach 120. Thereafter, if the Radar sends the robot's coordinates at a frequency of 5 Hz and an accuracy of 0% to the Referee System server, the Tracking Progress of the robot within 3 to 4 seconds at an interval of 0.2 seconds will be 119.2, 117.6, 115.2, 112, and 108 respectively. Thereafter, if the Radar does not send the Engineer's coordinates for 1 seconds, the Tracking Progress of the robot within 4 to 5 seconds will be 103.2 and 97.6 respectively (determined as two additional and consecutive errors based on the previous error). Thereafter, if the Radar sends the correct coordinates of the Engineer twice consecutively, the robot's Tracking Progress will be 98.6 and 100.6 respectively.

When the Tracking Progress of a team's robot is greater than or equal to 100 but less than 120, the other team's mini-map will display the actual location detected by that robot's UWB Positioning Module and mark it with a special tag. The marked Ground Robot will receive a 15% vulnerability effect. When the Tracking Progress of a team's Ground Robot reaches or exceeds 120, the marked Ground Robot will receive a 20% vulnerability effect. Otherwise, it will only show the location corresponding to the robot's coordinates sent by the team's Radar and will not mark the location with a special tag or have other effects.

When the Tracking Progress for a team's robot reaches 50 or higher, the team's mini-map will display the actual location detected by the UWB Positioning Module for that robot and mark it with a special tag. If the Tracking

Progress is less than 50, only the location corresponding to the robot's coordinates sent by the team's Radar will be displayed, without any special marking or additional effects.

When the radar accumulates a vulnerability effect on the opposing robot for 1 minute (When multiple robots are affected, the duration of the buff applied to these robots will be calculated separately), the Radar will receive a chance to trigger "double vulnerability". The Radar can send a command to the Referee System, consume the chance, and turn the -15% Defense Buff of all vulnerable robots into a -30% Defense Buff, which will last 30 seconds. This "double vulnerability" effect can be triggered twice at most for each round.

If the wireless environment of the competition site is poor, or a large part of the Positioning System Module is blocked, the coordinates detected by a robot's Positioning System Module may drift intermittently or continuously.

Radar Countermeasures

For details, see "5.6.3 Special Mechanisms of Drone".

Radar Analysis of Information Waves

During the match, the Battlefield will have electromagnetic waves containing opponent team information, as well as interference waves in nearby frequency bands.

Participating teams can design and install antennas and demodulation circuits on the Radar to analyze and recognize information waves and interference waves, providing information and advantages for their team.

Frequency, bandwidth, and strength of information waves are detailed in the table below:

Table 5-20 Frequency, Bandwidth, and Strength of Information Waves

Wave Source	Center Frequency (MHz)	Bandwidth (MHz)	Power Strength (dBm)
Red team broadcast source	433.2	0.5	-60
Red team Level 1 interference source	432.2	1	-10
Red team Level 2 interference source	432.6	0.6	10
Red team Level 3 interference source	433.2	0.4	-10

Wave Source	Center Frequency (MHz)	Bandwidth (MHz)	Power Strength (dBm)
Blue team broadcast source	433.92	0.5	-60
Blue team Level 1 interference source	434.92	1	-10
Blue team Level 2 interference source	434.52	0.6	10
Blue team Level 3 interference source	433.92	0.4	-10

The information wave will include the following details about the opponent team:

- Each robot's location
- Each robot's HP
- Remaining ammunition for each robot
- Remaining funds of the team
- Total funds of the team
- Buff Points status

At the start of the round, both sides' interference waves are set to Level 1 by default.

Interference waves do not contain specific information, but include a 6-character alphanumeric key custom-set by the opponent, with a default value of RM2026.

Radar can interpret the key and send it to the RoboMaster Champion server. If the interpretation is successful, the subsequent opponent interference wave difficulty increases by one level. When the difficulty level of the opponent interference wave difficulty is higher than that of the team, it will not cause vulnerability or accumulate "double vulnerability" progress even if the opponent Radar accurately marks the team's robots.

5.6.7 Energy Mechanism

In each round, every Hero, Infantry, or Sentry robot has an initial chassis energy of 20,000 and a chassis energy limit of 40,000. After the chassis energy is depleted to 0, robots will enter the power-saving mode. Robots in the power-saving mode will have their Chassis Power Limit reduced to 35 W. When the chassis energy is greater than or equal to 25,000, the base chassis power limit for robots will increase to 125% of the base value corresponding to the relevant Performance System and Level, but it will not exceed 200 W.

Chassis Energy Consumption

Each time the Power Management Module Chassis Port outputs 1 J of energy, the chassis energy decreases by 1. Chassis energy will not become negative.

Replenishing Chassis Energy

When occupying the Resupply Zone Buff Point, robots can recharge chassis energy wirelessly.

When the Supercapacitor Management Module Port (input) is transferring energy to the supercapacitor, the chassis energy increases by 8 for every 1 J difference between the energy output from the Supercapacitor Management Module Port (input) and the Power Management Module Chassis Port.

5.7 Custom Client

A Custom Client is a multi-purpose device developed by participating teams to receive competition-related information and robot video feeds sent by the Referee System's RoboMaster Champion server, and to transmit competition-specific interaction commands to the RoboMaster Champion server.

A typical Custom Client may consist of a display device, a computing platform (including memory, controller, and processor), and a set of input devices; it can also be integrated into a single unit, such as a handheld tablet device with a network port.

Example: A team developed a Custom Client capable of receiving robots' video transmission and displaying it on a self-made monitor, while simultaneously overlaying a custom UI shown via an unofficial data link. The robots can be controlled through a Custom Controller or a homemade joystick. As a result, the team can complete the competition without having to use the official Referee System player's client.

The Custom Client can connect to the RoboMaster Champion Server via the RJ45 interface configured in the Operation Room to send and receive relevant data, and can also communicate with the Custom Controller via wired connection.

A team's Custom Client can receive competition information accessible to the team from the Referee System player's client, such as both teams' HP, the team's economy, remaining match time, and announcements of key match events.

The Custom Client can send competition commands supported by the player's client to the Referee System, such as selecting a Performance System before the match, exchanging Projectile Allowance during the match, and initiating Air Support or other active commands.

5.8 Match Format and Winning Criteria

The official competition of RMUC 2026 consists of two stages: Group Stage and Knockout Stage. The Group Stage follows the BO2 or BO3 competition format, while for Knockout Stage it is BO3 or BO5.

The winning criteria for a single round are described below:

1. When a round has ended and if the Base of either team is destroyed, the team with the higher Base remaining HP will be the winner.
2. If a round has ended and the Remaining Base HP of both teams is the same and neither Outpost has been destroyed, the team whose Outpost has higher remaining HP will be the winner.
3. If a round has ended and the Remaining Base HP of both teams is the same, but only one team's Outpost has been destroyed, the team whose Outpost remains intact will be the winner.
4. If a round has ended and the Remaining Base HP of both teams are the same and the Outposts have the same remaining HP or have all been destroyed, the team that has inflicted the higher total Attack Damage will be the winner.
5. If a round has ended and the Remaining Base HP of both teams is the same, the Outposts have the same remaining HP or have all been destroyed, and the total Attack Damage inflicted by each team is the same, the team with the higher total remaining HP will be the winner.
6. If neither team fulfills these criteria, the round is considered a draw. In BO3 and BO5 rounds, any draw will result in an immediate tie-breaker round, until a winner emerges.

6. Competition Process

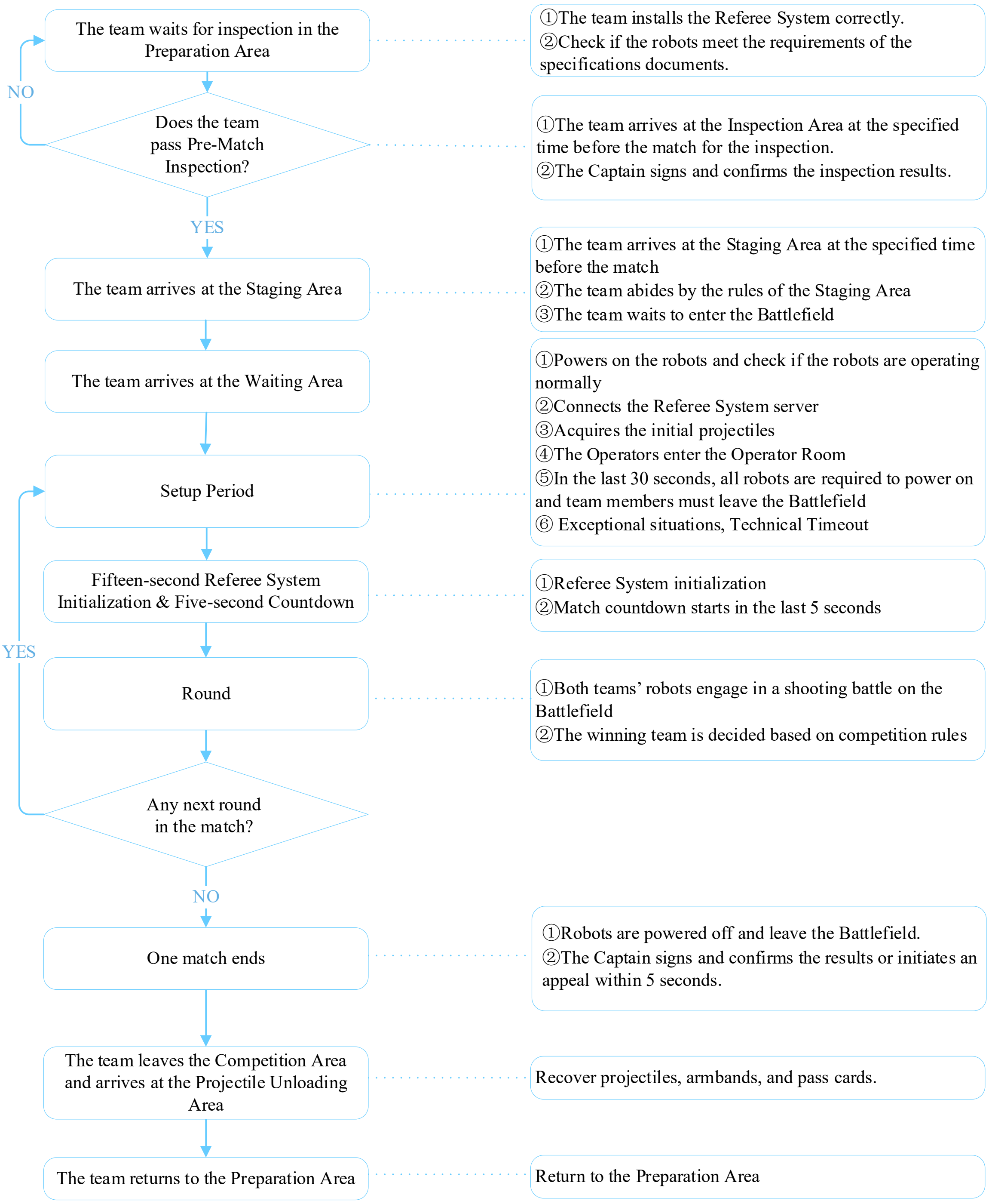


Figure 6-1 Process of a Single Match

6.1 Pre-Match Inspection

- The inspection results of the Mock Inspection and Practice Match are for reference only and are not taken into account for the inspection in the actual competition.
- The inspection results during the competition are only applicable to the current match.
- Passing of inspection only means that the robot meets the specifications at the time of inspection. Teams are required to ensure that their robots fulfill the requirements of the Building Specifications Manual at all times. During inspection, teams must proactively demonstrate all robot functions to inspection personnel.

To ensure that robots built by the participating teams meet the requirements outlined in the RoboMaster 2026 University Series Robot Specification Manual, participating teams are advised to enter the Inspection Area for Pre-Match Inspection 60 minutes before each match begins. If a participating team does not begin Pre-Match Inspection 55 minutes prior to the start of the match, their Backup Robot will lose eligibility to enter the field. If Pre-Match Inspection has not started 50 minutes before the beginning of the match, one Ground Robot will be disqualified from entering the field. Any robots that have not arrived at the Inspection Area 45 minutes before the start of the match will not be permitted to enter the Inspection Area. The inspection process is as follows:

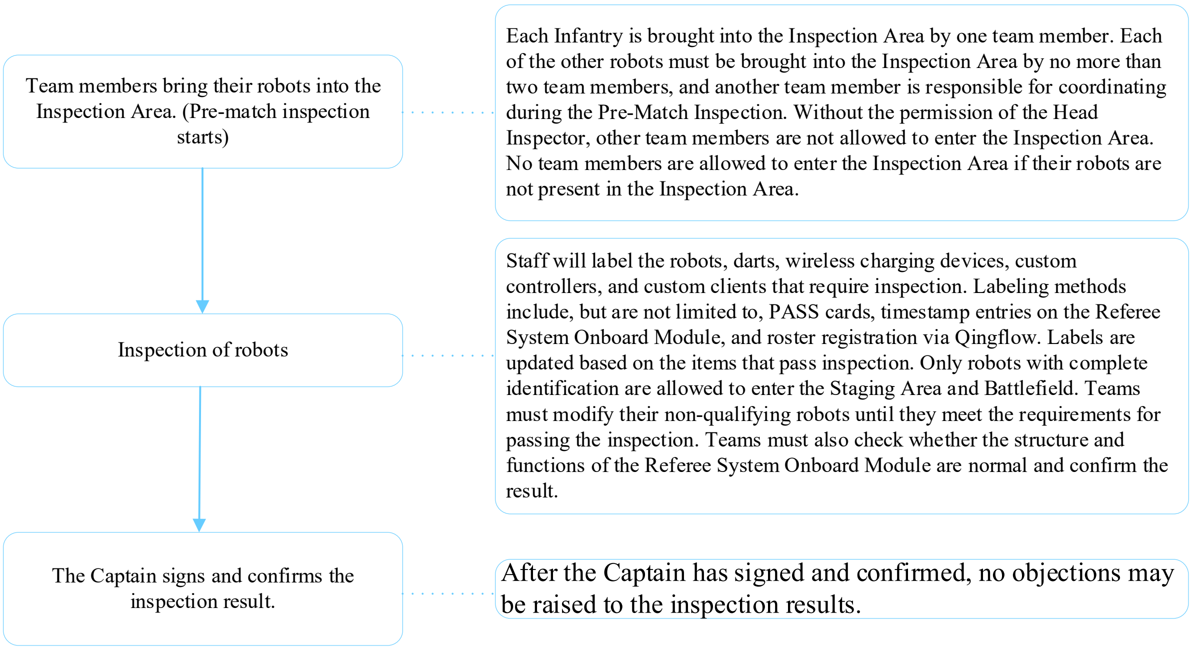


Figure 6-2 Pre-Match Inspection Process

The rules regarding backup robots are as follows:

- During the mock inspection, each team may register up to 10 robots. Throughout subsequent matches in the same competition stage, both the robots and backup robots of participating teams must be listed on the registered roster, and no substitutions are allowed within the same competition stage. Unique identifiers, such as tamper-evident labels or laser engraving (to be determined onsite), are used to mark the primary structures of registered robots. During competition inspections, these identifiers will be verified. If a marked robot requires maintenance during the competition, the unique identifier will be affected, and any changes must be reported through the designated channel.

- Each team can have a maximum of two backup robots for each match. A maximum of four backup Darts are allowed in each BO2 and BO3 match, while a maximum of eight backup Darts are allowed in each BO5 match.
- Team members are required to declare the types of backup robots they are carrying during the Pre-Match Inspection. Backup Hero and Engineer robots must have an Armor Sticker affixed in the Inspection Area. If the backup robot needs to enter the field as an Infantry or Sentry, the Pit Crew must promptly obtain the appropriate Armor Sticker from the Referee. Armor stickers must be affixed in accordance with the requirements stated in the RoboMaster 2026 University Series Robot Specification Manual.
- Teams can borrow the Referee Systems for no more than two backup robots.

6.2 Staging

Teams must arrive at the Staging Area 10 minutes before each match. The staff at the Staging Area will verify the Pass Cards of participating robots and details of Pit Crew, and issue armbands. Each team is allowed to have a maximum of 20 Pit Crew members, which must include up to 19 Regular Members, a Team Advisor, and one Supervisor. One Pit Crew Member should wear the "Captain" armband and undertake the captain's role. If any team needs to repair its robots after entering the Staging Area, they must obtain the permission of the referee. A robot may leave the Staging Area for repair only after the staff at the Staging Area have removed the Pass Card on the robot. When repair is finished, the robot needs to be brought back to the Inspection Area for another Pre-Match Inspection before re-entering the Staging Area. If the team is unable to arrive at the Staging Area in time as a result of this delay, the robot will not be able to enter the match, and the team will bear the consequences.

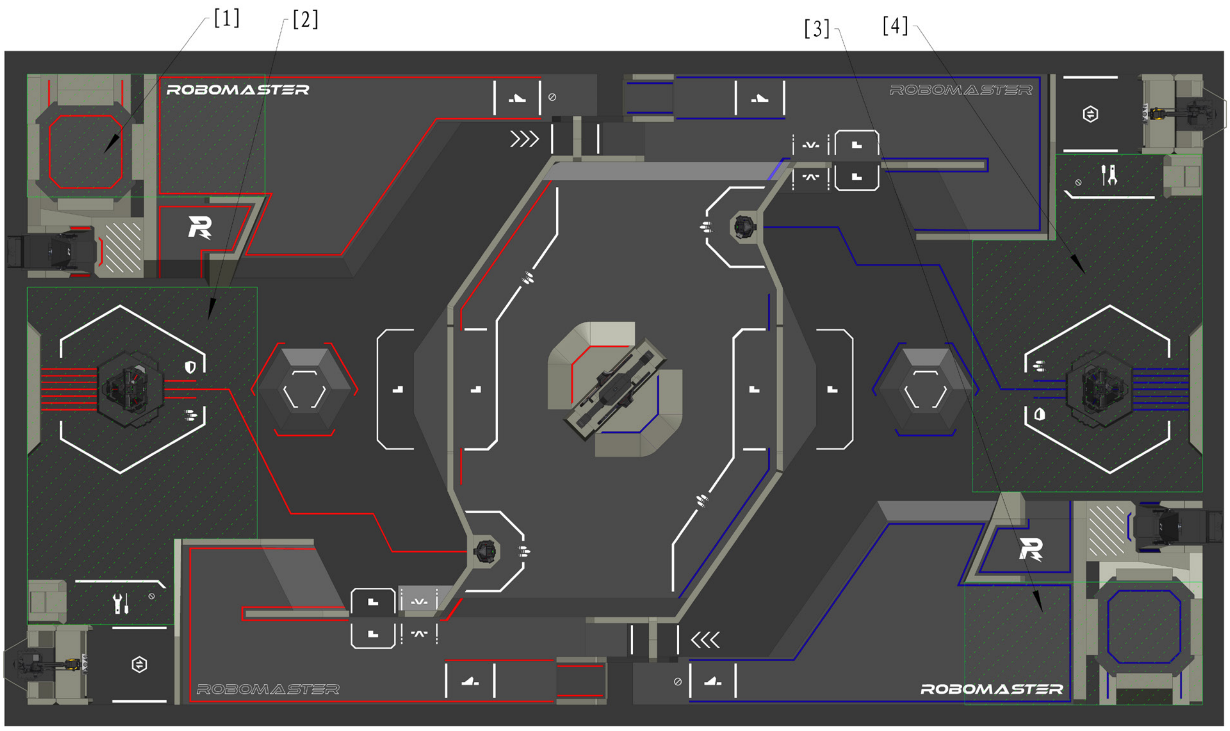
Captain Armband: Any Regular Member wearing the Captain Armband will assume the Captain role during the match. The Captain's duties include managing the team's competition process, confirming results, and raising appeals.

After leaving the Staging Area, participating teams will enter the Staging Area to place their robots. With the permission of the referee, participating teams will wait at the entrance of the Battlefield with their robots for further instructions. The referee will follow the competition process and open the door and lead the team members into the competition area. The countdown for the Three-Minute Setup Period begins once the door is open.

6.3 Three-Minute Setup Period

- After the end of the second and fourth rounds of a BO5 match, both teams have 10 minutes to debug their robots. When 10 minutes run out, the Three-Minute Setup Period of the next round begins.
- When participating teams arrive at the Competition Area, they may collect all the required 42 mm Luminous Projectiles from the designated location. For information on the projectile quantity limit, see “4.6.2 Projectile”. For example, under the BO3 Match Format, a participating team may claim up to 300 42 mm Luminous Projectiles at once.
- After a match has ended and the Radar has been removed from the Radar Foundation, participating teams may place the Radar on the Radar Foundation in preparation for the next match.

During the Three-Minute Setup Period, the Pit Crew will place robots in their respective initial zones, check whether the Referee System is operating normally, pre-load their robots with projectiles, load Darts into the Dart System, mount the Radar on the Radar Foundation, and connect the Drone Safety Rope to the Drone. The Pit Crew may repair robots or replace equivalent parts, provided the requirements of the specifications documents are met. The Pit Crew are required to commission their robots near their team’s Initial Zone. Only one Pit Crew member from each team is allowed to leave the zone to commission their robots, and they must not cross the Central Elevated Ground. The initial zones are shown in the figure below:



[1] Red team Landing Pad and its initial zone [2] Red team Base and its initial zone

[3] Blue team Landing Pad and its initial zone [4] Blue team Base and its initial zone

Figure 6-3 The Debug Area during the Setup Period

- Equivalent Part: Standard modules or components having the same material, form and functions, for example, motors of the same model and self-built friction wheel modules. The main structure supporting the robot's body, such as a robot's chassis or the body of a Drone, cannot be replaced as equivalent parts.
- If the equivalent part includes the Referee System Module, the participating team must ensure that the Referee System Onboard Modules work properly, including version number, sensor calibration, and Mounting Specifications. When the Referee System Robot Onboard Modules in the equivalent part experience an abnormality, all consequences arising therefrom will be borne by the participating team and it will not trigger an Official Technical Timeout.

When there are 1 minute and 30 seconds remaining in the Three-Minute Setup Period, it is recommended that the Operator enter the Operation Room to complete the debug of the keyboard and mouse (both of which may be brought in). Any USB keyboard or mouse connected to the computer in the Operation Room must be produced by a certified manufacturer and comply with the USB HID protocol specifications. Check and confirm that robots are operating properly and that official equipment is functioning normally. If any official equipment in the Operation Room cannot operate normally, the Pit Crew must raise the issue before the final 15 seconds of the Three-Minute Setup Period. Otherwise, the referee will not announce technical timeout. During the Three-Minute Setup Period, only one person, in addition to the operators and Gimbal Operators for the deployed robots, is allowed to enter the Operation Room to make tactical deployments. Personnel other than the operators and Gimbal Operators for the deployed robots must leave the Operation Room before the Three-Minute Setup Period ends.

When there are 30 seconds remaining in the Three-Minute Setup Period, all robots on the Battlefield must be powered on and personnel must leave the Battlefield in an orderly manner; any robots not powered on when 15 seconds remain must be removed from the Battlefield. After the Three-Minute Setup Period ends, the Pit Crew must place the Sentry and Radar Remote Controllers in the designated area at the Battlefield entrance or the Operation Room entrance.

6.3.1 Official Technical Timeout

During the Three-Minute Setup Period, if the Referee System, official equipment, or other modules related to the Referee System experience any faults, or a robot needs to be inspected urgently (see below for details), the Head Referee may announce an Official Technical Timeout and pause the setup countdown. The starting time of the Timeout is decided by the Head Referee based on the situation.

During an Official Technical Timeout, Pit Crew members cannot fix the malfunctions of the Referee System or other official equipment unless the fix is required by the referee. If a replacement is needed for a malfunctioning Referee System, replace the Referee System Onboard Modules within the time limit specified in the RoboMaster 2026 University Series Robot Specification Manual. When the relevant malfunctions of the Referee System or official equipment have been eliminated and the Head Referee has resumed the countdown, Pit Crew members are required to follow the set procedures for the Three-Minute Setup Period and leave the Battlefield within the specified time.

Table 6-1 Malfunctions

Regulation	Description
1	A fault occurs with the official equipment in the Operation Room, and any key competition component in the Battlefield experiences structural damage or functional irregularity.
2	During the Three-Minute Setup Period of the first round, a Referee System Onboard Module experiences faults, such as damage to the Armor Module, Speed Monitor Module disconnection, etc.
3	During the Three-Minute Setup Period, the main controller of the Referee System is unable to connect to the server or a robot cannot transmit images to the Operation Room.

Regulation	Description
4	Other situations where the Head Referee deems it necessary to call an Official Technical Timeout.

As it is impossible to determine whether the malfunction stems from the Referee System module itself, flaws in the robot's electrical or structural design, or damage caused during previous matches, an Official Technical Timeout will not be called for regular battle damage, except for the situations mentioned in Rule 3 above. The referee will provide backup Referee System modules. Participating teams may request a Team Technical Timeout to repair their robots.

If the referee determines that the malfunction referred to in Rule 2 and 3 above is caused by the team, the referee will explain the situation and end the Official Technical Timeout.

6.3.2 Team Technical Timeout

If the mechanical structure of a robot, a software system, the keyboard or mouse in the Operation Room or other equipment experiences any faults, the Pit Crew may make a request to the referee in the Battlefield or Operation Room for "Team Technical Timeout" only before the 15-second countdown in the Three-minute Setup Period, where they should also provide the reasons for the request. Team Technical Timeout once requested and conveyed to the Head Referee, this Timeout cannot be canceled or revised.

After the Team Technical Timeout is confirmed by the Head Referee, the Referee will notify both teams at the same time regardless of which team initiated the timeout. During the Team Technical Timeout, both teams may continue to debug and repair their robots as usual.

The Head Referee may end the Technical Timeout once they determine that the teams are ready. Even if the participating team does not enter the Battlefield or ends the Technical Timeout early, the opportunity consumed is still the opportunity corresponding to the time declared by the participating team when applying for the timeout.

To ensure that subsequent matches begin on time, only one Team Technical Timeout is allowed in each Three-Minute Setup Period on a first-come-first-served basis. The Technical Timeout usage is recorded in the Match Results Confirmation Form.

The Team Technical Timeout arrangements for different phases in a competition stage are as follows:

Table 6-2 Team Technical Timeout Arrangements

Competition Stage	Phase	Arrangement
Regional Competition	Group Stage	2 Technical Timeouts for 2 minutes each

Competition Stage	Phase	Arrangement
Regional Competition	Knockout Stage	1 Technical Timeout for 3 minutes; Technical Timeout opportunities not used in the Group Stage can be carried over to the Knockout Stage.

6.4 15-Second Referee System Initialization Period

After the Three-Minute Setup Period ends, the competition enters a 15-Second Referee System Initialization Period. During the Initialization Period, the Referee System server automatically detects the connection status of the Referee System player's client, the status of Referee System Modules, the status of Battlefield Components, and reset the HP of all robots to ensure their HP is full when the match officially begins.

6.5 Five-Second Countdown

After the 15-Second Referee System Initialization Period, the match enters a Five-Second Countdown. At this time, the Referee System player's client will not respond to control commands from robots (including Custom Controllers and Custom Client). Once the countdown ends, the Referee System player's client will resume responding to control commands from robots, and the competition starts.

6.6 Seven-Minute Round

During the Seven-Minute Round, robots from both teams will engage in a shooting battle on the Battlefield.

6.7 End of Round

When the time of one round of competition runs out or one team triggers the winning criteria in advance, this round of competition ends. See "5.8 Match Format and Winning Criteria". The match is over when a winner has emerged or all rounds have ended.

6.8 Result Confirmation

During a match, the referee will record on the Match Results Confirmation Form the penalties issued for each round, the key competition data at the end of the match, the winning teams, the use of Technical Timeout opportunities by the teams, and other relevant details.

Within five minutes after the end of a match, the Captains of both teams must sign and confirm the match results. If a team Captain does not sign and confirm the results within 5 minutes or has not requested an appeal, it is deemed that the team agrees with the match results.

6.9 Exit the Field

After a match is over, members from both teams must power off all their robots, remove them from the Battlefield, and proceed to the Projectile Unloading Area to unload their projectiles. At the Projectile Unloading Area, teams must follow the instructions from the staff and return all armbands and pass cards, empty the projectiles in their robots, and return all projectiles used in the competition.

7. Violations and Penalties

In order to ensure the fairness of the competition and maintain competition discipline, the participating teams, participants, and participant robots must strictly follow the competition rules. If there is a violation, the referee will issue a corresponding penalty. Some violation penalties issued before the official start of the competition will be enforced after the official start of the competition. Serious violations and all appeals in the competition will be publicized.

Penalty for violation stated in this section will be determined by the Head Referee according to the actual situation. If an event that affects competition fairness happens during the competition but it does not involve penalty rules or serious violations, the Head Referee will make a decision based on the actual situation.

If a team's actions have directly caused the other team to commit a violation, the other team is not penalized but must cease its violating behavior immediately.

7.1 Penalty System

7.1.1 Forms of Penalties

During the competition, the Referee System or referees will issue penalties against participants and robots that violate competition rules. The forms of penalties are as follows:

Table 7-1 Forms of Penalties

Form of Penalty	Description
Automatic penalties by the Referee System	All HP losses in "5.1HP Loss and Penalty for Exceeding Limit", except those caused by attacks or collision and the penalty damage, are automatic penalties by the Referee System.
Manual penalties via the Referee System	Penalties issued by the referee using the server against robots for violation of rules.
Manual penalties by human referees	Where penalties cannot be issued via the Referee System, for example, issuing a verbal warning or disqualifying a team.

7.1.2 Types of Penalties

The types of manual penalties are as shown below:

Table 7-2 Types of Penalties

Type of Penalty	Description
Verbal Warning	Verbal Alert
Yellow Card	● One team receives a Yellow Card: ➢ If the offending robot is a Sentry, its chassis is powered off for 2 seconds while the operating interface of other alive robots is blocked for 2 seconds. ➢ If the offending robot is not a Sentry, the operating interface for the offending robot is blocked for 5 seconds and those for other robots are blocked for 2 seconds. ➢ The Referee System will automatically deduct the offending robot's HP by 15% of its current Maximum HP, while the remaining alive robots will have their HP deducted by 5% of their current Maximum HP. If the robot receives a Yellow Card again within 30 seconds after it receives a Yellow Card, the percentage of their current Maximum HP lost will be twice that of the previous loss for that robot, and 5% for the other alive robots on the team. ➢ In each round, an Engineer that has received two Yellow Cards will also receive a Red Card. For all other robots, a Red Card will be issued when a total of three Yellow Cards have been received. ● Both teams receive a Yellow Card: The interface of all operators is blocked for two seconds and the HP of all robots loses by 5% of their Maximum HP, without taking into account the cumulative number of Yellow Cards received. ● If a robot receives successive Yellow Cards, the time that the operation interface is blocked will add up accordingly. The 30-second period will be reset. ● If a robot's remaining HP is less than or equal to that needs to be deducted from penalty, this robot's HP reduces to 1. ● The duration of chassis power off, blockage of the operation interface, and other such situations caused by violation penalties is independent from the Competition Mechanism and will not be combined.
Red Card (Ejection)	● Ejection of robots:

Type of Penalty	Description
	➢ If a robot is ejected before entering the 15-Second Referee System Initialization Period, the offending robot will not be allowed to enter and must be removed from the Battlefield, nor can it be replaced by other robots in any round of the current match. ➢ If a robot is ejected during the 15-Second Referee System Initialization Period, the Red Card should be issued after the competition starts. ➢ If a Drone is ejected during the competition, its Launching Mechanism will be powered off, image transmission data link will be disconnected, the Pilot cannot start the Drone's propellers, and the Gimbal Operator cannot call for air support; if the Drone is flying, the operator must immediately land it onto the designated location. ➢ If the Dart System is ejected during the competition, the Dart launching button will be hidden from view, and the gate of the Dart Launching Station can no longer be opened; if the gate is already open, it will close immediately. ➢ If Radar is ejected during a match, its Referee System Serial Port will be disconnected, and its laser launch device and actuator will be powered off. ➢ If a robot other than a Drone, Dart, and a Radar is ejected during the competition, the robot's HP will become zero and the transmitted images will become monochrome. ● Pit Crew members ejected by a referee must leave the Competition Area immediately and cannot be replaced by other Pit Crew members for all rounds in the current match. If an operator is ejected, all robots controlled by them must also be ejected for the current round and will not be allowed to join the Battlefield nor can they be replaced by other robots for all rounds in the current match. If a Gimbal Operator, Drone, and Dart System are all ejected, they will no longer be allowed to compete, nor can they be replaced by other robots in all rounds of the current match.
Forfeiture	● If a Forfeiture is issued for a round (hereinafter referred to as "Round Forfeiture"), the following rules will apply. ➢ If a Forfeiture is issued before the Seven-Minute Round, including the Three-Minute Setup Period and 15-Second Referee System Initialization Period, the HP of the offending team's Base will be zero, and the HP of the team's other robots will remain full. The HP of the other team's Base and robots will be full.

Type of Penalty	Description
	➢ If a Forfeiture is issued during the Seven-Minute Round, the round will end immediately. The offending team's Base and other robots maintain their HP level at the end of the round. The HP of the other team's Base and robots will remain at the level when the round ended. ➢ If a Forfeiture is issued after the Seven-Minute Round, the HP of the offending team's Base will become zero. The team's other robots will maintain their HP level at the end of the round. The HP of the other team's Base and robots will remain at the level when the round ends. ● If a Forfeiture is issued in a match (hereinafter referred to as "Match Forfeiture"), it applies to all rounds in the match, and the HP for each round should be calculated according to the above descriptions.
Exclusion from Awards	● Participants are excluded from awards. ● Participating teams are excluded from awards.
Disqualification from the Competition	● Participants are disqualified from the competition and excluded from awards. ● The team is disqualified from the competition and excluded from the awards, but its results so far in this season will be maintained as a basis for other teams' advancement.

7.2 Penalty Rules

This section outlines the Penalty Rules for violations. The rules numbered R# clearly specify the regulations that teams, participants, and robots must follow.

7.2.1 Personnel

7.2.1.1 General Specifications

R1 Participating teams must comply with the requirements outlined in the RoboMaster 2026 University Championship Participant Manual.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R2 Participants and their actions must not interfere with the normal operation of the Official Equipment, competition processes, or the regular work of the RMOC staff.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R3 Participating teams are not allowed to set up wireless devices or use walkie-talkies in any competition-related areas, including but not limited to the Preparation Area, Inspection Area, Staging Area, and Competition Area.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R4 Participating teams must not damage any official equipment, including but not limited to equipment in the Competition Area, Staging Area, Preparation Area, and Inspection Area.

The highest penalty that can be imposed on the offending team is disqualification and the offender must compensate as per the price.

R5 Apart from Pit Crew who have entered the Staging Area and Competition Area due to match-related reasons, no participants are allowed to enter either area without special reasons.

Penalty: The highest penalty that can be imposed on the offender is disqualification from the competition.

R6 Any Pit Crew member who has entered the Staging Area and Competition Area for competition may not leave either area or be replaced by another Pit Crew member without the permission of the referee.

Penalty: Offenders are not allowed to enter the Staging Area and Competition Area. The highest penalty that can be imposed on the offender is disqualification from the competition.

R7 Except for the projectiles preloaded in the Inspection Area, participating teams are not allowed to bring any projectiles intended for use in the competition into the Staging Area or Competition Area.

Penalty: Projectiles will be confiscated. The highest penalty that can be imposed on the offender is disqualification from the competition.

R8 After a match is over, the Pit Crew must power off all their robots, remove them from the Competition Area and empty all projectiles inside the robots at the projectile unloading area.

Penalty: The offending robot will be detained in the Projectile Unloading Area, until its projectiles are cleared.

R9 After a match ends, Pit Crew must return all projectiles used in the competition to the Projectile Unloading Area.

Penalty: Confiscation of projectiles and disqualification of the offender from subsequent matches in the current division. The highest penalty that can be imposed on the offender is disqualification from the competition.

R10 Except for emergencies, teams must be present at the Inspection Area at least 55 minutes before the start of each match for Pre-Match Inspection. Teams must stand by at the Staging Area 10 minutes before each match.

Penalty: The highest penalty is an immediate Round Forfeiture.

R11 Without the permission of the referee, a team may not power on and commission or repair a robot in areas outside the Battlefield from the time the robot passes an Inspection to the end of a match.

Penalty: The highest penalty is an immediate Round Forfeiture.

R12 The identities and number of personnel of each team entering designated areas such as the Preparation, Inspection, Staging, and Competition Areas must meet the relevant requirements.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R13 Pit Crews must wear armbands which must not be covered. One member must wear the “Captain” armband.

Penalty: The highest penalty that can be imposed on the offender is disqualification from the competition.

R14 Without permission from the referee, Pit Crew entering the Competition Area must not communicate with anyone from the outside.

Penalty: The highest penalty that can be imposed on the offender is disqualification from the competition.

R15 Except for the Radar and wireless charging devices placed on the Battlefield, Pit Crew are not allowed to use official equipment power supply to supply power to personal devices within the Competition Area. However, they may carry their own power supply.

Penalty: The highest penalty that can be imposed on the offender is disqualification from the competition.

R16 Except for special circumstances, Pit Crew are prohibited from wearing slippers into the Competition Area.

Penalty: The offender may receive a Red Card as the highest penalty.

R17 The team that makes an appeal must provide valid evidence or the specifications or rules in the official documents they are citing.

Penalty: The highest penalty that can be imposed on the offender is disqualification from the competition.

7.2.1.2 Battlefield Regulations

R18 Pit Crew must wear safety goggles when inside the Battlefield.

Penalty: A Verbal Warning will be issued, and the offender will be removed from the Battlefield. If the warning is ineffective, the offender will be issued a Red Card.

R19 During an Official Technical Timeout, Pit Crew members are not allowed to fix faults other than those in modules related to the Referee System.

Penalty: Verbal Warning. If the warning is ineffective, the offender will be issued a Red Card.

R20 After the end of the Three-Minute Setup Period, Pit Crew must return to the designated area outside the Battlefield. During the competition, Pit Crew are not allowed to leave the area without the permission from the referee.

Penalty: Verbal Warning. If the warning is ineffective, the offender will be issued a Red Card.

R21 After the Three-Minute Setup Period, the remote controllers for the Sentry and Radar must be placed in the designated area at the Battlefield entrance or the Operation Room entrance. Remote controllers for other robots must be placed inside the Operation Room or at the designated location at the Battlefield entrance.

Penalty: If the incident occurs before the Seven-Minute Round, a Verbal Warning will be issued. If the warning is ineffective, a Red Card will be issued to the offending robot. If the incident occurs during the Seven-Minute Round, a Red Card will be issued immediately to the offending robot.

R22 After the Five-Second Countdown, the Pit Crew must not operate remote controllers located outside the Operation Room that correspond to deployed robots.

The offending robot will be issued a Red Card, with the highest penalty being a Round Forfeiture.

R23 Pit crew members must ensure their robots are operating safely and will not cause harm to any person or equipment in the Competition Area.

Penalty: The offending team must bear the corresponding responsibility.

7.2.1.3 Operation Room Regulations

R24 During the Three-Minute Setup Period, only one person, in addition to the operators and Gimbal Operators for the deployed robots, is allowed to enter the Operation Room. Personnel other than the operators mentioned above must leave the Operation Room before the Three-Minute Setup Period ends.

Penalty: Verbal Warning. If the warning is ineffective, the offender will be issued a Red Card.

R25 Operators must remain in their respective Operation Room to operate the relevant robot control equipment and wear the required headset during the 15-Second Referee System Initialization Period and the Seven-Minute Round, unless otherwise permitted by the referee. After the Three-Minute Setup Period ends, operators may not change their location.

Penalty: Verbal Warning. If the warning is ineffective, the offender will be issued a Red Card.

R26 The operator does not arrive at the Operation Room before the five-second countdown or leaves the Operation Room during the Seven-Minute Round.

Penalty: The offender will receive a Red Card.

R27 During the Seven-Minute Round, Gimbal Operators may use gimbal remote controllers or custom controllers for Drones, and remote controllers for Darts at the same time, while a Pilot may use only one remote controller. Besides, each operator must configure a control device and custom client according to the requirements for the robot’s control method.

Penalty: Verbal Warning. If the warning is ineffective, the offender will be issued a Red Card.

R28 The use of personal headphones or computers is prohibited in the Operation Room, except for custom clients.

Penalty: Verbal Warning. If the warning is ineffective, the team will be issued a Round Forfeiture.

R29 The Operation Room provides three USB Type-A receptacle ports, and expansion is prohibited. USB keyboards and mice that are produced by unqualified manufacturers, modified, homemade, or do not comply with the USB HID protocol are not allowed to be connected to the Operation Room computer.

Penalty: Verbal Warning and prohibition of using the violating device. If the warning is ineffective, the offender will be issued a Red Card.

R30 During the competition, communication with those outside the Operation Room is forbidden.

Penalty: Verbal Warning. If the warning is ineffective, the team will be issued a Round Forfeiture.

R31 During the match, Pilots operating Drones must obtain Pilot qualification and wear a Pilot wristband.

Penalty: The offender will receive a Red Card. The highest penalty will be disqualification the offending team from the competition.

7.2.2 Robots

7.2.2.1 General Specifications

R32 Robots and related equipment to be deployed in a match must pass the Pre-Match Inspection.

Penalty: Round Forfeiture.

R33 In the first round of a match, the robots must meet the minimum battle team lineup.

Penalty: Match Forfeiture.

R34 Robots must meet the requirements in the RoboMaster 2026 University Series Robot Specification Manual.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

- The RMOC will conduct random checks on robots.
- Any report made against a robot for not complying with the robot building specifications manual must be supported by the relevant evidence.
- If the UWB Positioning Module, Speed Monitor Module, or Armor Module is not installed according to requirements during the match, the corresponding module will be considered offline.

R35 In the event of a dispute, teams are obligated to show their robot’s mechanisms, circuit design drawings and relevant code documents to the RMOC and answer relevant technical questions.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R36 Before the 15-Second Referee System Initialization Period, armor stickers that meet the requirements in the specifications documents must be attached to robots.

Penalty: Verbal Warning. If the warning is ineffective, the offending robot will be issued a Red Card.

R37 When waiting in the Staging Area, team members are not allowed to bring robots out of the Staging Area without permission.

Penalty: Verbal Warning. If the warning is ineffective, the offenders and robots will be issued a Red Card, with the highest penalty being disqualification.

R38 Robots must be free of safety issues including but not limited to short circuits, crashing, creating fumes or lighting flames, parts falling to the ground, and gas cylinder explosions. If a safety issue is identified or presents itself, team members must execute the relevant operations in accordance with the referee’s instructions.

Penalty: If it happens before the start of a match, the Pit Crew need to resolve the safety issue as required by the referee. Otherwise, the offending robot will not be allowed to appear on the Battlefield. If it is during the competition, a Verbal Warning will be issued. If the warning is ineffective, the offender or the offending robot will be issued a Red Card. Any incident involving serious safety hazards must be handled by the Head Referee in accordance with 8Irregularities.

R39 Robots are not allowed to fire projectiles out of the Battlefield.

Penalty: Verbal Warning. If the warning is ineffective, the offending robot will be issued a Red Card.

R40 Dart Systems are not allowed to fire Darts out of the Battlefield.

Penalty: The offending robot will be issued a Red Card.

R41 During the Three-Minute Setup Period, the 15-Second Referee System Initialization Period, and the Five-Second Countdown, robots in the Battlefield are not allowed to leave their respective initial zones.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

R42 If any projectile needs to be fired during the Three-Minute Setup Period, it must be launched into the projectile clearance bag.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

R43 During the Three-Minute Setup Period, the replacement modules or parts used on robots must meet the requirements for Equivalent Parts.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

7.2.2.2 Ground Robot

R44 During the competition, no alive robot is allowed to block any of its Armor Modules with its body. No robot is allowed to block more than one Armor Module on any other alive robot in its team. When an Engineer is grabbing or carrying a Mobile Component, any one of its armors is allowed to be blocked by the carried Mobile Component and its relevant body structure, and the Armor Module obstructed can be different each time, but multiple Armor Modules are not allowed to be obstructed at the same time.

- Hero, Infantry, and Sentry robots are not allowed to obstruct their Armor Modules when carrying a Mobile Component.
- If the obstructed Armor Modules are disconnected, the robot will not be deemed in violation of rules.

Warnings are issued against the offending team based on their subjective intention. If the blocking was intentional, a Yellow Card will be issued along with a Verbal Warning. If the warning is ineffective, a Red Card will be issued. If the blocking was passive in nature, the offender will be issued a Yellow Card. If more than one Armor Module is blocked, a Yellow Card and a Verbal Warning will be issued when a violation occurs. If the warning is ineffective, a Red Card will be issued.

7.2.2.3 Drone

R45 The remote controller used by a pilot is solely intended for functions related to the takeoff, landing, calibration, and other controls of the Drone. It must not be used for any other purposes.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

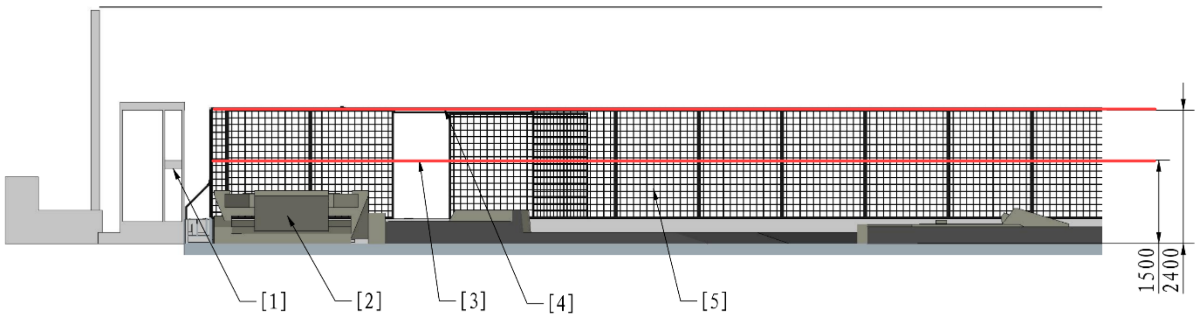
R46 During the Three-Minute Setup Period, Pit Crew members may debug the Drone near the Landing Pad, but must not start the propeller.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

R47 The hook of the Drone Safety Rope must be attached to the rigid ring of the Drone’s vertical rigid safety rod.

Penalty: The offending robot will be issued a Red Card. Drones are not allowed to enter the other rounds in the same match and all matches after this competition stage.

R48 During the competition, the distance between the lowest point of a Drone and the Battlefield ground must not be less than 1.5 m, and no part of the Speed Monitor Module (17mm Projectile) carried by the Drone’s Gimbal Launching Mechanism can exceed the highest point of the Perimeter Wall of the Flight Zone.



[1]	Pilot Operation Zone	[2]	Landing Pad	[3]	Minimum Flight Altitude	[4]	Maximum Height of Perimeter Wall	[5]	Perimeter wall

Figure 7-1 Flight Altitude Restrictions

Penalty: A gesture or Verbal Warning is given to the Pilot, to remind him or her to adjust the flight altitude. If a warning is ineffective, the offending robot will be issued a Red Card and forbidden from entering any rounds in the same match.

R49 During the competition, Drones are forbidden from flying outside the Battlefield.

The offending robot will be issued a Red Card, and the pilot will be disqualified. The Drone is not allowed to enter the other rounds in the same match.

R50 During the competition, a Drone is prohibited from launching projectiles while in contact with the Landing Pad.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

R51 If a Drone experiences technical faults or is damaged due to the unreasonable design of the propulsion system or power supply system during the competition, it must be checked by the referee and must be cleared by the Head Referee as hazard-free before being allowed to return to the match.

Penalty: The offending robot will be issued a Red Card. The offending robot is not allowed to enter the other rounds in the same match.

- If a Drone crashes in the Battlefield, the referee may, depending on the situation, retrieve the robot to the Landing Pad and any consequences arising from the retrieval process must be borne by the participating team. If the robot has suffered severe structural damage, the Drone Safety Rope hook has fallen off, or the competition is about to end, the referee may not retrieve the robot.

- For safety reasons, if a Drone is flying erratically during the competition, the Head Referee will eject the robot, and the Pilot must land the robot and stop its propeller, and is no longer allowed to operate the robot.

7.2.2.4 Other Robots

R52 The Dart System must not remain in a ready-to-launch state other than during the Seven-minute Round.

Ready-to-launch state: The energy storage component used for providing initial kinetic energy for Darts is in a tense, inflated, or rotating state. Energy storage component includes but is not limited to rubber band, cylinder, friction wheel, etc.

Penalty: Verbal Warning. If the Verbal Warning is ineffective, the offending robot will be issued a Red Card.

7.2.3 Interaction

7.2.3.1 Interaction between Robots

R53 A robot may not use any of its body structures to strike an opponent robot in collision. If a defeated robot is blocking a key path, the robot can be slowly pushed away.

- In any collision between a Drone and a Ground Robot while the Drone is flying, the Drone will be deemed the offending robot.
- In any collision between a Drone and a Ground Robot after the Drone has landed on the Battlefield or while it is being retrieved, the Ground Robot will be deemed the offending robot.
- In any collision between two Ground Robots, the offending robot will be the one deemed by the referee to be the initiator.

Penalty: Warnings will be issued against the offending robot based on their subjective intention and the degree of collision.

Table 7-3 Collision Violation Penalty Standards

Violation Level	Description
Yellow Card	● Actively causing frontal and high-speed collision ● Active pushing that causes the other team's robot to move noticeably ● Active pushing that impedes the normal movement of the other team's robot

Violation Level	Description
Red Card	● Actively causing high-speed, repeated, and intense frontal collision ● Active pushing that causes the other team's robot to move for a longer distance ● Active pushing that seriously impedes the normal movement of the other team's robot

If any collision has caused serious consequences to the other team's robot or rendered the competition unfair, in the next round of the same competition stage, the offending robot (identified by its number) will be barred from the Battlefield and cannot be substituted by another robot.

R54 A robot must not get entangled with any other robot (except for robots that cannot be respawned) due to active interference, blocking, or collision.

Penalty: Counting from when an entanglement is determined, warnings should be issued against the offending robot based on the length of the violation. If it exceeds 10 seconds, a first Yellow Card will be issued. Thereafter, each 20 seconds will incur a further Yellow Card. This should carry on until the offending robot is ejected. Regardless of whether the offending robot is alive, if the violation lasts longer than 90 seconds, the offending team may be issued a Round Forfeiture, which is the highest penalty, based on their subjective intention.

R55 No robot may attack the Drone, Dart Launcher, Radar, or wireless charging devices.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R56 No robot and its actions are allowed to block an opponent robot's entry into its Resupply Zone.

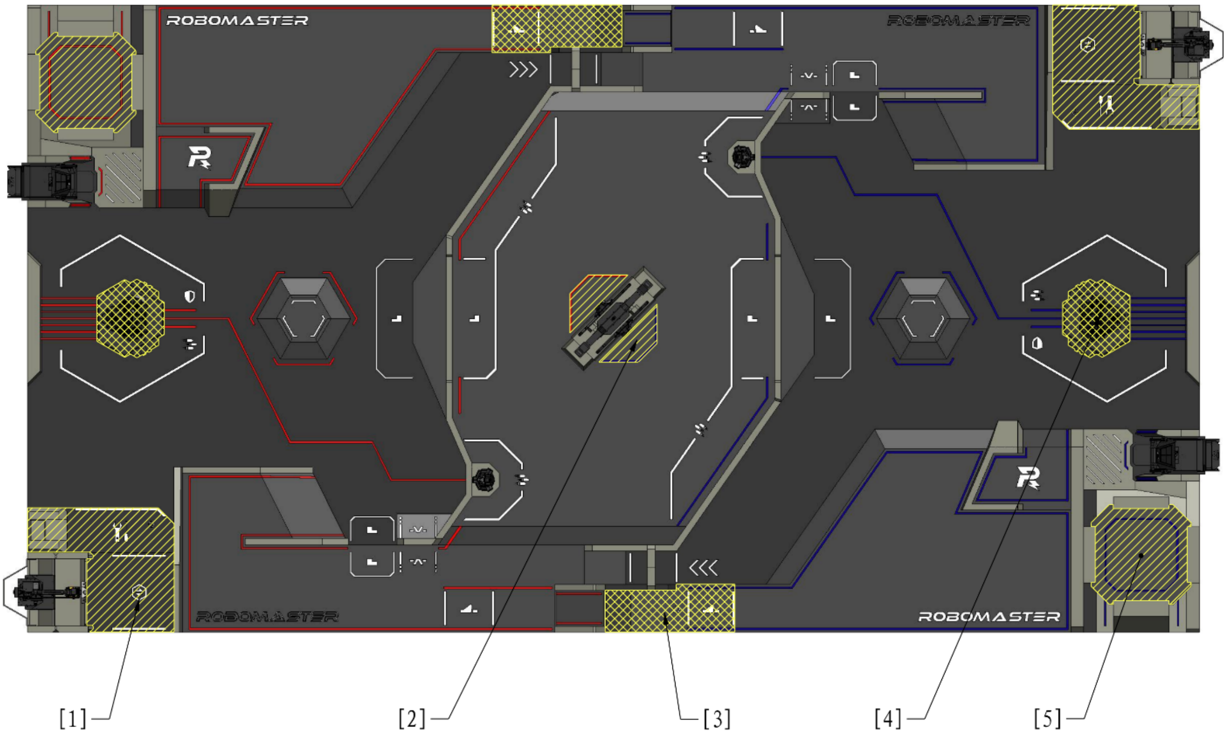
Penalty: The offending robot will be issued a Yellow Card. If the warning is ineffective, the offending robot will be issued a Red Card.

R57 Robots are not allowed to intentionally carry opponent Darts.

Penalty: Verbal Warning. If the Verbal Warning is ineffective, the offending robot will be issued a Red Card.

7.2.3.2 Interaction between Robots and Battlefield Components

Multiple Penalty Zones are set in the Battlefield. The entry of a team's robot (including any Battlefield Component carried by the robot) into the range of the Penalty Zone will be deemed as the robot's entry into the Penalty Zone. The Penalty Zone is shown below.



[1]	Resupply Penalty Zone	[2]	Assembly Penalty Zone	[3]	Road Penalty Zone
[4]	Base Penalty Zone	[5]	Landing Pad Penalty Zone		

Figure 7-2 Battlefield Penalty Zone

R58 Ground Robots are forbidden from entering the Base Penalty Zone, Road Penalty Zone, or Landing Pad Penalty Zone.

- If a robot completely traverses the 17° slope from the bottom to the top of the Launch Ramp and then enters the Road Penalty Zone, losing mobility and being unable to leave the Penalty Zone, it will not be considered a violation. In all other cases, it will be regarded as a violation.
- A robot is not deemed in violation of rules if it is unable to leave the Road Penalty Zone from a Terrain Crossing Buff Point (Launch Ramp) after a failed attempt at the Launch Ramp, because the opponent's robot is also on the Buff Point.

Penalty: Warnings will be issued against the offending robot based on how long the robot remained in the Penalty Zone and the impact of the violation. If it exceeds 3 seconds, a Yellow Card is issued. Thereafter, every 10 seconds will incur a further Yellow Card. This carries on until the offending robot is ejected. An offending robot that causes serious damage to an opponent robot by remaining in a Penalty Zone will be issued a Red Card.

R59 Robots are not allowed to enter the Resupply Penalty Zone or the Assembly Penalty Zone.

If a robot is defeated or ejected in any Penalty Zone, the referee may activate the robot temporarily as required and guide the robot's operator in controlling the robot to leave the Penalty Zone.

Penalty: Warnings will be issued against the offending robot based on how long the robot remained in the Penalty Zone and the impact of the violation. If it exceeds 3 seconds, the first Yellow Card will be issued. Thereafter, every 5 seconds will incur a further Yellow Card. This will carry on until the offending robot is ejected. A defeated or abnormally disconnected robot that stays in a Penalty Zone longer than 20 seconds may be imposed a Round Forfeiture as the highest penalty.

R60 Robots are not allowed to bring Mobile Components into both teams' Road and Base Penalty Zones, and their team's Resupply Penalty Zone, and the other team's Dart Launching Station.

A Yellow Card will be issued against the offending robot. If any subsequent Mobile Component has a decisive impact on the other team's Launch Ramp, projectile supply, Dart launches, and target hits, or affects the normal operation of any Core Component, the offending robot will be issued a Red Card.

R61 During the competition, robots may only use projectiles supplied by the RMOC.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R62 During the competition, robots are not allowed to damage the Battlefield, Battlefield Components, or projectiles, nor affect the normal function of the Battlefield Components.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R63 Teams are not allowed to hit a Power Rune with 42 mm projectiles or Darts.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

7.3 Serious Violations

The following actions are considered serious violations. The highest penalty the RMOC may impose on an offending team for serious violations is disqualification. In the event of any violation of local laws and regulations, the RMOC will fully cooperate with the relevant authorities to hold the offender legally responsible.

Table 7-4 Types of Serious Violations

Regulation	Type
1.	Malicious destruction of the Battlefield, Battlefield Components, other official equipment, or the robots or equipment of other teams.
2.	Falsification, assumption of a false identity, or any other behavior determined as cheating.

Regulation	Type
3.	Tampering with or damaging the Referee System, or interfering with any detecting function of the Referee System by technical means.
4.	Circumstances that violate the specifications documents and are determined by the Chief Referee as a serious violation.
5.	Disobedience over penalties, refusal to cooperate, deliberate delay, disrupting the competition, forfeiting without valid reasons, boycotting, or other behavior that hinders the competition.
6.	Match throwing or manipulation.
7.	Providing property to others or illegally soliciting or accepting property from others for the purpose of obtaining an unjust competition outcome or improper benefits.
8.	Uncivilized and immoral conduct involving defamation, verbal abuse, rude gestures, malicious heckling, or malicious throwing of objects.
9.	Publishing, spreading, or disseminating to the media false or irresponsible remarks.
10.	Deliberately attacking or colliding with others in a manner that endangers themselves or others.
11.	Carrying hazardous items or contrabands.
12.	Other behavior that violates the spirit of the competition and are deemed a serious violation.
13.	Other conduct that violates core socialist values, sports ethics, public order and norms, the culture and discipline of the competition, laws and regulations, or that causes an adverse impact on society.

8. Irregularities

There will be a certain delay in the referee's manual penalties and handling of abnormal situations. If it has a major impact on the result of the competition, the Chief Referee will determine the final result according to the actual situation.

If any of the following abnormalities occur, it shall be handled according to the corresponding process, to which both teams cannot object. The handling process is as follows:

- When a serious safety hazard or abnormality has occurred, such as: a battery explosion, stadium power outage, explosion of a compressed gas cylinder, or interpersonal conflict, the Head Referee will notify both teams' operators after discovering and confirming the emergency, and eject all robots through the Referee System. The result of the round will be invalidated. The round will restart after the safety hazard or abnormality has been eliminated. The RMOC will give priority to addressing safety issues and will make every effort to protect Robots within reasonable limits. However, it cannot guarantee that Robots will be entirely unaffected, and any loss or impact resulting from safety measures shall be borne by the participating teams.
- If non-key Battlefield Components are damaged during a match (damage to the PVC flooring, the light indicators on the competition area, or the light effects on the Base), which do not affect the fairness of the match, the match will proceed as usual.
- The competition will carry on despite any anomaly with a robot's armor light effects or light indicator effects or any damage to an Armor Module Sticker.
- If functional abnormalities or structural damage occur to key Official Equipment and Battlefield Components during the competition, thus rendering it unfair, for example, network issues causing robots to go offline, failure to trigger the Buff Effect after striking the Power Rune, or Battlefield Components malfunctioning, the referee will manually address such issues using the Referee System. If the failure cannot be dealt with manually, the referee will notify the operators of both sides and eject all robots at the same time. The competition will end immediately, and the result of the competition will be invalid. When problems are solved, there will be a rematch.
- During a match, if key Official Equipment and Battlefield Components experience functional abnormalities or structural damage that affects the fairness of the competition, and the Head Referee did not confirm and end the game in time, leading to a situation where a game that should have ended continues and has a winner, the results for the round shall be invalidated once the Head Referee has made a determination to that effect within 5 minutes after the end of the round, and a rematch shall be held.
- In the case of a serious violation that would clearly have triggered a penalty of forfeiture, and the Head Referee did not confirm and execute it in time, the results for the round shall be invalidated once the Head Referee has

made a determination to that effect within five minutes after the end of the round, and the offending team will be issued a forfeiture.

- During the competition, if any situation has occurred that may affect the fairness of the competition, the Chief Referee will notify the Captains of both teams of the situation and suspend the results confirmation process within five minutes after the end of the match, and will make a determination within 60 minutes and notify both Captains of the final course of action. The handling outcome is final and cannot be challenged by both teams.

9. Appeal

Every team has one appeal opportunity during each of the Regional Competition, Wild Card Competition, and Final Tournaments. Appeal opportunities cannot be used cumulatively across competitions. If an appeal is successful, the team involved retains its right to appeal again in future matches. If it is unsuccessful, the team will have exhausted its one opportunity to appeal. When a team has exhausted its opportunity to appeal, the Arbitration Commission will no longer accept any appeal from the team. After the appeal is accepted, the Arbitration Commission will deliberate on the appeal materials and relevant evidence. On behalf of the Arbitration Commission, the Chief Referee will then communicate the appeal decision to both teams. The Arbitration Commission has the right to final interpretation of the appeal decision.

The following situations do not constitute a basis for appeal:

- Verbal Warnings and Yellow and Red Cards issued as penalties for violations.
- The types and processes of Technical Timeouts initiated.
- Regular battle damage occurred at the Referee System Onboard Module.

No appeal is allowed five minutes after a Match Results Confirmation Form has been signed or a match has ended.

9.1 Appeal Process

Teams filing an appeal need to follow the procedures as shown below:

Figure 9-1 Appeal Process

9.2 Appeal Materials

Each file submitted by participating teams as appeal materials must not exceed 500 MB in size. No more than 10 files can be submitted.

9.3 Appeal Decision

Appeal Decisions include: maintaining the original match results, forfeiture against the respondent, and rematch between both teams. After the Arbitration Commission confirms the appeal decision, neither team is allowed to challenge it.

- Successful appeal: forfeiture against the respondent, rematch between both teams.
- Appeal failed: maintaining the original match results.

If the communicated appeal decision is a rematch, but neither team is willing to accept a rematch, the appeal will be deemed failed and the original match results will be maintained.

- Provided it does not affect the schedule of the entire competition, the rematch will in principle be held on the same day after all the other matches, depending on the actual situation.
- The flow of the rematch will be the same as regular matches. Both teams are required to compete according to the time stipulated by the RMOC and comply with relevant rules.

If the round involved in the appeal decides the final winner of a match, the match winner must be re-decided.

Example: In a BO3 match with a result of 1:2, the Red Team wins the first round, while the Blue Team wins the second and third rounds. After the end of round, the Red Team files an appeal regarding the second round, and the appeal decision is a rematch between both teams. If the Red Team wins the rematch round, the final match result becomes 2:0; if the Blue Team wins the rematch round, the final result remains 1:2.

Appendix I: Pre-Match Procedures

This season does not have a separate Sentry Mapping section.

Mock Inspection

The RMOC will do its best to assist participating teams in checking robot issues during the mock inspection and practice match. However, the inspection results do not represent the official inspection results. If a robot fails the official inspection before the match, it will not be allowed to enter the field.

Unique identification registration is only allowed during the mock inspection. After the mock inspection concludes, only robots with existing unique identification can complete replacement registration. If a team needs to repair or replace components with unique identification, they must apply to the RMOC for re-registration and re-labeling. Each team may submit up to six applications within the same division. Unauthorized replacement of uniquely identified components will disqualify the robot from entering the field.

- Non-emergency registration procedure: If more than two hours remain before a team's next match, or if it is outside RMOC working hours, the team may replace the relevant parts independently. They must submit either detailed before-and-after photos of the robot or a complete video of the replacement process through the Robot Unique Identifier Supplementary Registration portal. Teams must also book an inspection slot via the same link and report to the second Inspection Area at the scheduled time, where an RMOC official will carry out the review. Robots that pass the review will receive a new unique identifier.
- Emergency reporting procedure: If less than two hours remain before a team's next match and it falls within the RMOC's working hours, the team must contact the referee in the second Inspection Area to request urgent repairs. After the RMOC staff confirm the request, they will accompany the team to the Preparation Area and supervise the team's repair process. In this case, robots are not required to have all unique identifiers in order to complete the inspection. However, once the participating team has finished its next match, the robots must be brought to the second Inspection Area for a follow-up review and to assign any missing unique identifiers.

Battlefield Component Training

The battlefield component training area is equipped with components and wooden structures similar to those used in official matches, providing participating teams with a space for testing. A pre-match inspection is not required before battlefield component training. Participating teams must ensure that their robots' referee system are functioning properly. Any time spent addressing referee system issues on-site will be counted as part of the team's training time.

All participating teams are requested to arrive at the Staging Area for battlefield component training at least 20 minutes in advance. If you do not arrive on time, your allotted battlefield component training time will proceed as scheduled and will not be extended.

Each team is allotted two 30-minute training sessions with battlefield components, including time for entering and exiting the field. All teams must strictly adhere to their scheduled time slots and are not permitted to take up training time allocated to other teams. The first battlefield component training session will be scheduled by the RMOC, while the second session can be booked by teams.

A maximum of 12 people from each team may enter the battlefield component training area, and all personnel must wear goggles.

The RMOC will provide only two operator computers for use on the video transmission channel. If you require additional player's clients for more robots, you may bring your own computers, but they are not allowed to connect to the competition network.

The RMOC will provide only mobile components for testing and will not supply 17 mm or 42 mm projectiles. Teams must bring their own projectiles if testing is required. When testing with 42 mm projectiles, teams must not intentionally target the Power Rune.

Due to venue limitations, ambient lighting may affect the training area, and teams should make the necessary adjustments.

If a team fails to follow the instructions of the on-site referee or commits other violations, its component training privileges will be revoked and the team must leave the competition area immediately. Serious violations will result in disqualification from the practice match as well.

Practice Match

The practice match gives teams an opportunity to debug before the official match. Its primary purpose is to help teams become familiar with the match environment and competition process, and to support them in testing and tuning their robots.

Due to the tight schedule of the practice match, which is primarily intended for debugging, matches will generally not be suspended unless a critical issue occurs with official equipment. In the event of a malfunction in the Referee System Onboard Module, an official technical timeout of up to two minutes may be granted. This pause will be included in the practice match duration. Team technical timeouts are not permitted during the practice match.

Robots that fail check-in due to lateness or safety hazards are not permitted to participate in the practice match. Robots that fail check-in due to serious non-compliance with check-in standards may only take part in the free debug phase of the practice match. The number of robots participating in the practice match must not exceed the maximum allowed in the robot line-up.

During the free debug phase, launching projectiles within the match area is prohibited, except when striking the Power Rune. If projectile testing is required, teams may request a clear projectile bag from the referee and launch into the bag. When debugging the Power Rune with projectile launching, teams must inform nearby personnel and follow safety precautions.

During the free debug phase, Drones must not start their propellers, and launching darts is prohibited. During the free debug phase, all debug props must be requested from the referee.

During the free debug phase, the match status is set to “Not Started,” and teams must adapt to the match stage on their own.

Appendix II: Future Rule Planning

In future seasons, the RMOC will place greater emphasis on guiding and encouraging the robot capabilities, scenario adaptability, and technical directions outlined in the table below.

Core Robot System	Subsystem	Capability/Technology	Typical Robots
Perception System	External Sensor	Multi-sensor data fusion on a single platform	All
	Inter-robot Communication	Inter-robot information fusion	All
	Guidance System	Self-motion status acquisition	Dart System
Decision-making System	Computing Platform	Low-power operation	All
	Decision-making Algorithm	● Decision-making capability under minimal or no external influence (especially advanced semantic analysis and robust decision-making in extreme conditions) ● High-efficiency algorithm design with low time complexity under limited computational power	Sentry robots and radar systems
	Guidance System	Target localization and automated recognition	Dart System
Actuation System	Mobility System	● Cross narrow terrain by folding or transforming ● Stably traverse rugged, uneven, and harsh terrain	Ground Robot

	Projectile Launching Mechanism	● New types of launching mechanisms beyond friction wheel systems (such as pneumatic or electromagnetic launching) ● Irregular object recognition and high-precision launch ● Beyond-visual-range strike	Hero, Infantry, and Sentry
	Object Manipulation Mechanism	● Multi-degree-of-freedom motion mechanism ● Multi-degree-of-freedom terminal executive mechanisms	Engineer
	Flight and Power System	● High payload-to-weight ratio ● Folding and transformation for compact transport	Drone
	Guidance Mechanism	● Equipped with propulsion, capable of autonomously altering its trajectory (non-aircraft), with particular emphasis on terminal maneuverability. ● Non-parabolic trajectory design	Dart System
	Modular Mechanism	Designed as a functional module that can operate independently of the overall robot system to perform a specific task, while also allowing rapid integration into the system, for example, an independent projectile launching mechanism.	Ground Robot

	Complete Unit	Low-power operation	All
	Complete Unit	● The ability to make effective decisions in complex environments ● Collaboration between automatic robots and manual robots	Sentry robots and radar systems
Human-Robot Interaction	Complete Unit	● Robots efficiently and accurately convey the information necessary for humans to make decisions. ● Make robots easier to operate, while highlighting the irreplaceable value of human decision-making.	All

E-mail: robomaster@dji.com
Forum: https://bbs.robomaster.com
Website: https://www.robomaster.com
Tel: +86 0755-84357613 (GTC+8, 10:30AM-7:30PM, Monday to Friday)
Address: T2, 22F, DJI Sky City, No. 55 Xianyuan Road, Nanshan District, Shenzhen, China